

# **Big Data**

## **Extra**



**Dinh-Thang Duong – TA**  
**Truong-Binh Duong – STA**

# Outline

- **Introduction**
- **Hadoop**
- **Spark**
- **Question**

# Getting Started

## ◆ Objectives



In this session, we will discuss about:

- Introduction to big data.
- Introduction to Hadoop.
- How to install virtual machine.
- How to install and use Hadoop on virtual machine.
- Introduction to spark and pyspark.

# Introduction



# Introduction

## Getting Started

### 3 Important Statistics About How Much Data Is Created Every Day



#### 1 How much data is generated every minute?

Source: Domo

**41,666,667**

messages shared  
by WhatsApp users

**1,388,889**

video / voice calls made  
by people worldwide

**404,444**

hours of video streamed  
by Netflix users

**347,222**

stories posted by Instagram users

**150,000**

messages shared by Facebook users

**147,000**

photos shared by Facebook users

#### 2 Estimated Data Consumption from 2021 to 2024

Source: IDC / Statista



#### 3 Data Growth in 2021

Sources: TechJury, Internet Live Stats, Cisco, PurpleSec

**2 TRILLION**

searches on Google by the end of 2021

**1.134 TRILLION MB**

volume of data created every day

**3,026,626**

emails sent every second, 67% of which are spam

**278,108 PETABYTES**

global IP data per month by the end of 2021

**230,000**

new malware versions created every day

**82%**

share of video in total global internet  
traffic at the end of 2021

| UNIT           | VALUE           |
|----------------|-----------------|
| bit            | 1 bit           |
| byte (B)       | 8 bits          |
| kilobyte (KB)  | 1024 bytes      |
| megabyte (MB)  | 1024 kilobytes  |
| gigabyte (GB)  | 1024 megabytes  |
| terabyte (TB)  | 1024 gigabytes  |
| petabyte (PB)  | 1024 terabytes  |
| exabyte (EB)   | 1024 petabytes  |
| zettabyte (ZB) | 1024 exabytes   |
| yottabyte (YB) | 1024 zettabytes |

# Introduction

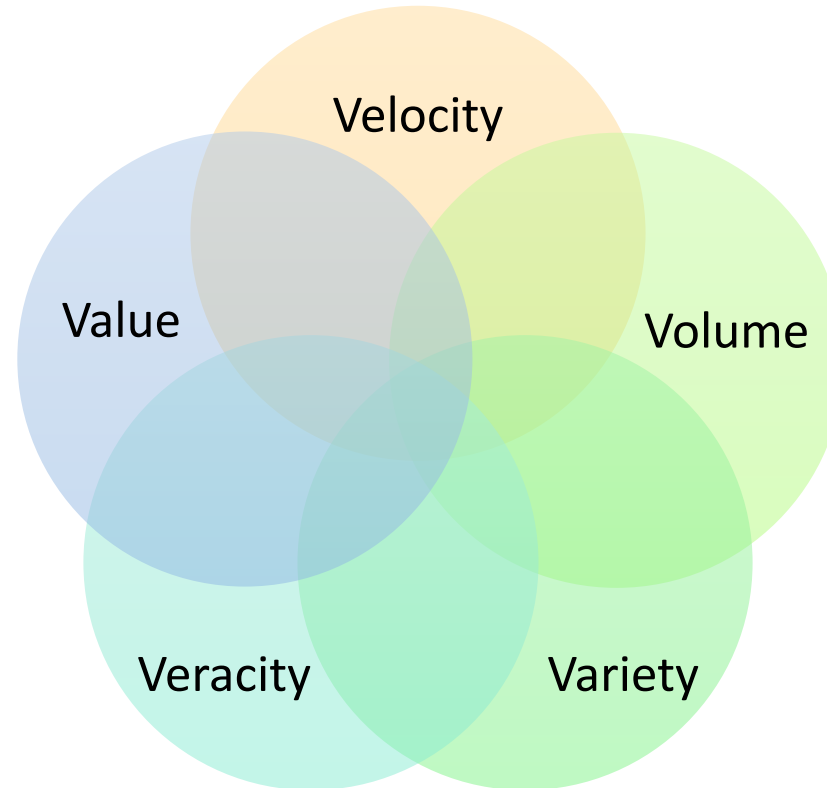
## ❖ Big Data Definition



**Big Data:** A collection of datasets that is large, complex that make it very difficult to process using traditional data processing applications.

# Introduction

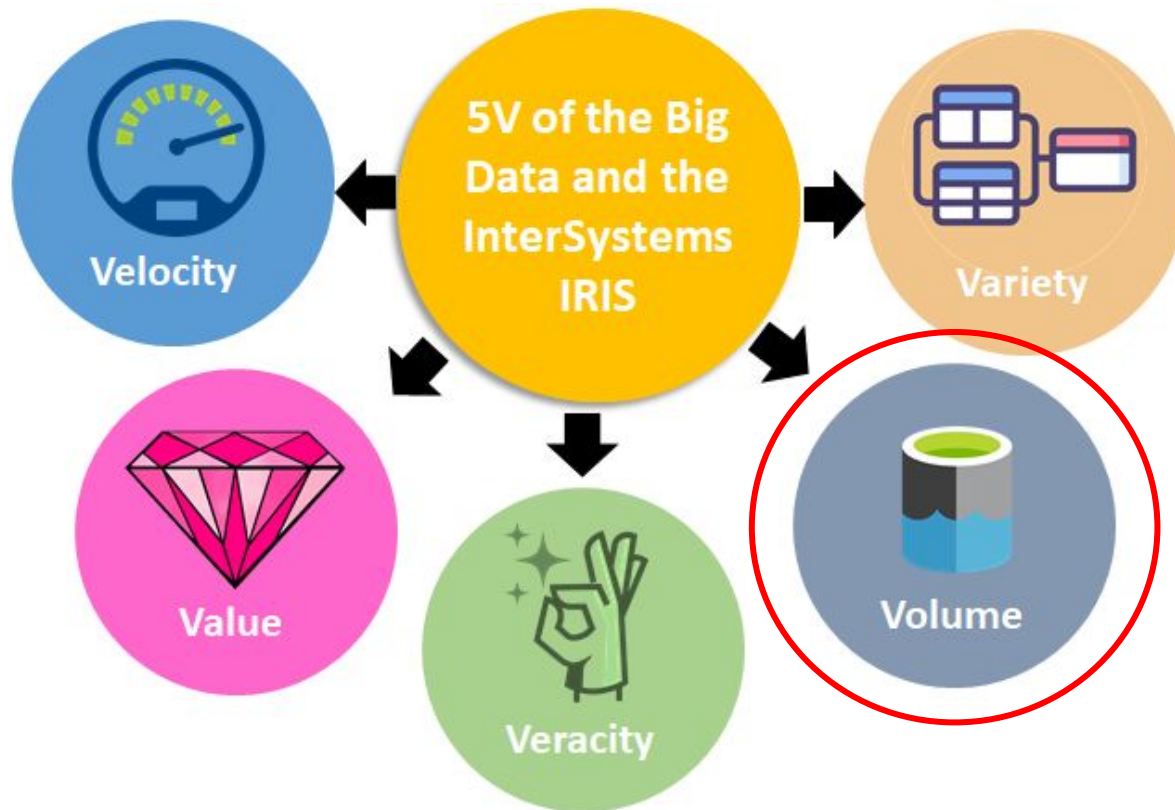
## ❖ Big Data Characteristics



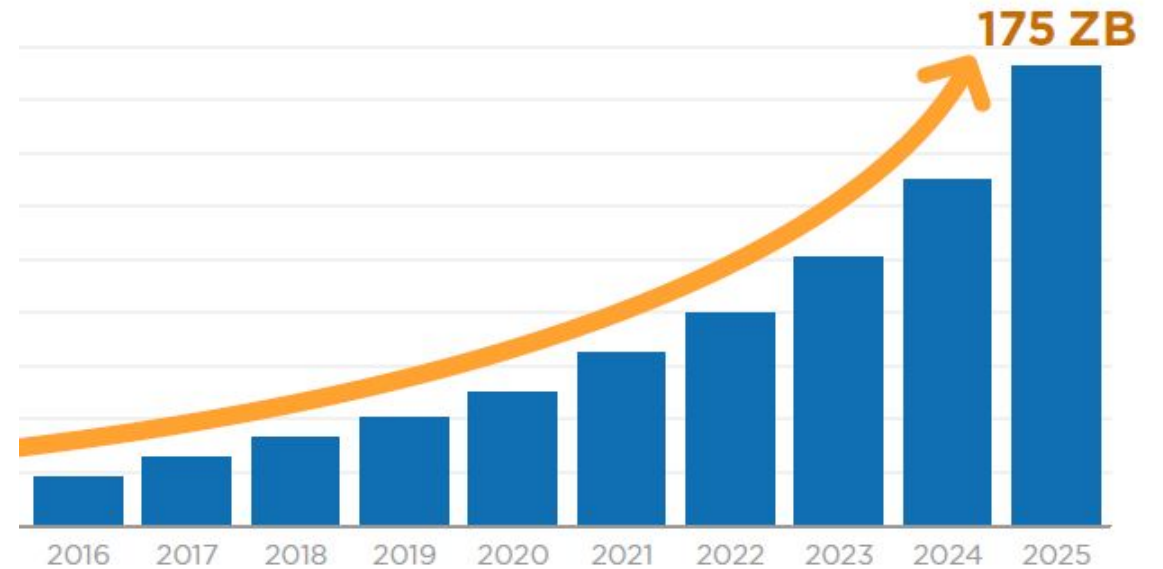
5V of Big Data

# Introduction

## ❖ Big Data Characteristics



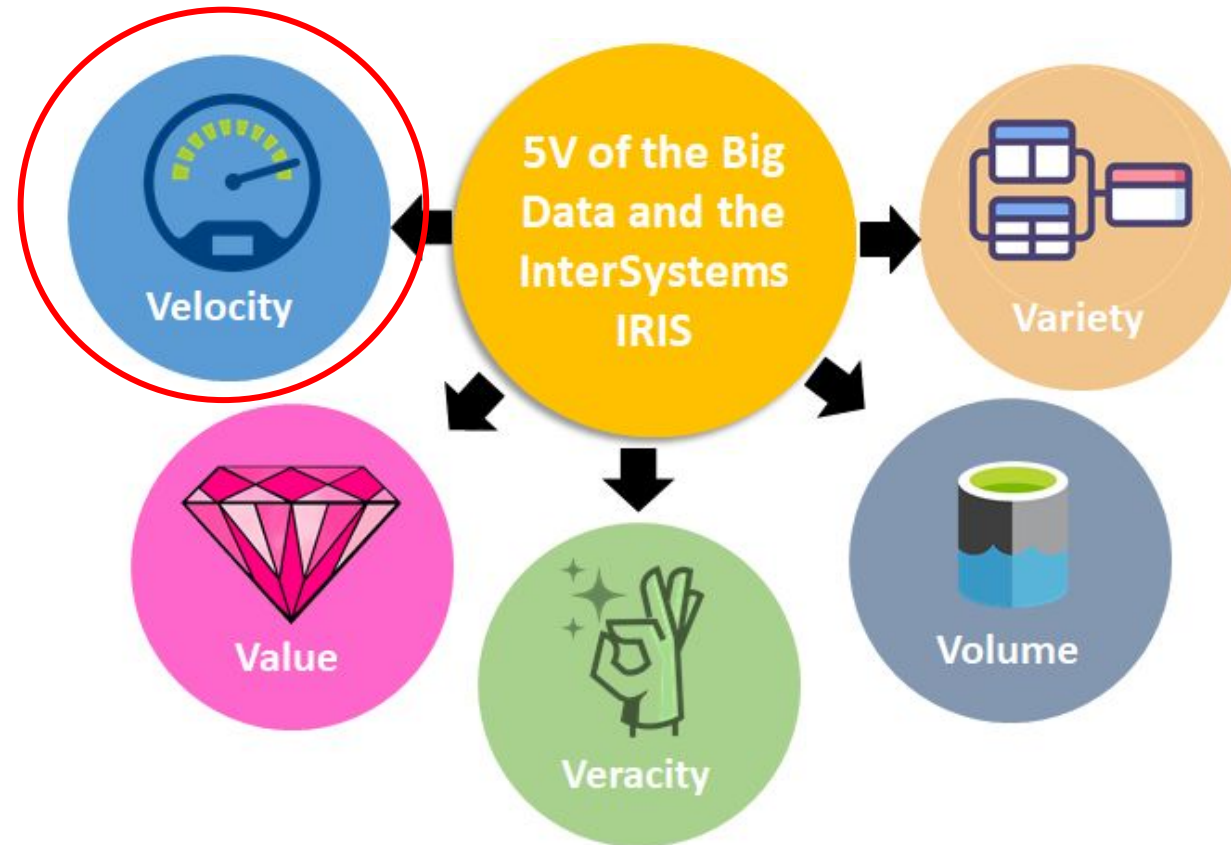
**Volume:** The amount of data



The amount of data generated on Internet each year

# Introduction

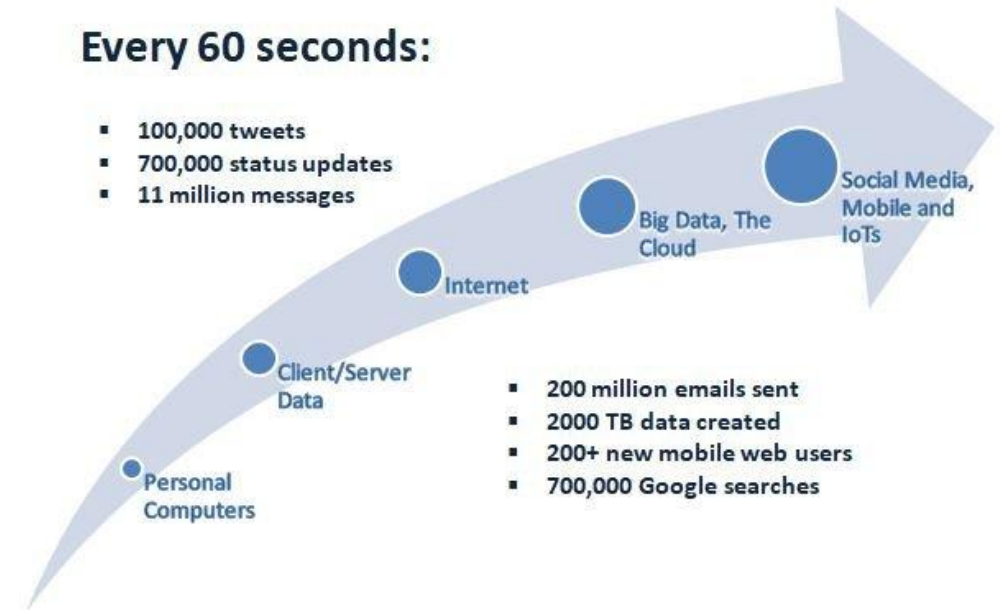
## ❖ Big Data Characteristics



**Velocity:** The growth of data

### Every 60 seconds:

- 100,000 tweets
- 700,000 status updates
- 11 million messages

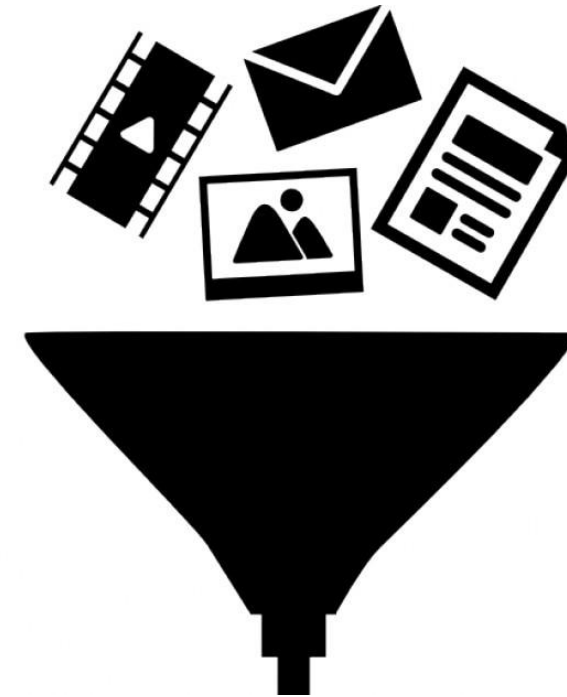
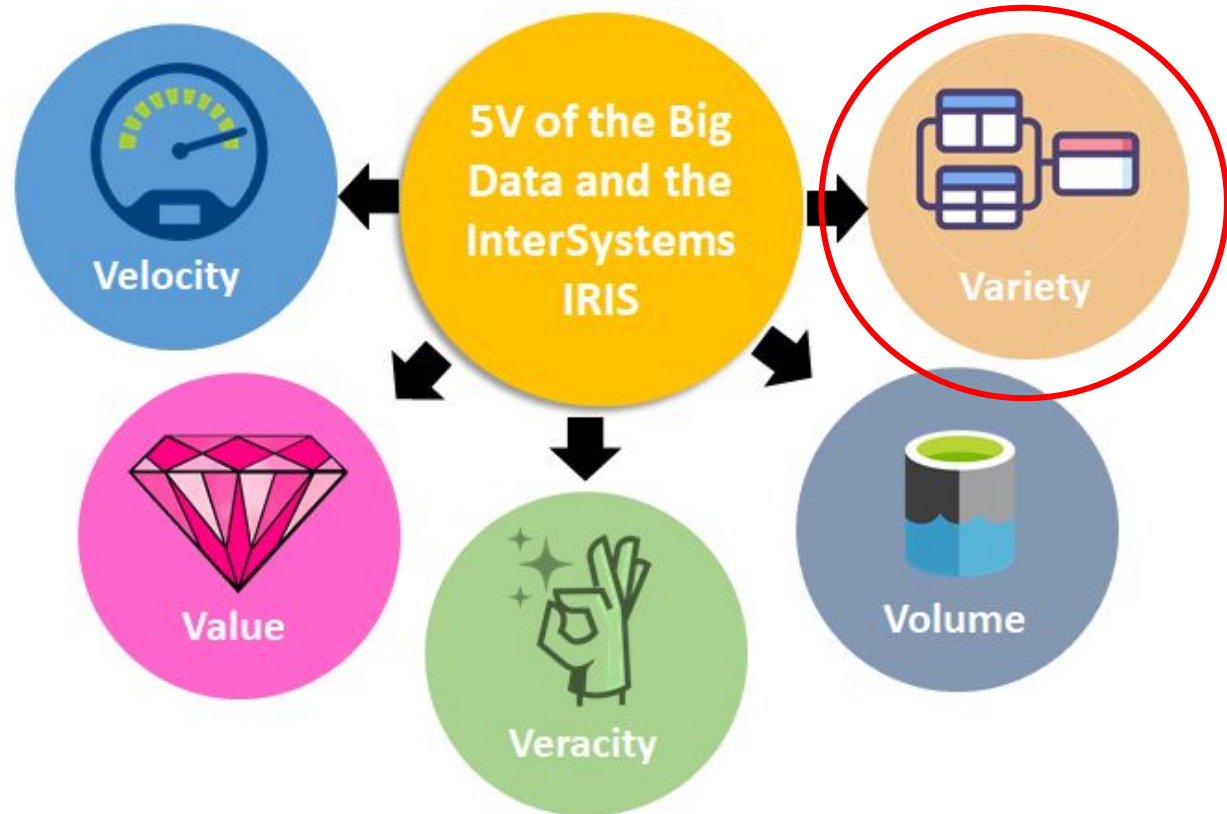


Data generated on Internet every 1 minute



# Introduction

## ❖ Big Data Characteristics

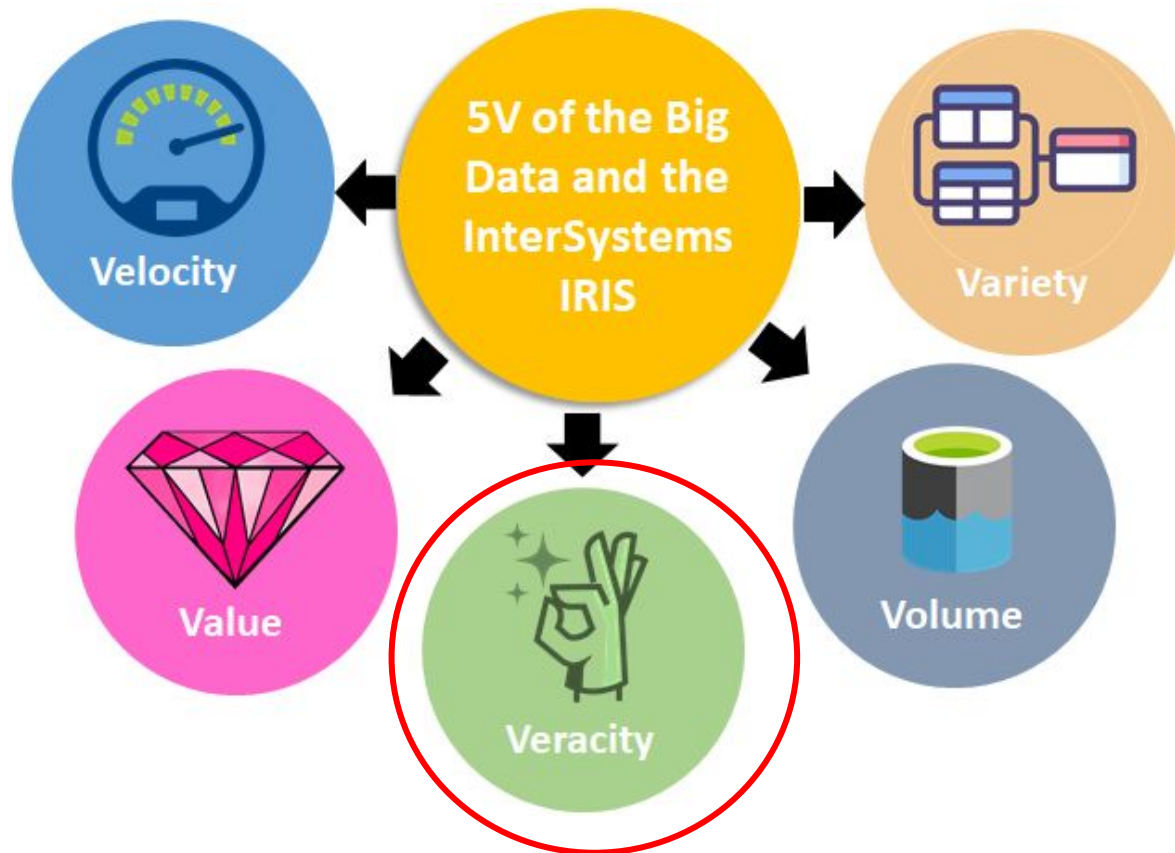


Lots of types of data to be processed

**Variety:** The diversity of data

# Introduction

## ◆ Big Data Characteristics



**Veracity:** The accuracy, reliability, and truthfulness of the data

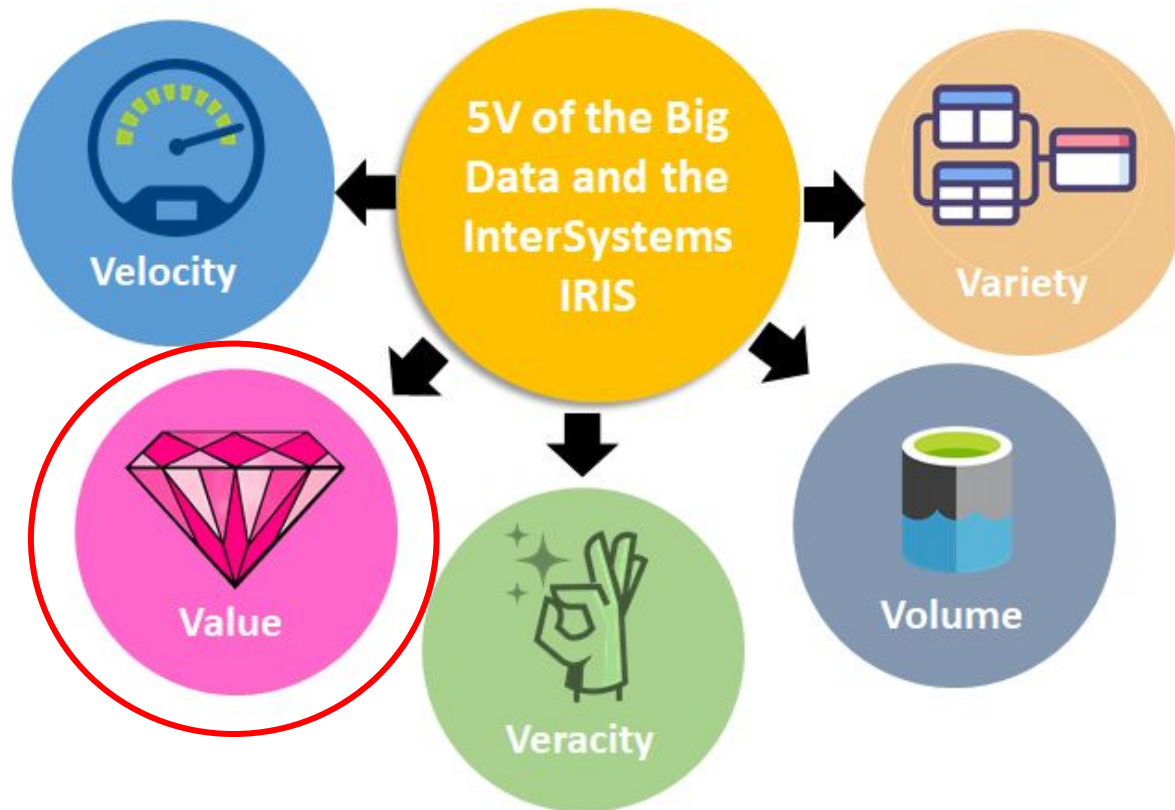
sciforce  
Sources of Data Veracity



It's not just the quality of the data that is important, but how trustworthy the source, the type, and processing of the data are.

# Introduction

## ❖ Big Data Characteristics



**Value:** The beneficial insights gained from data analysis.



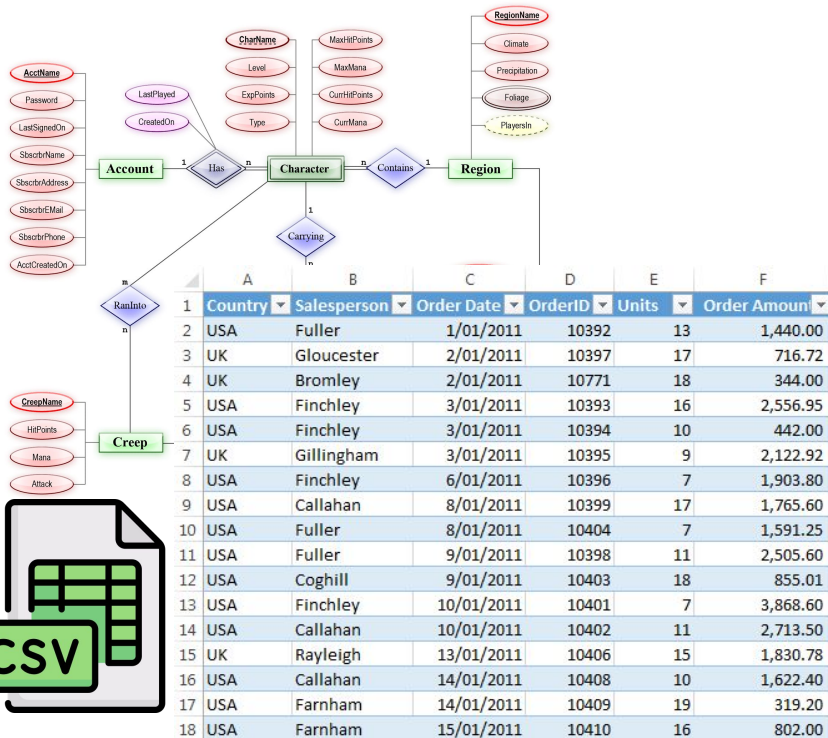
Value in big data is essential for deriving meaningful insights and driving decisions.



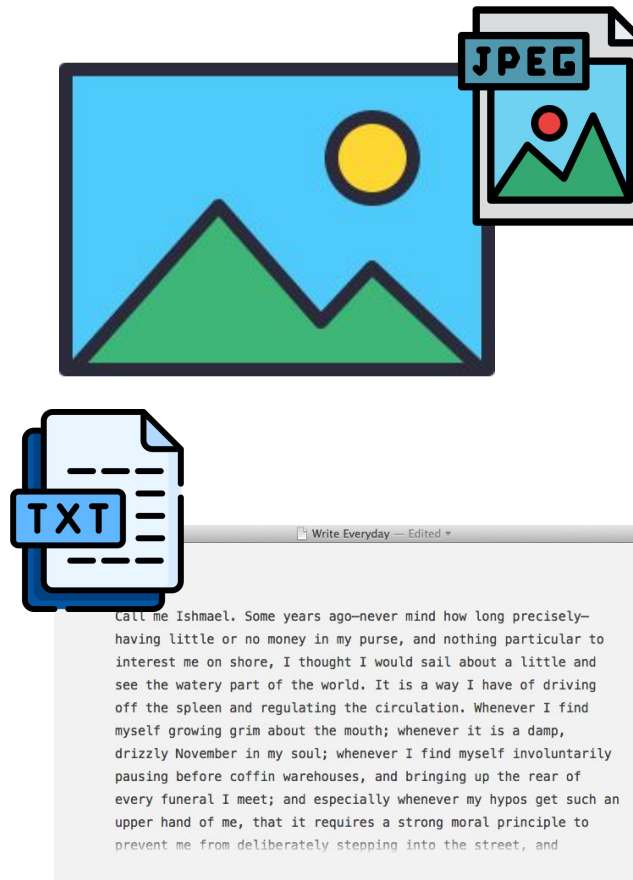
# Introduction

## Types of big data

### Structured Data



### Unstructured Data



### Semi-structured Data

The diagram shows a file explorer view of a project named 'MySimpleApp' and a code editor showing HTML and XML code. The file explorer shows folders: .nyc\_output, css, gif, node\_modules, src, and test. The code editor shows the following code:

```

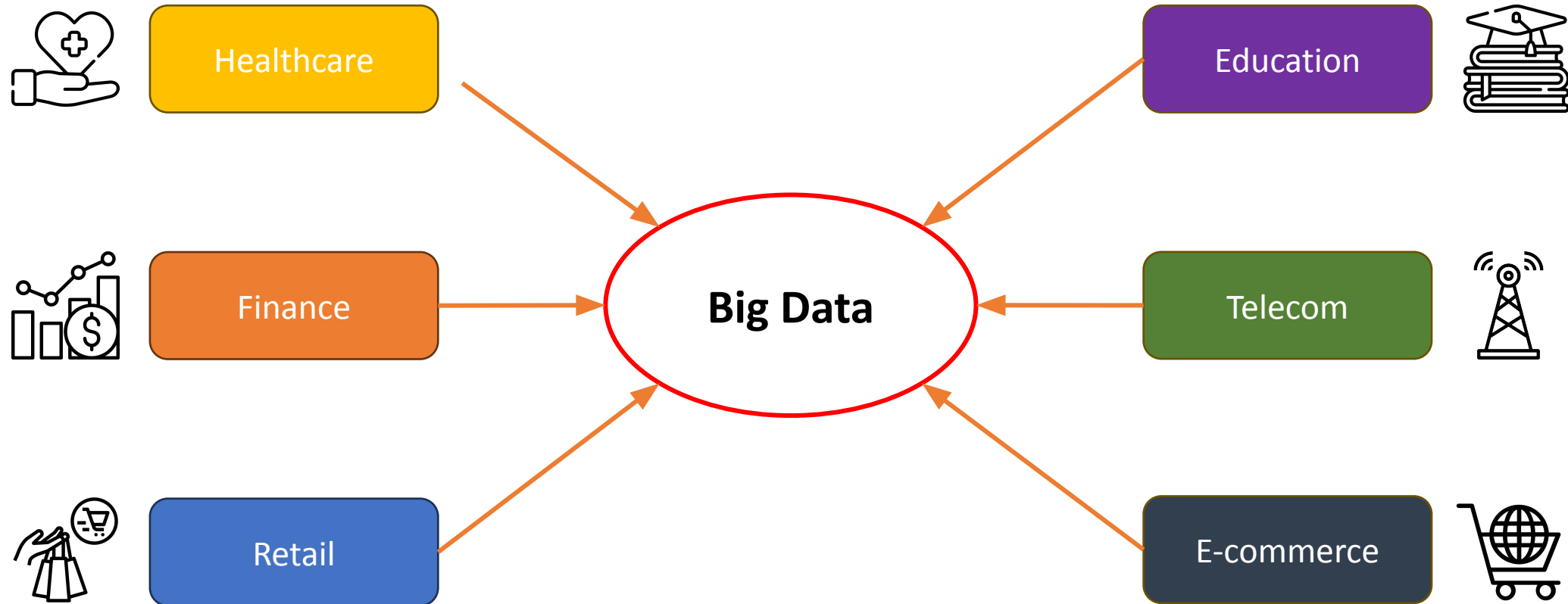
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Numbers</title>
  <link rel="stylesheet" href="css/my_file.css">
</head>
<body>
</body>
</html>
  
```

```

<?xml version="1.0" encoding="UTF-8"?>
- <EmployeeData>
  - <employee id="34594">
    <firstName>Heather</firstName>
    <lastName>Banks</lastName>
    <hireDate>1/19/1998</hireDate>
    <deptCode>BB001</deptCode>
    <salary>72000</salary>
  </employee>
  - <employee id="34593">
    <firstName>Tina</firstName>
    <lastName>Young</lastName>
    <hireDate>4/1/2010</hireDate>
    <deptCode>BB001</deptCode>
    <salary>65000</salary>
  </employee>
</EmployeeData>
  
```

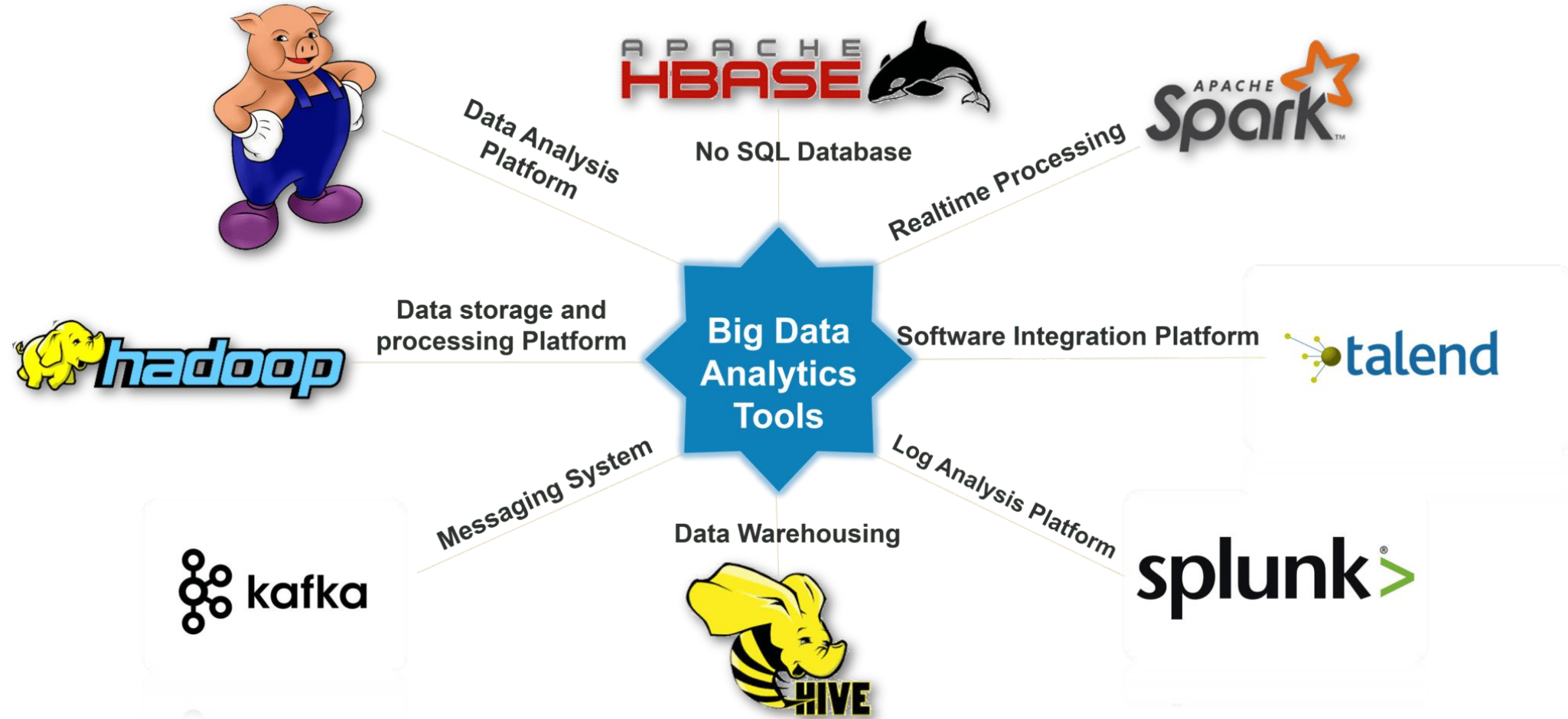
# Introduction

## ◆ Big Data Applications



# Introduction

## ◆ Tools for Big Data



# Hadoop

# Hadoop

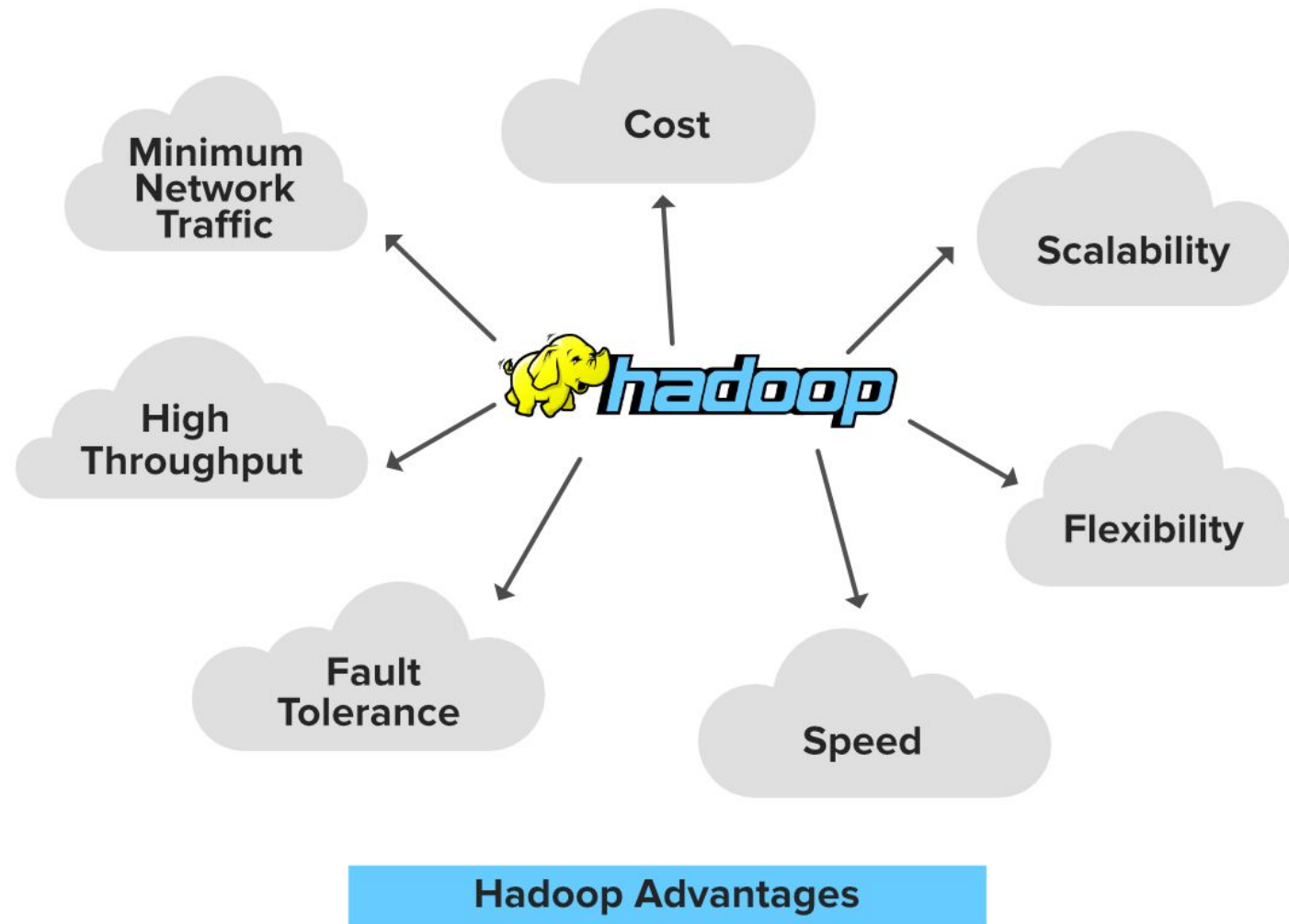
## ◆ Introduction



**Apache Hadoop:** an open-source framework that is used to **efficiently store and process large datasets** ranging in size from gigabytes to petabytes of data.

# Hadoop

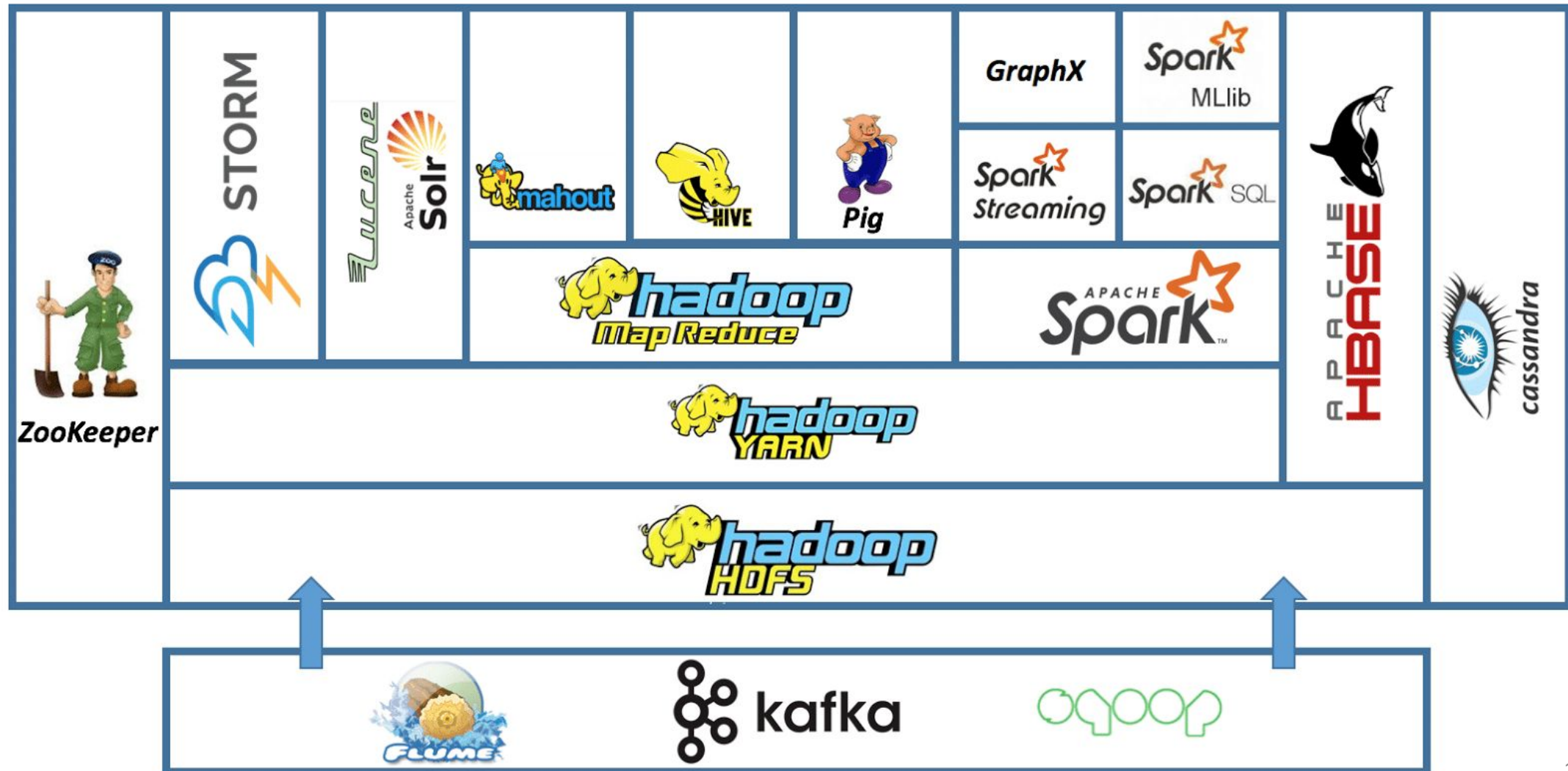
## ❖ Introduction





# Hadoop

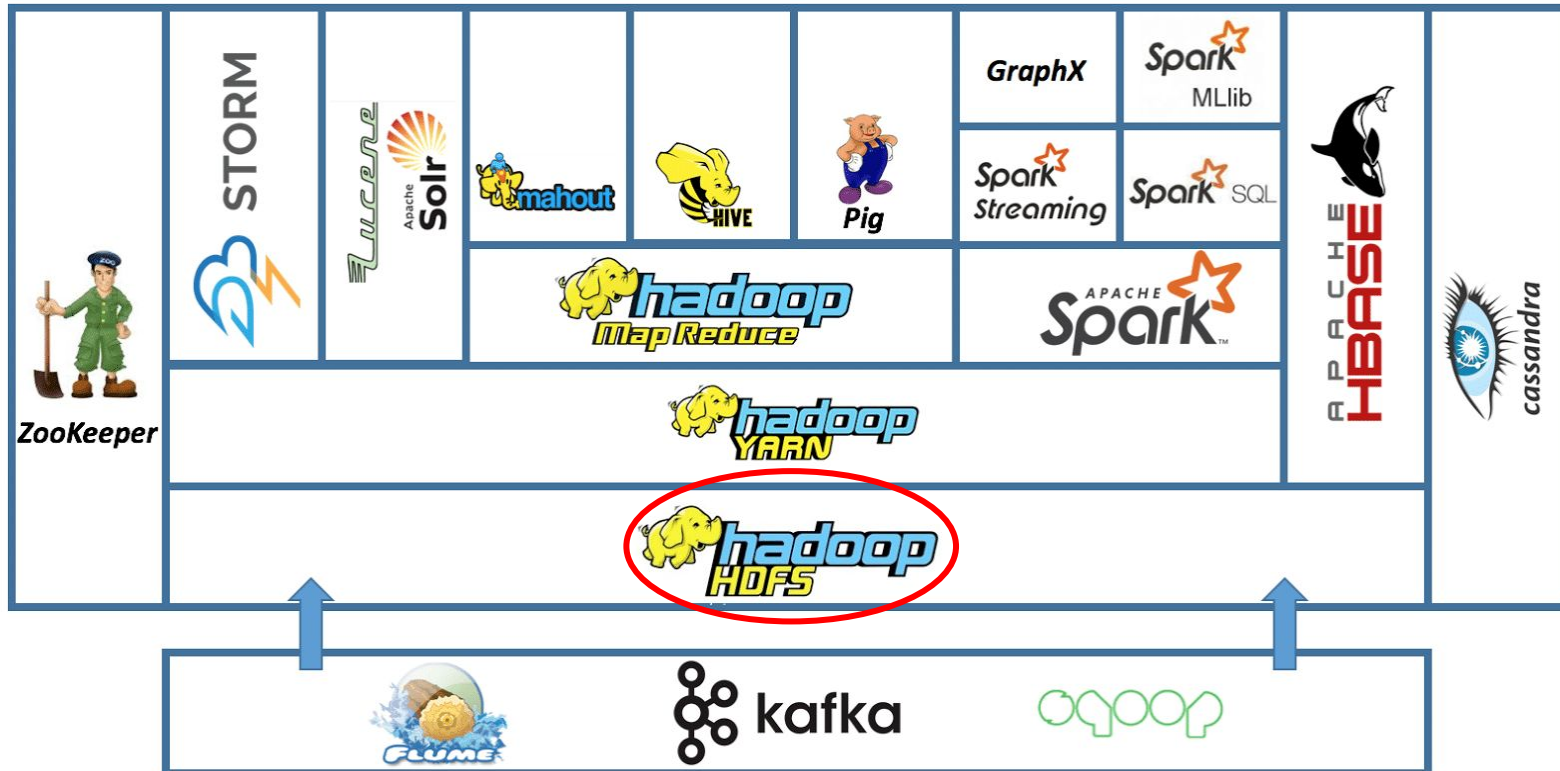
## ❖ Hadoop Ecosystem





# Hadoop

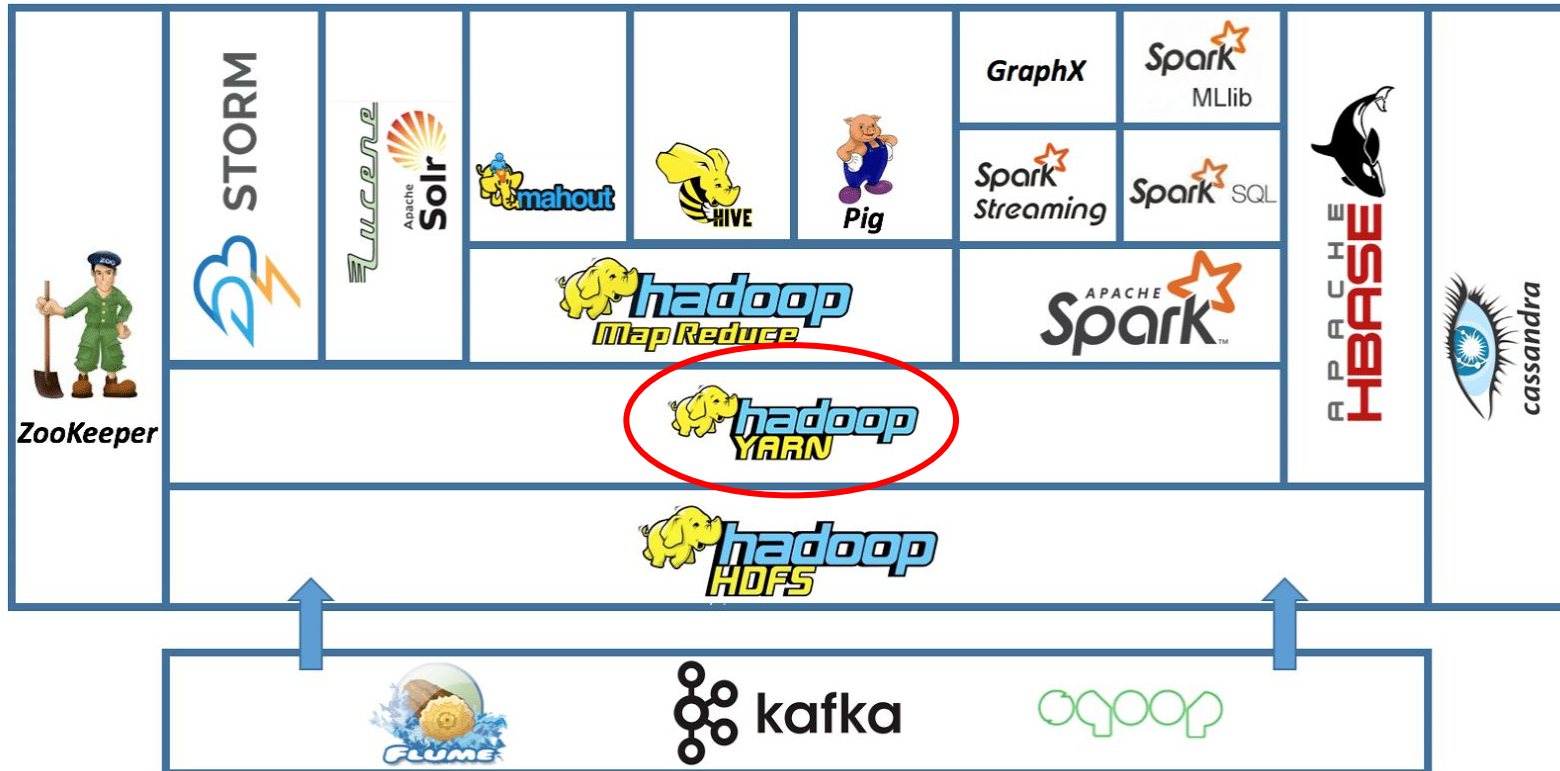
## ❖ Hadoop Ecosystem: Hadoop HDFS



**Hadoop HDFS (Hadoop Distributed File System):** A distributed file system that handles large data sets running on commodity hardware.

# Hadoop

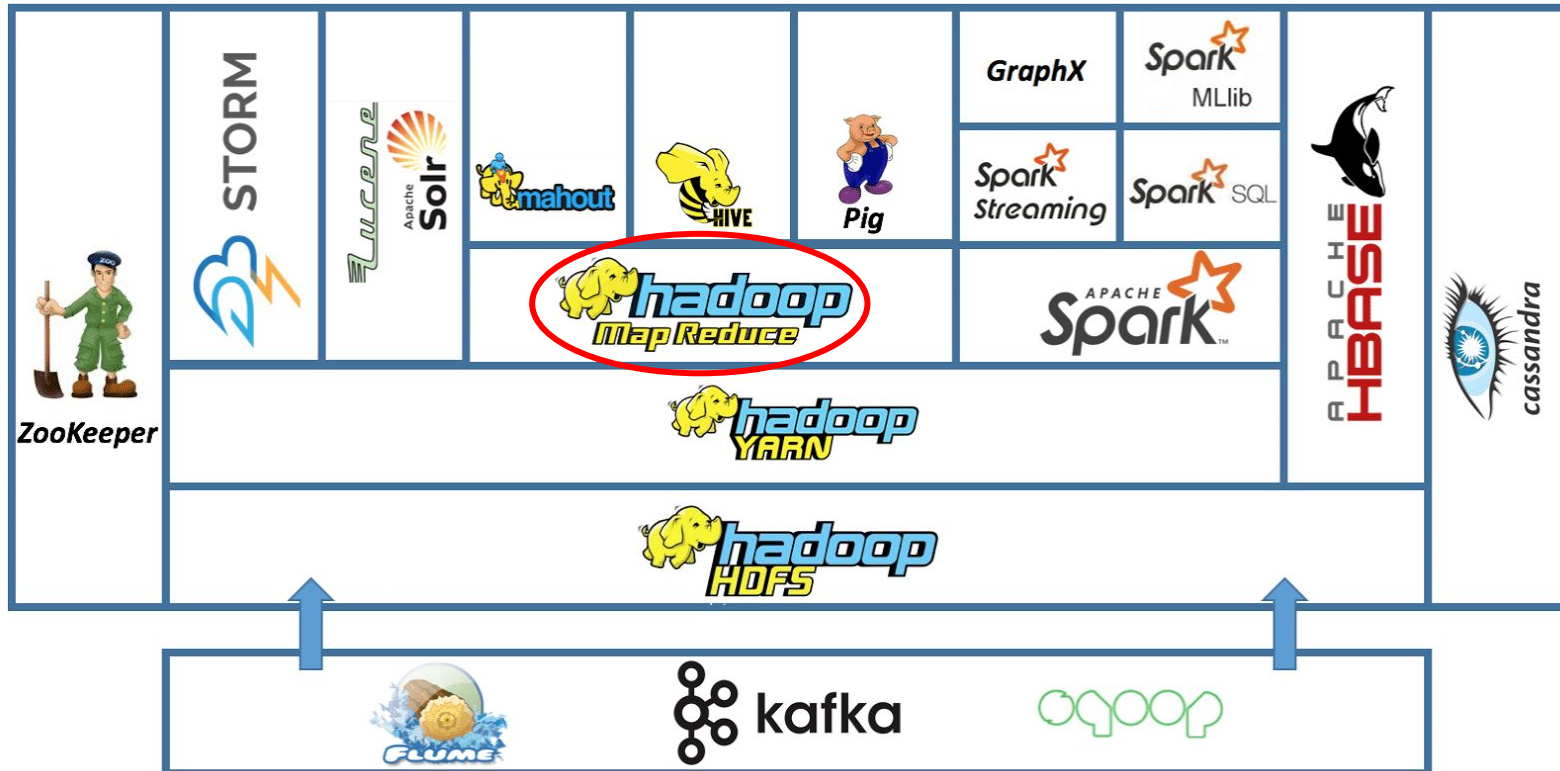
## ❖ Hadoop Ecosystem: Hadoop YARN



**Hadoop YARN (Yet Another Resource Negotiator):** the resources management and job scheduling technology.

# Hadoop

## ◆ Hadoop Ecosystem: Hadoop MapReduce

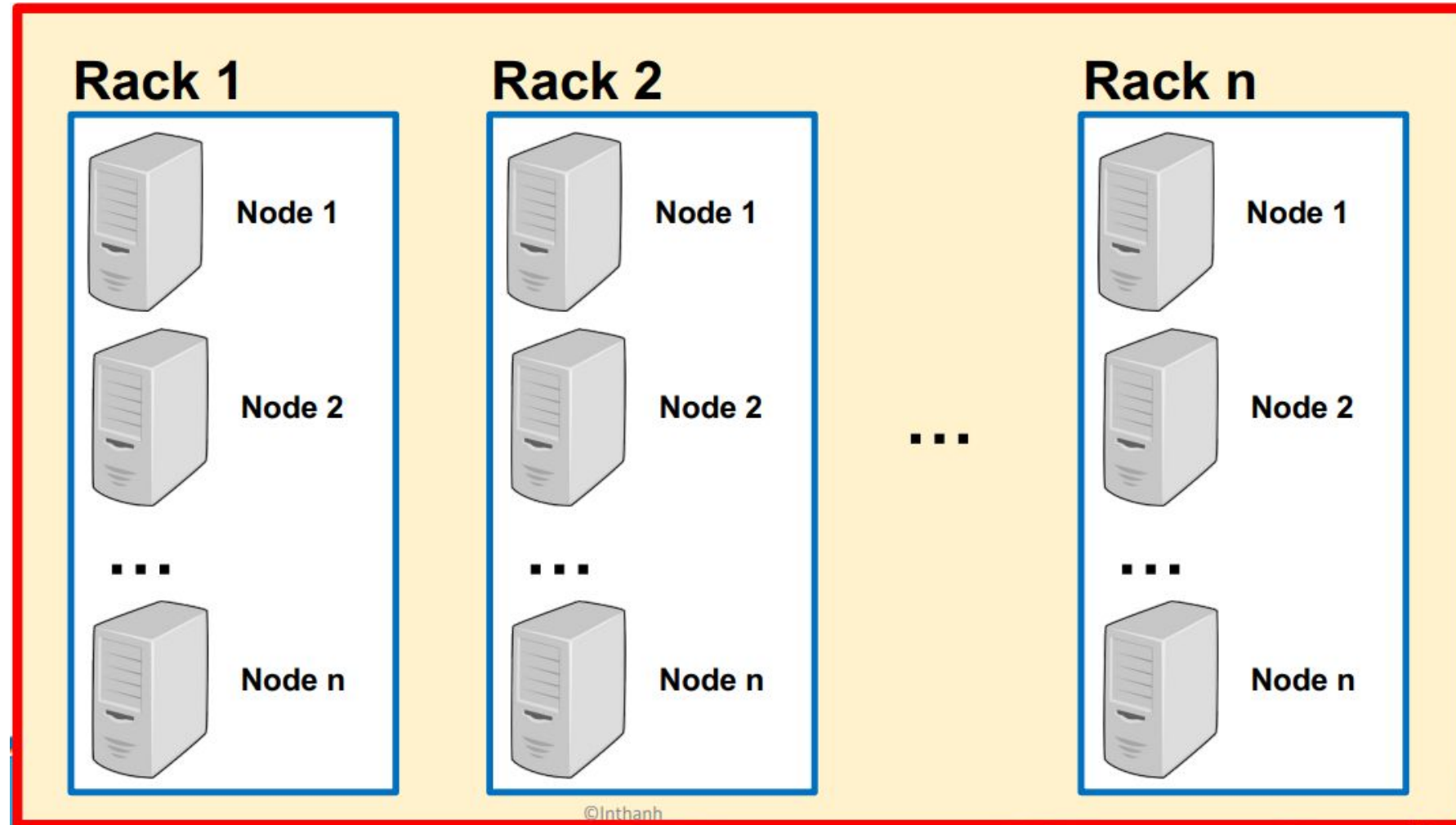


**Hadoop MapReduce:** A highly sufficient methodology for parallel processing of huge volumes of data.

# Hadoop

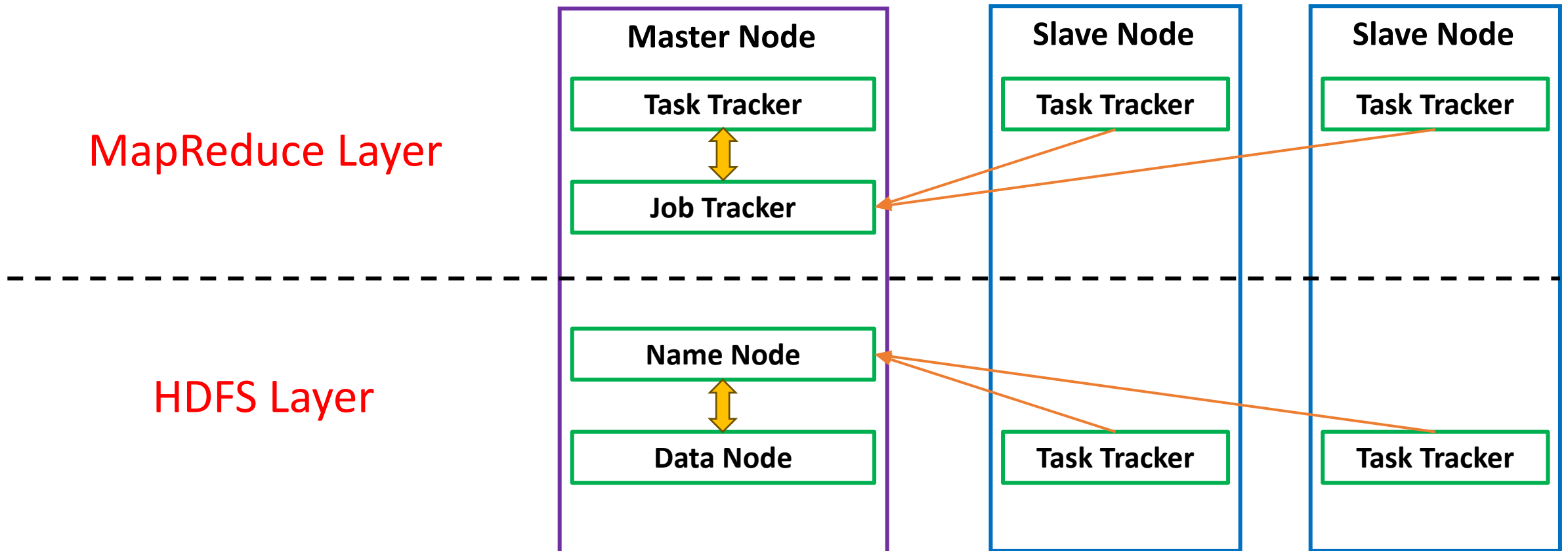
## ◆ Terminology

- Node
- Rack
- Cluster



# Hadoop

## ❖ Hadoop Architecture



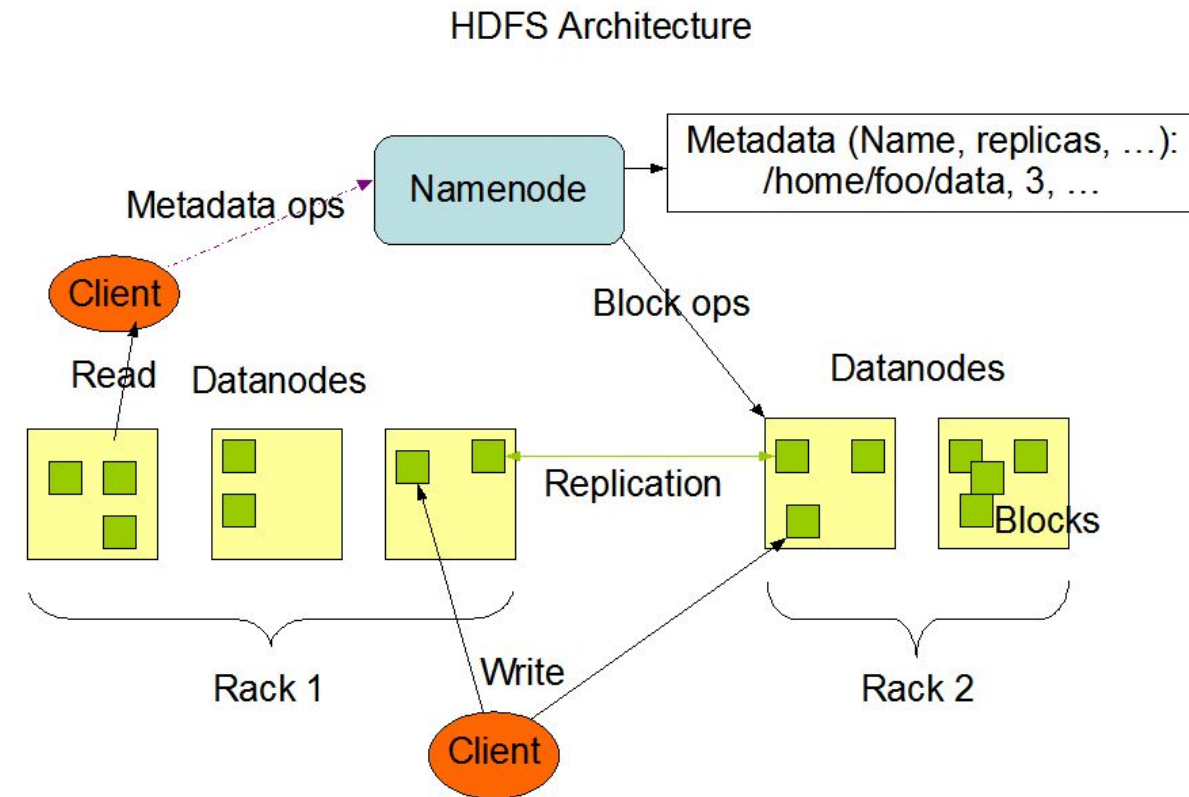
High-level Architecture of Hadoop

# Hadoop

## ❖ Hadoop HDFS



**HDFS (Hadoop Distributed File System):** A distributed file system designed to store and process large amounts of data across multiple machines in a Hadoop cluster, providing high fault-tolerance and scalability.

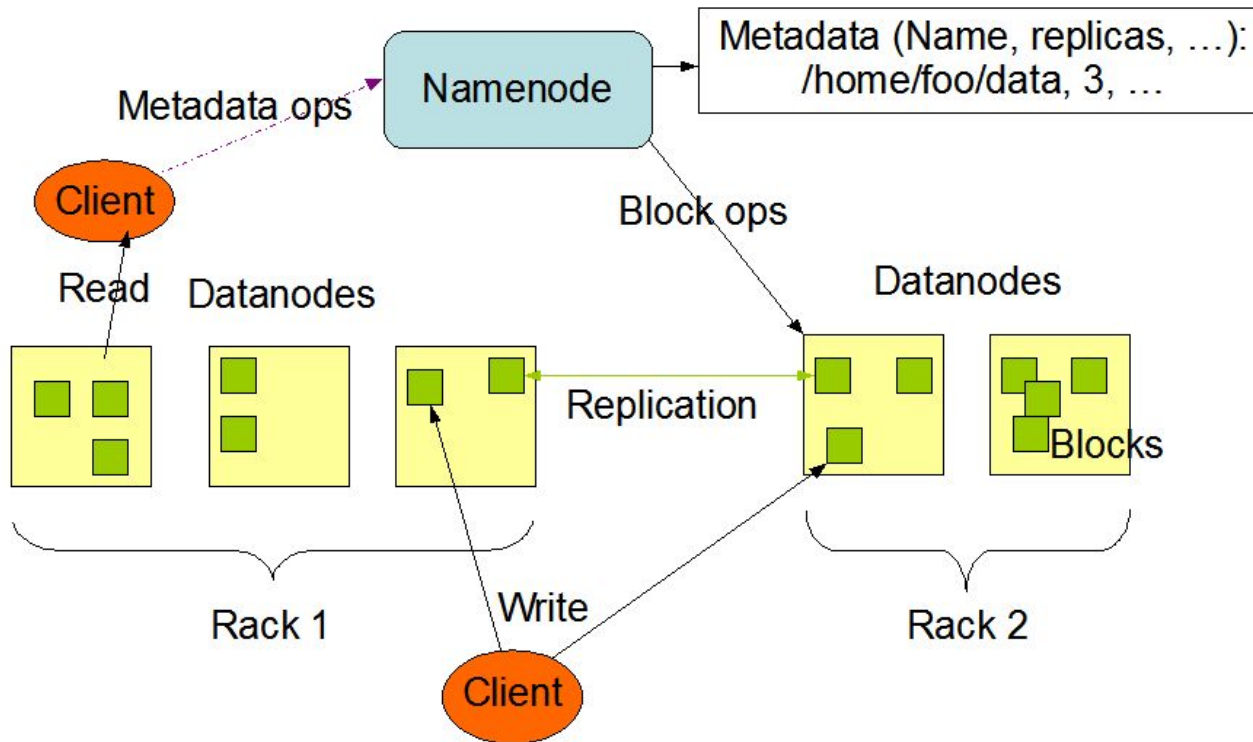




# Hadoop

## ❖ Hadoop HDFS

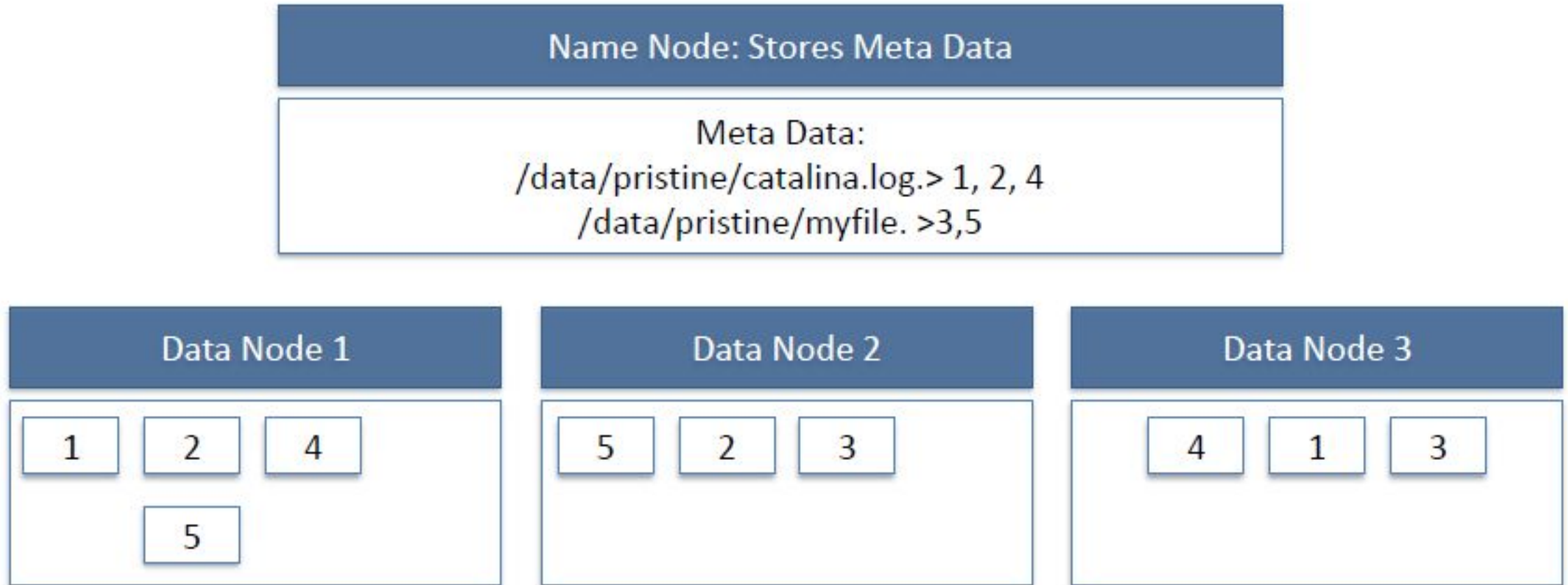
HDFS Architecture



- **Namenode:** Master node in a Hadoop cluster, managing file system's metadata.
- **Datanode:** Worker node in a Hadoop cluster, storing data blocks.
- **Blocks:** A fixed-size chunk of data. Default is 128MB (or 64MB) size.

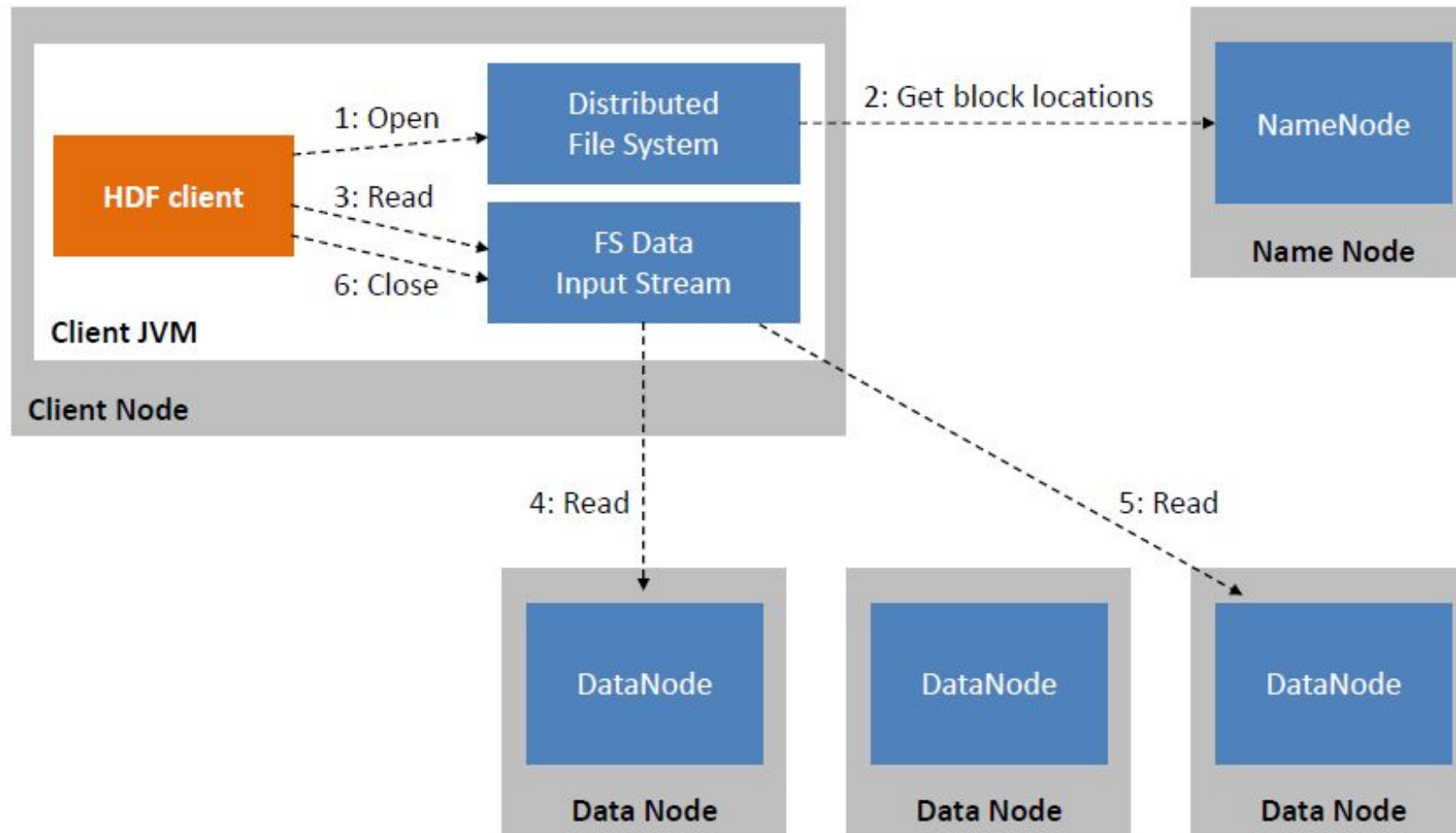
# Hadoop

## ❖ Hadoop HDFS



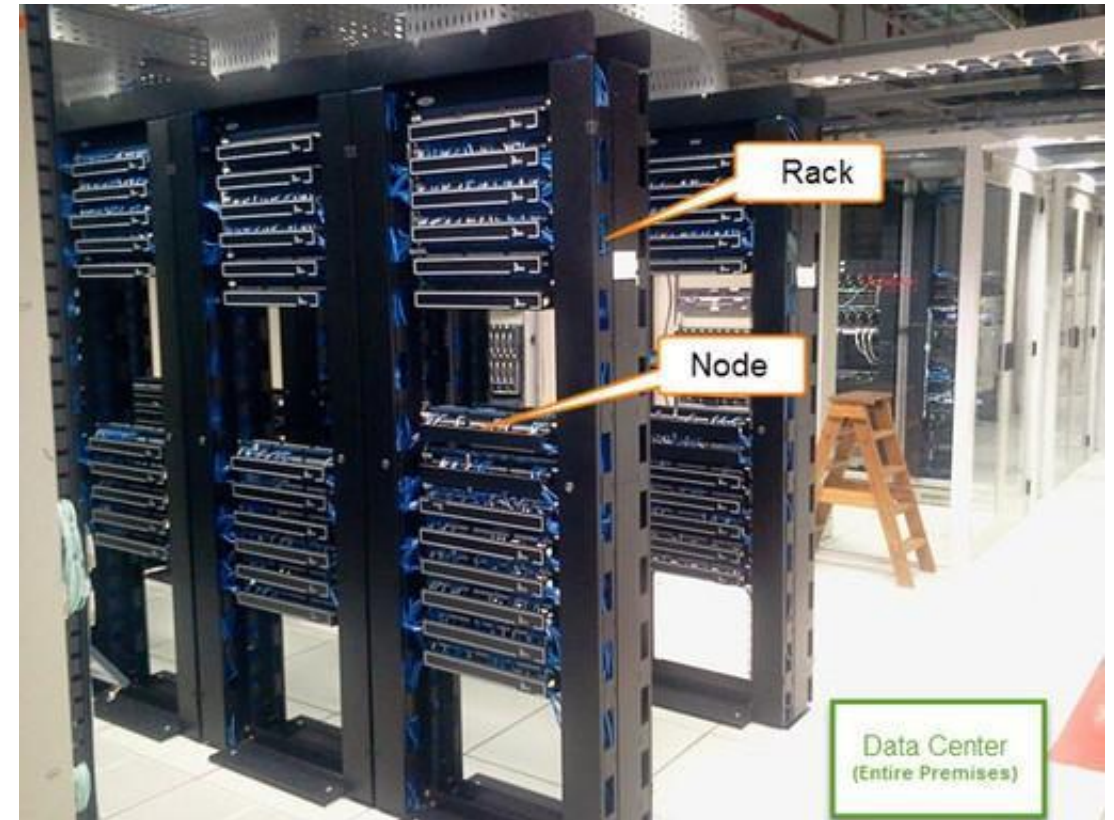
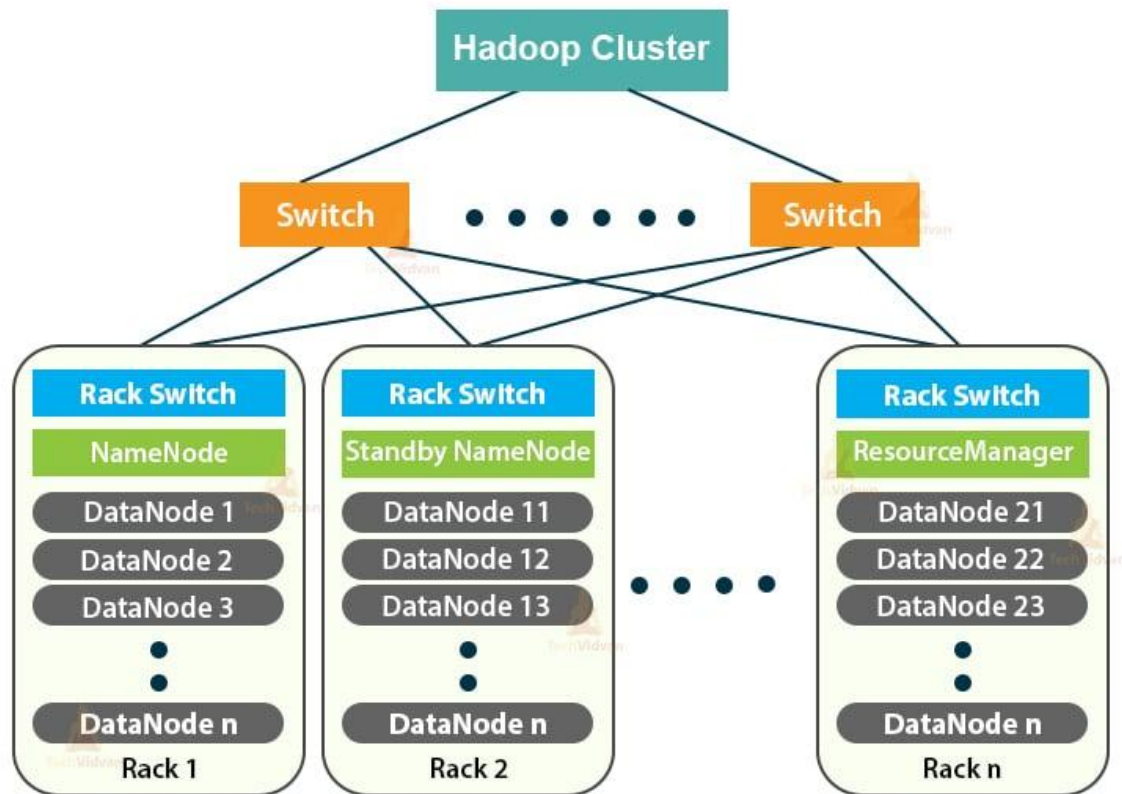
# Hadoop

## ❖ Hadoop HDFS



# Hadoop

## ❖ Hadoop HDFS

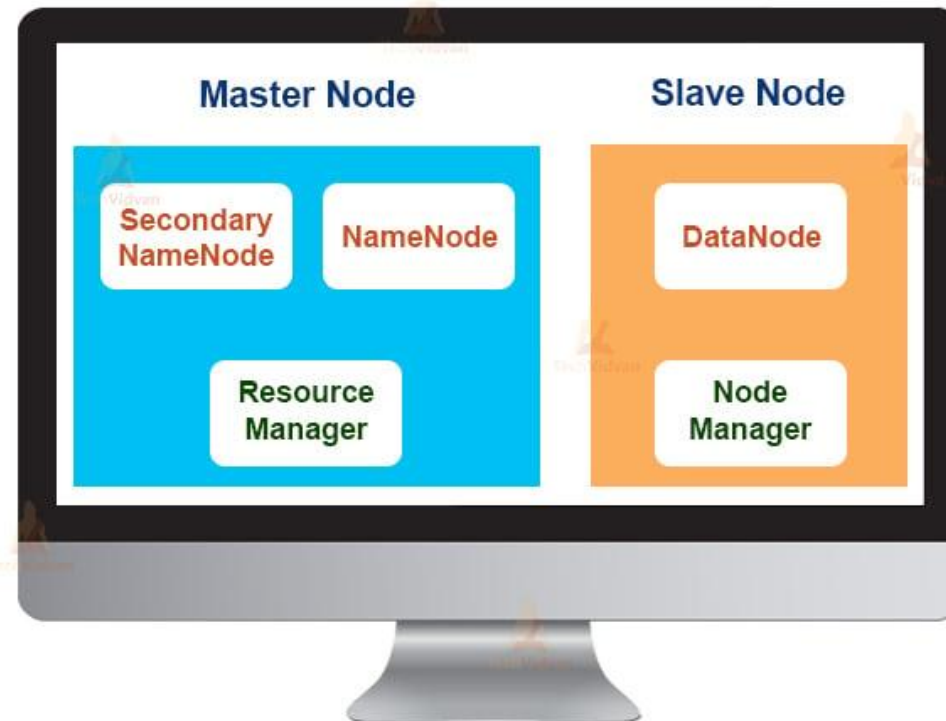


Hadoop Cluster Architecture

# Hadoop

## ❖ Hadoop HDFS

### Single Node Hadoop Cluster



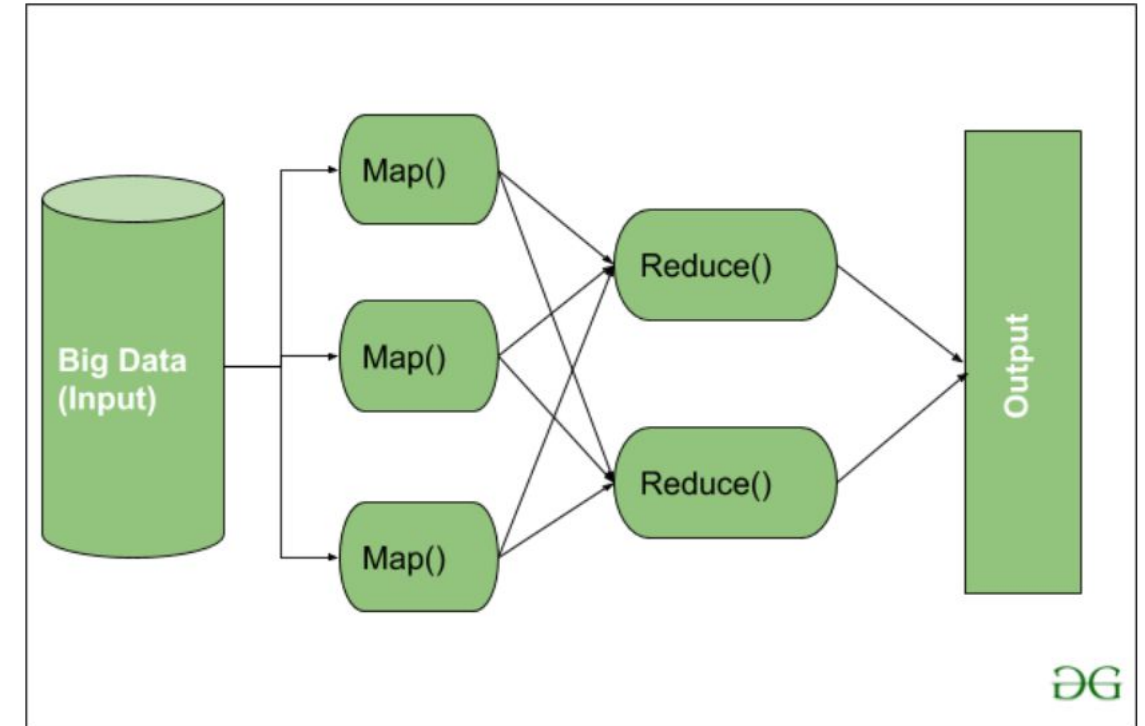
Hadoop Single Node Cluster Architecture

# Hadoop

## ❖ Hadoop MapReduce



**Hadoop MapReduce:** A distributed programming model and framework that breaks down large-scale data processing tasks into smaller, parallelizable operations (map and reduce), allowing for efficient processing and analysis of big data across a cluster of machines.

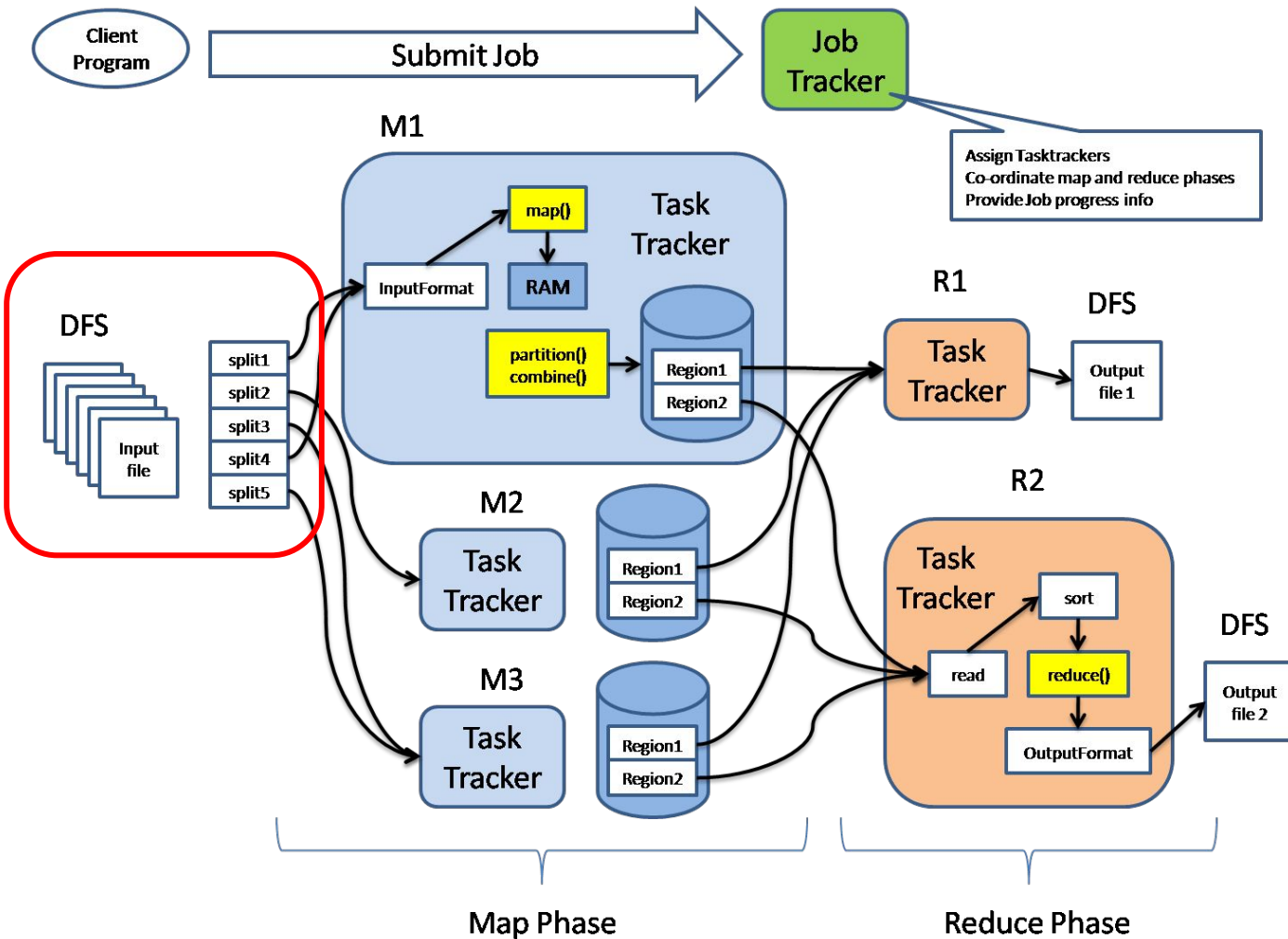


<https://www.geeksforgeeks.org/hadoop-architecture/>



# Hadoop

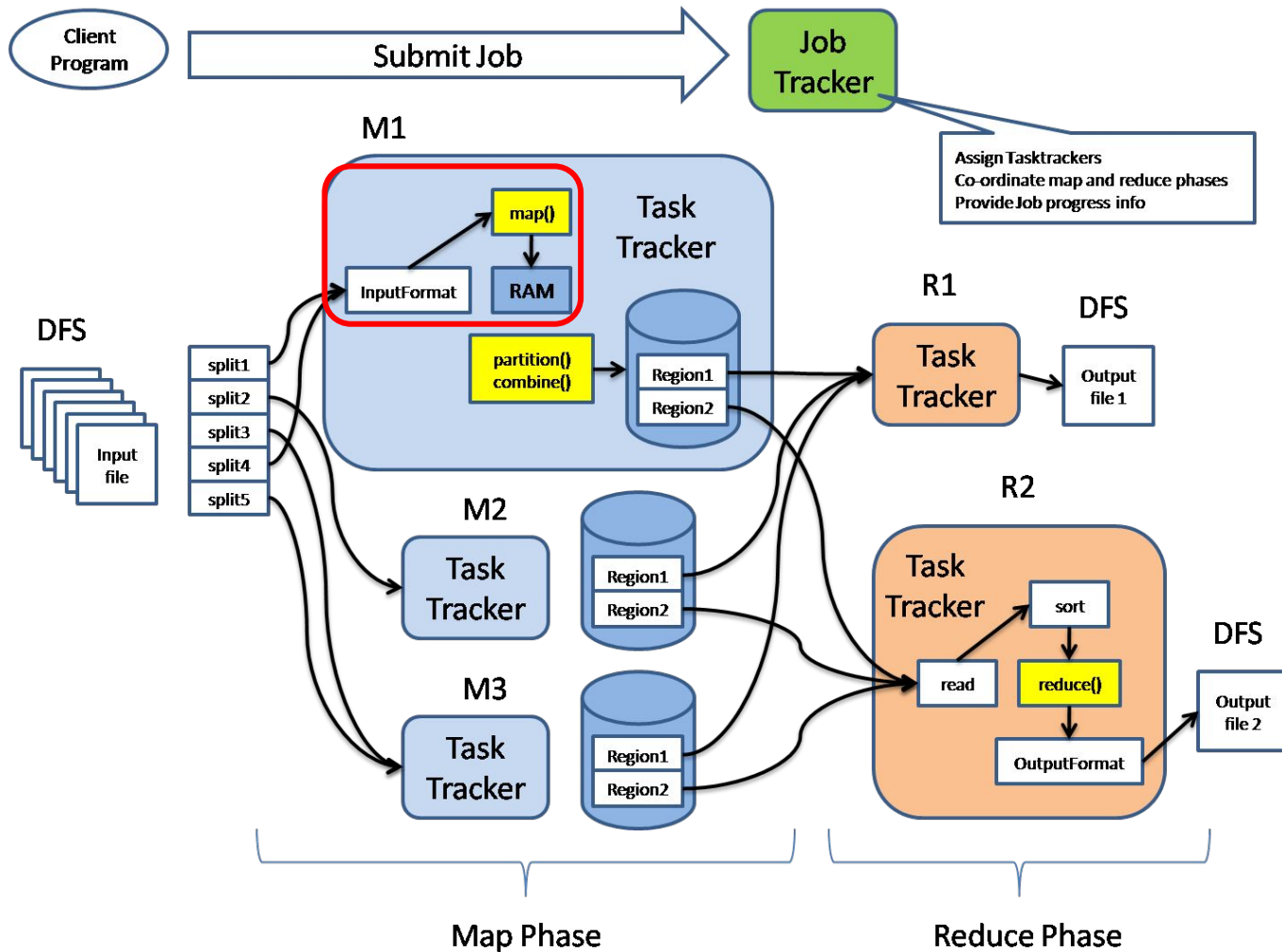
## ◆ Hadoop MapReduce





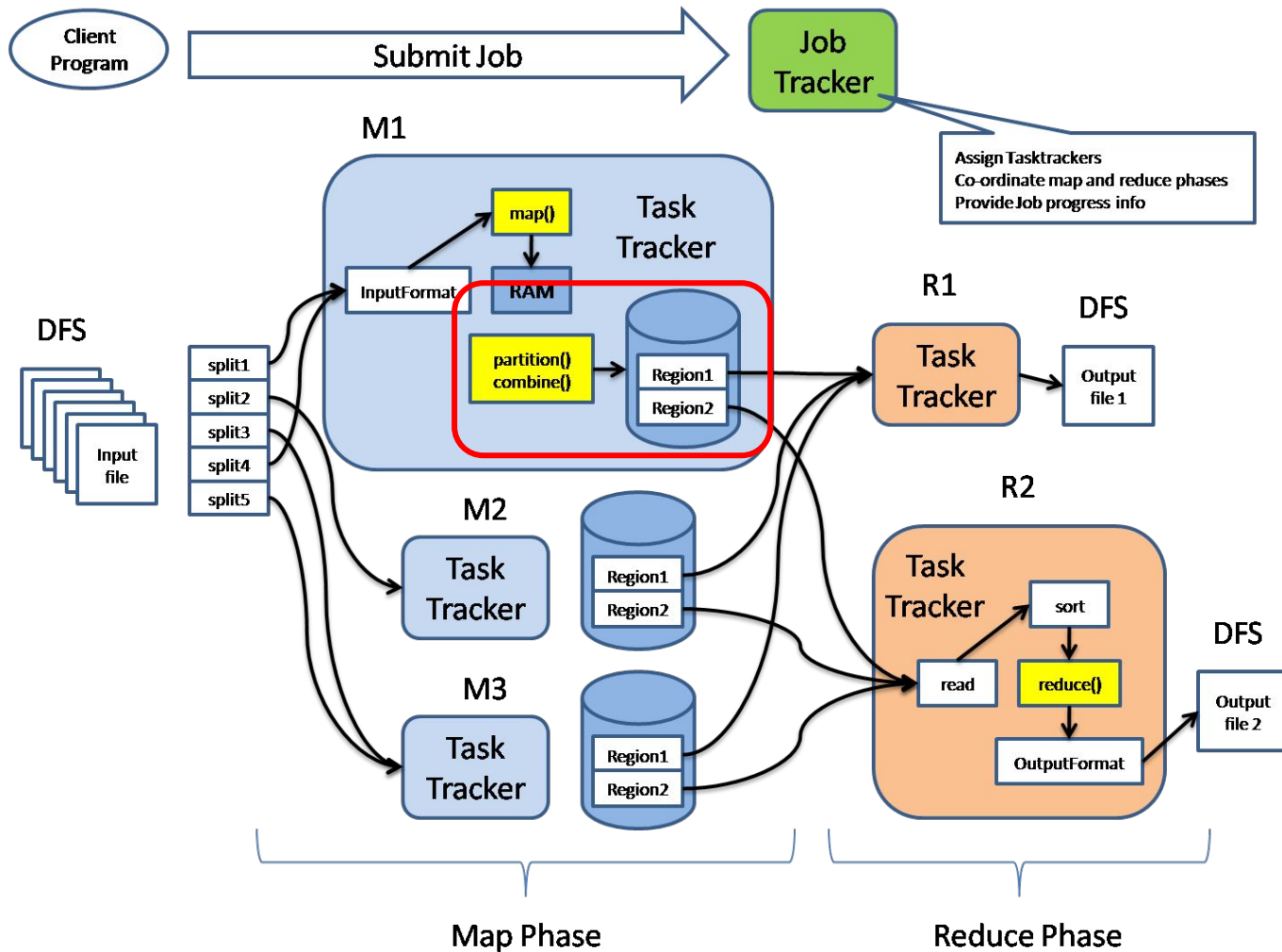
# Hadoop

## ❖ Hadoop MapReduce



**MapFunction:** Process the key-value pairs and generate the corresponding output key-value pairs.

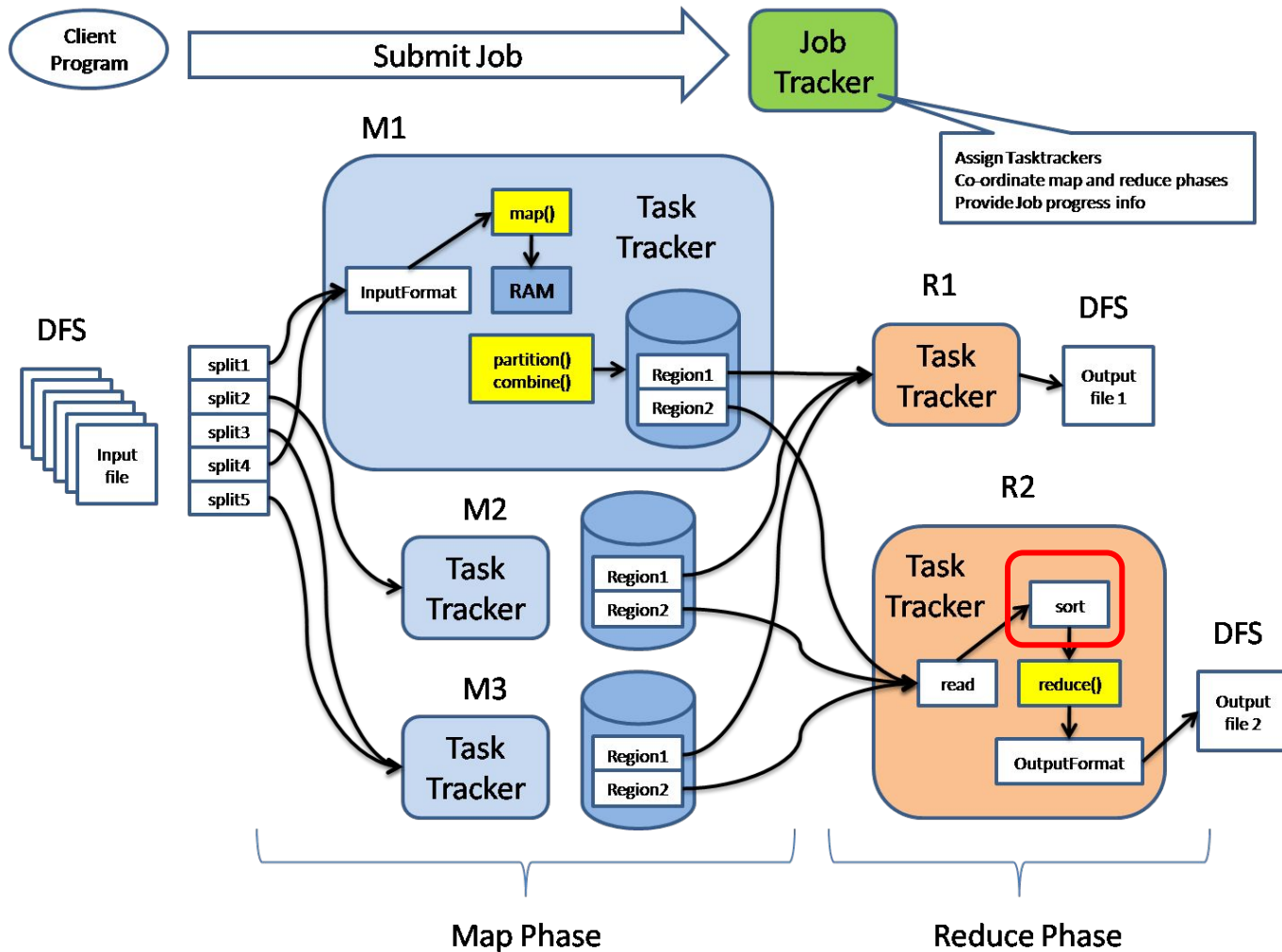
## ◆ Hadoop MapReduce



**PartitionFunction:** Assign the output of Mapping to the appropriate Reducer.

# Hadoop

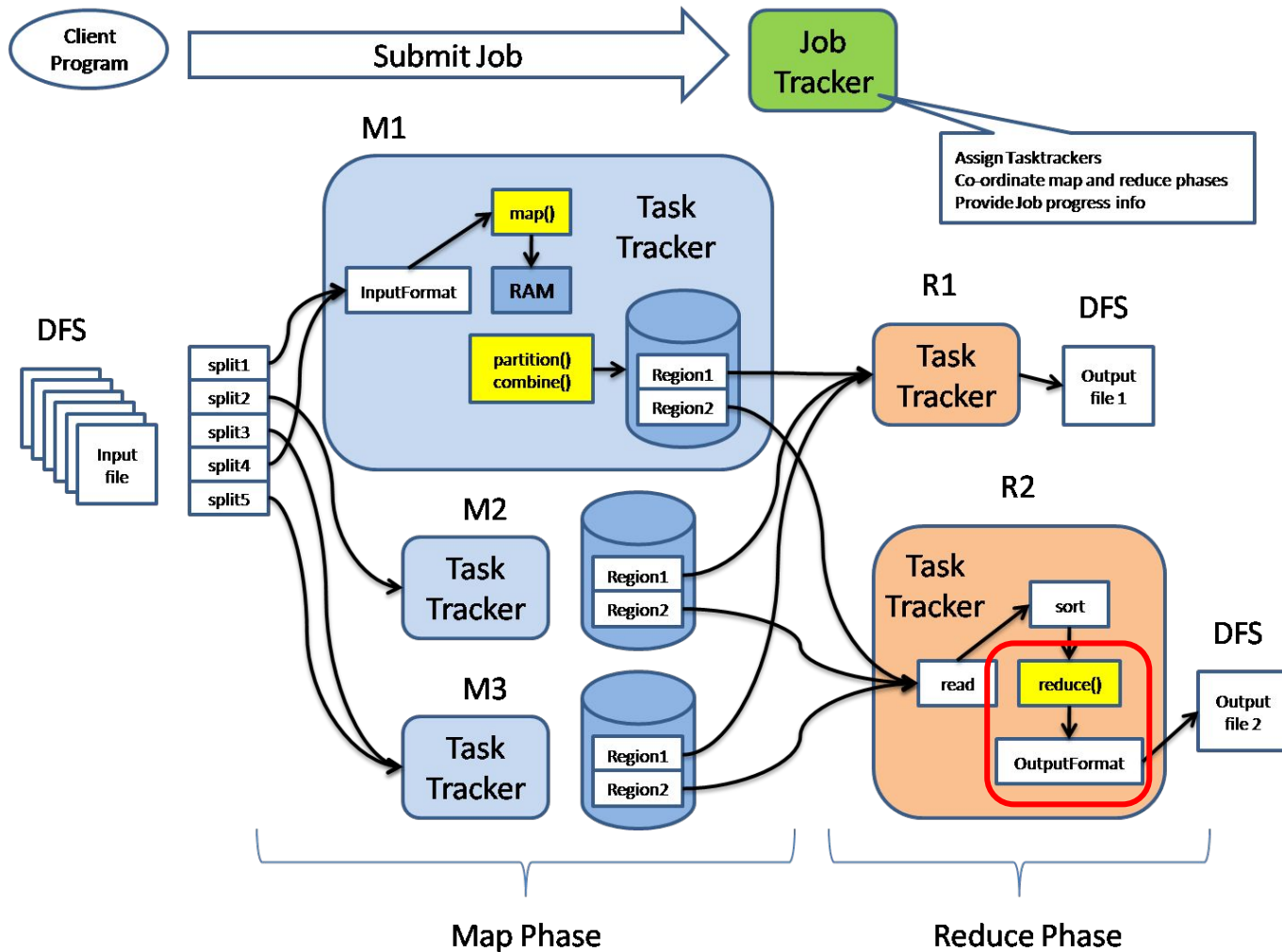
## ❖ Hadoop MapReduce



**ShufflingAndSorting:** Get the results of mapping and sorting

# Hadoop

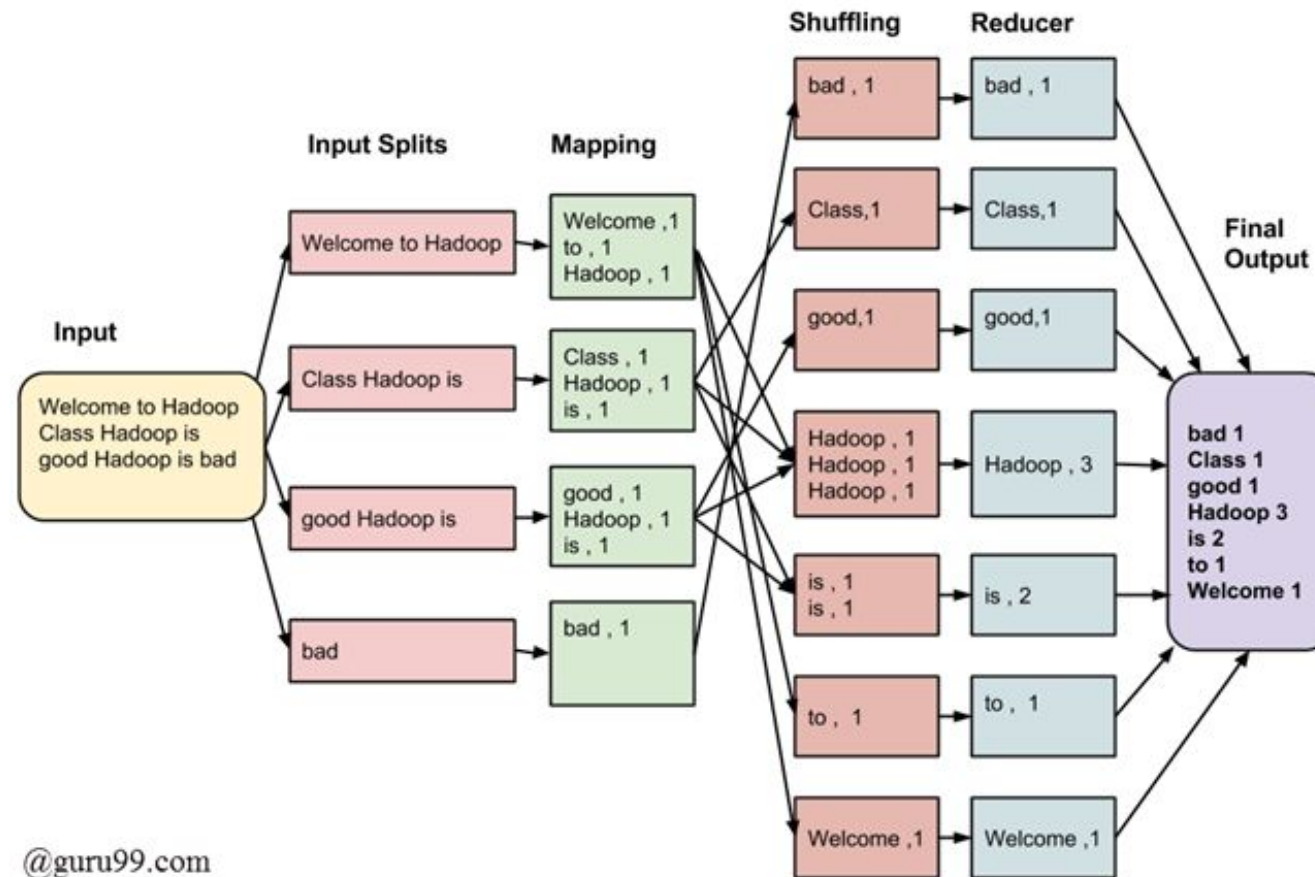
## ◆ Hadoop MapReduce



**ReduceFunction:** Grouping and processing the intermediate key-value pairs produced by the map phase.

# Hadoop

## ❖ Hadoop MapReduce: WordCount Example



@guru99.com

# Hadoop

## ❖ Break-quiz

**1. What does HDFS stand for in Hadoop?**

- A) Hadoop Distributed File System.
- B) High-Definition File Storage.
- C) Hadoop Data File System.
- D) Hierarchical Distributed File Structure.

**2. Which programming model does Hadoop use for processing data?**

- A) SQL.
- B) MapReduce.
- C) Machine Learning.
- D) Graph Processing.

**3. What is the primary function of the NameNode in HDFS?**

- A) Storing actual data.
- B) Managing metadata and file system namespace.
- C) Processing MapReduce jobs.
- D) Coordinating data replication.

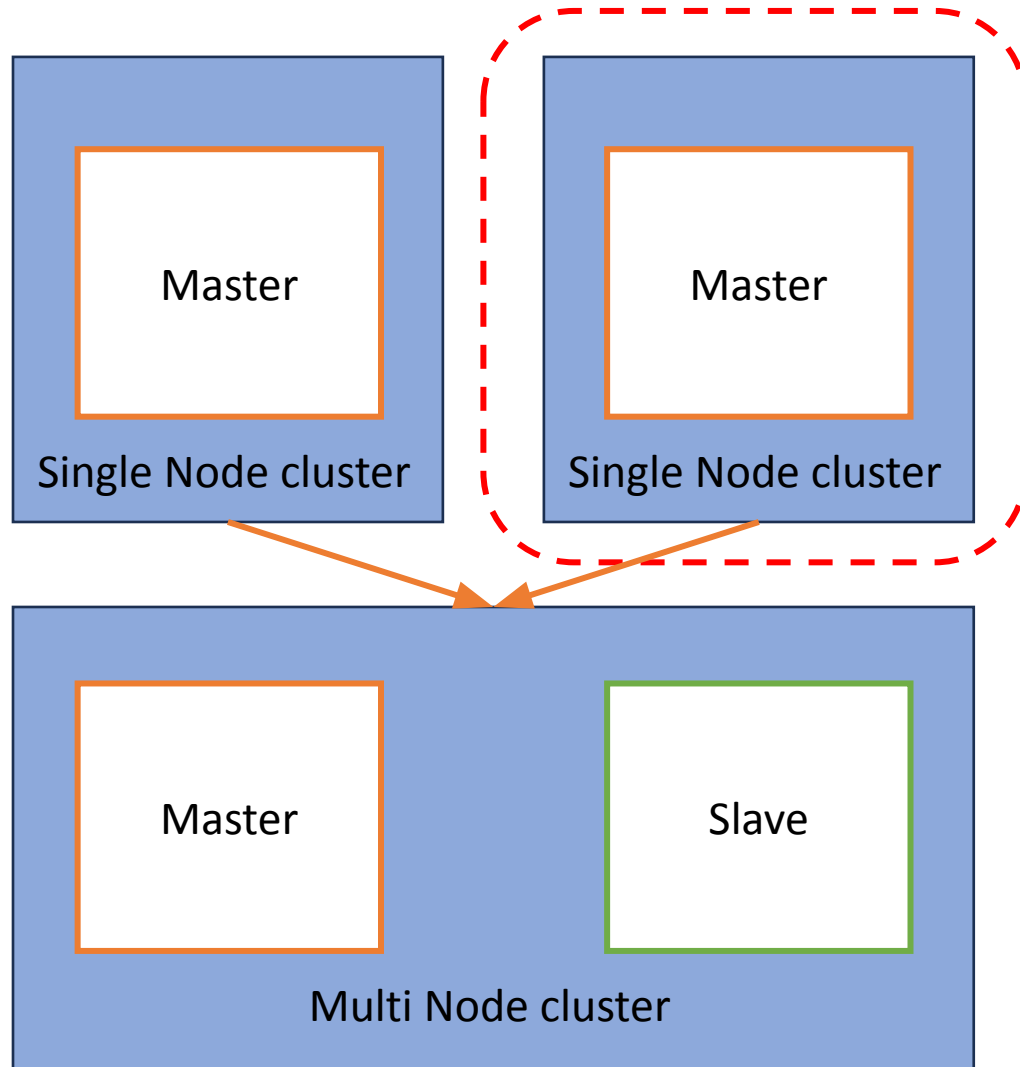
**4. How many copies of data does HDFS typically maintain by default?**

- A) 1.
- B) 2.
- C) 3.
- D) 4.



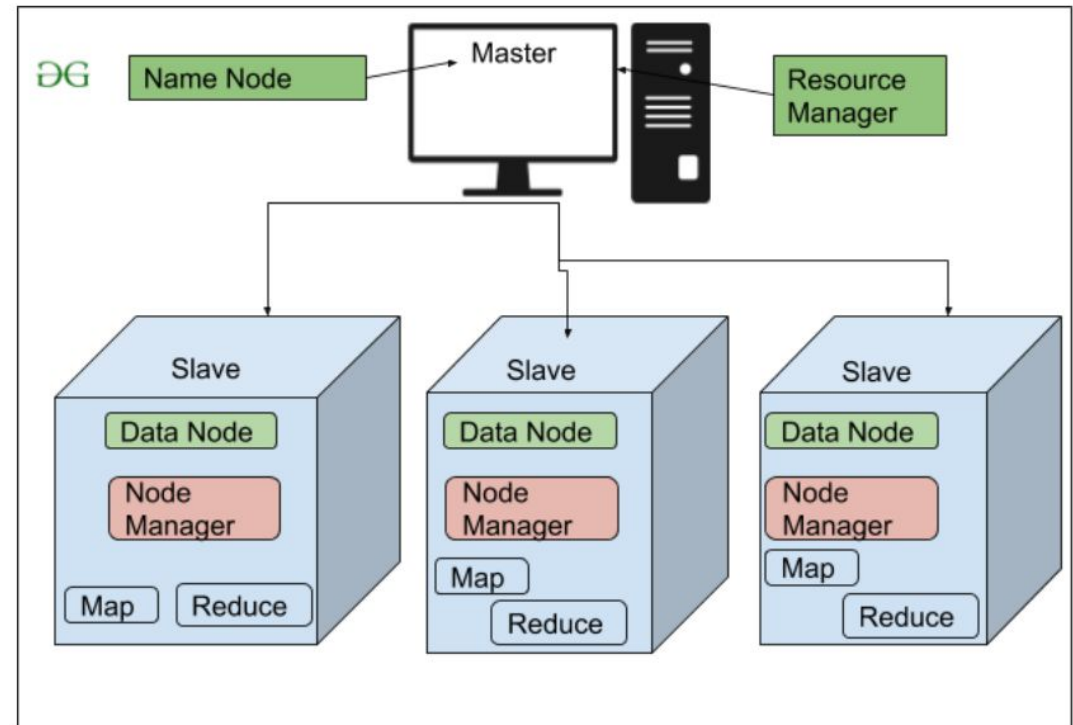
# Hadoop

## ❖ Installation



### Hadoop Single Node (Hadoop pseudo-distributed mode):

A configuration option in Hadoop that allows us to run Hadoop on a single machine.



<https://www.geeksforgeeks.org/hadoop-architecture/>

# Hadoop

## ❖ Installation: Requirements

### 1. Virtual Machine



VirtualBox



VMWare



Cloudera

### 2. CentOS



CentOS

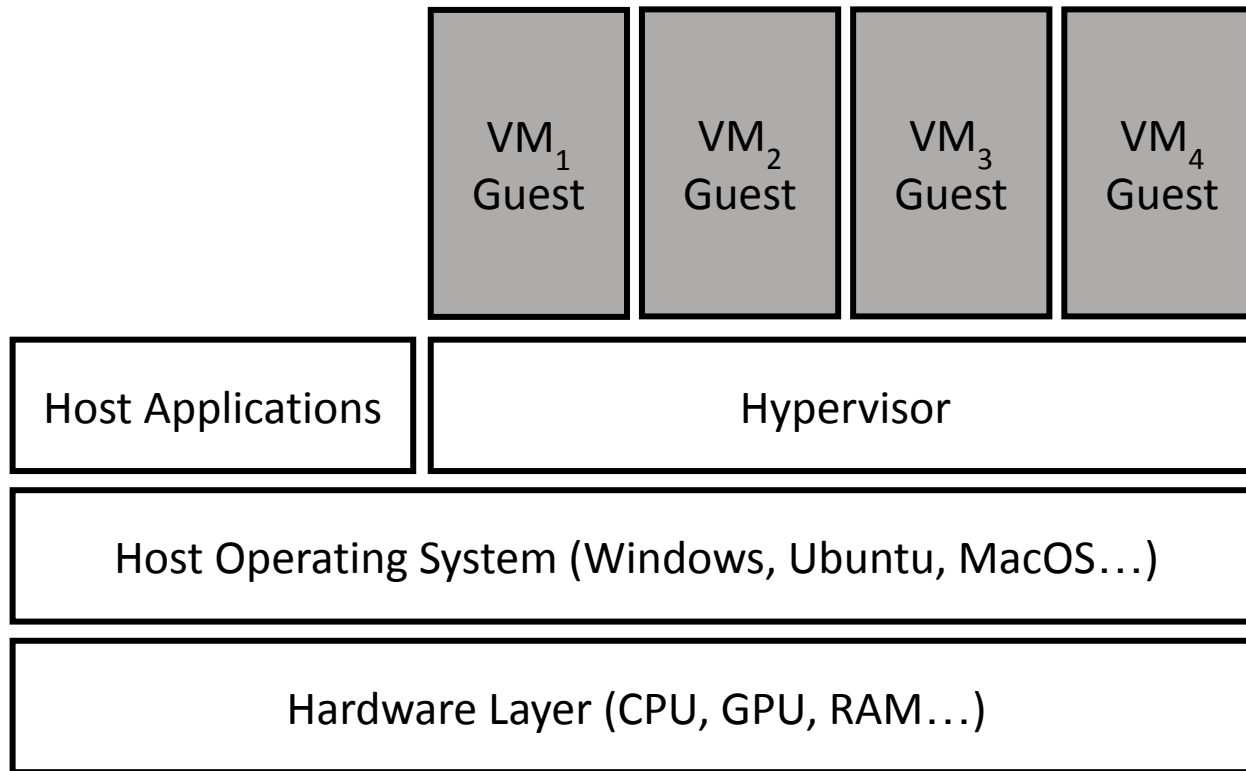
### 3. Hadoop



**Note:** The following installation demo is on Mac M1. You need to choose proper version for your computer.

# Hadoop

## ❖ Virtual Machine



**Virtual Machines (VMs):** An emulation of a computer system, where these machines use computer architectures to provide the functionality of a physical computer.



**Example:** A MacOS computer running two Windows VMs

## ❖ Installation: VMware



**VMware:** A software providing machine virtualization

## Download VMware Workstation Pro

[VMware Workstation Pro](#) is the industry standard desktop hypervisor for running virtual machines on Linux or Windows PCs. Start your free, fully functional 30-day trial today.

BUY ONLINE

**For Windows/Linux:** Use [VMware Workstation](#)

Run Windows and More on Mac

## VMware Fusion

Harness the full power of your Mac when you use VMware Fusion to run Windows, Linux, containers, Kubernetes and more in virtual machines (VMs) without rebooting.

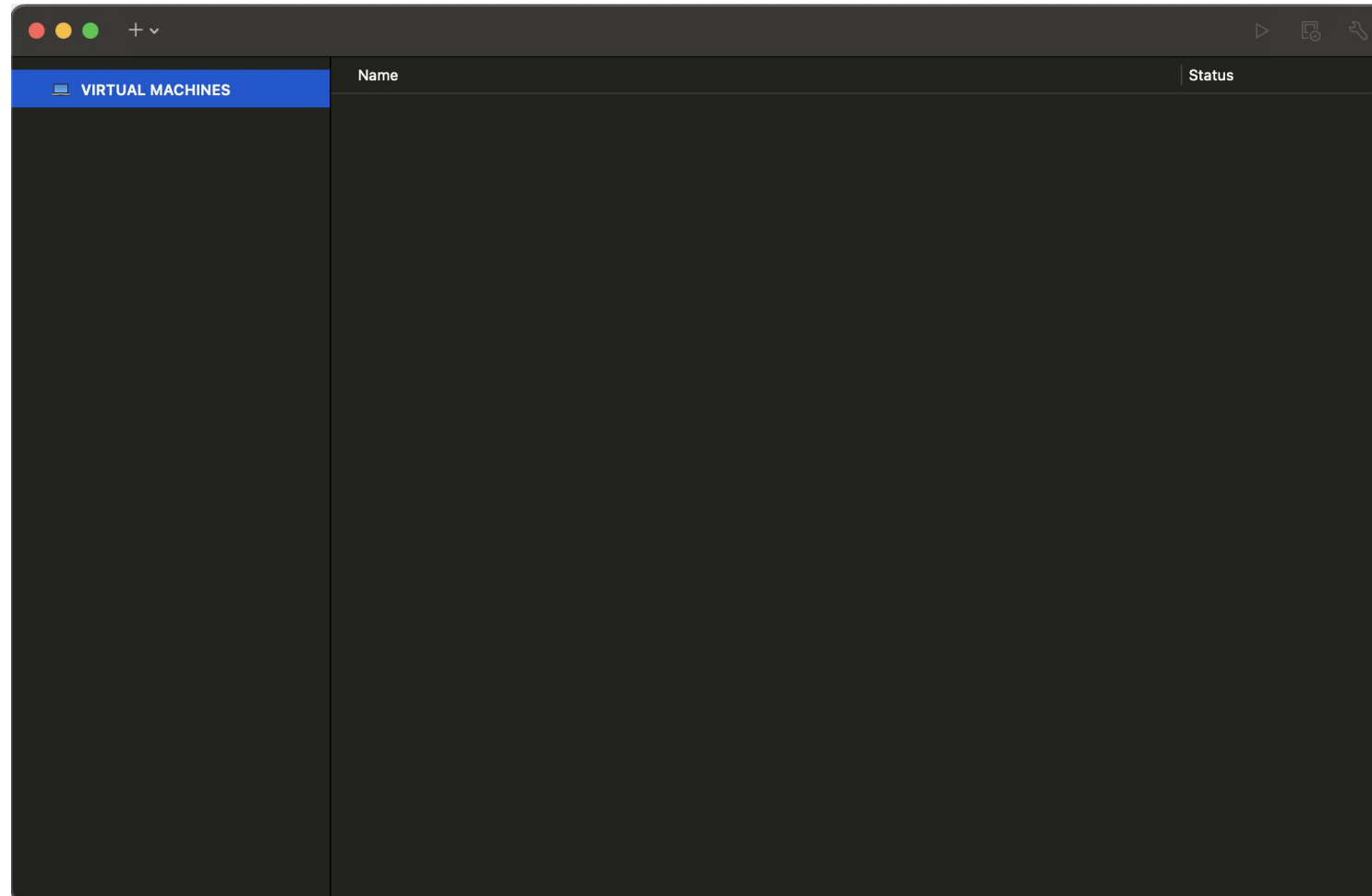
BUY ONLINE

TRY FOR FREE

**For MacOS:** Use [Vmware Fusion](#)

# Hadoop

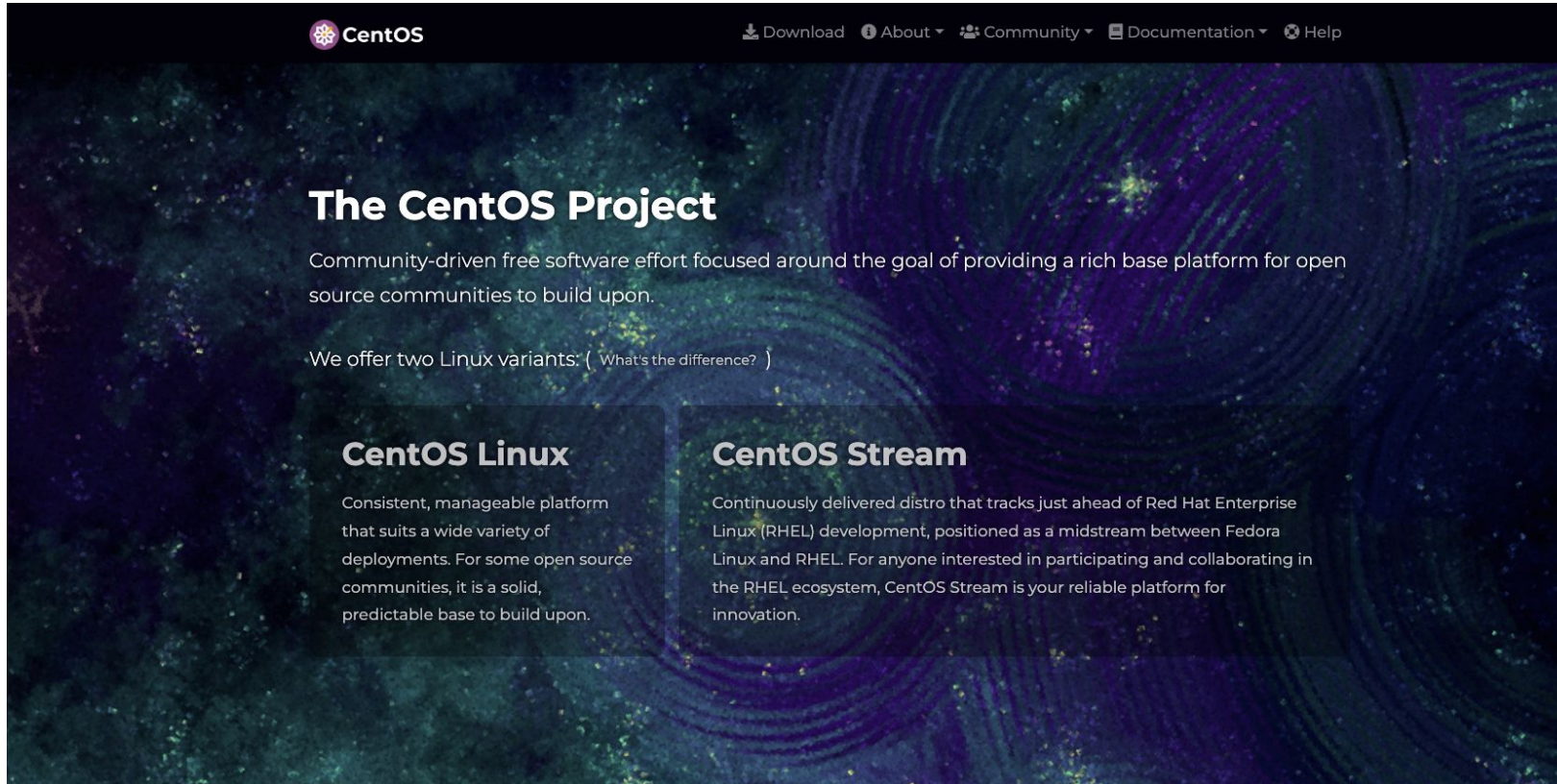
## ❖ Installation: VMware



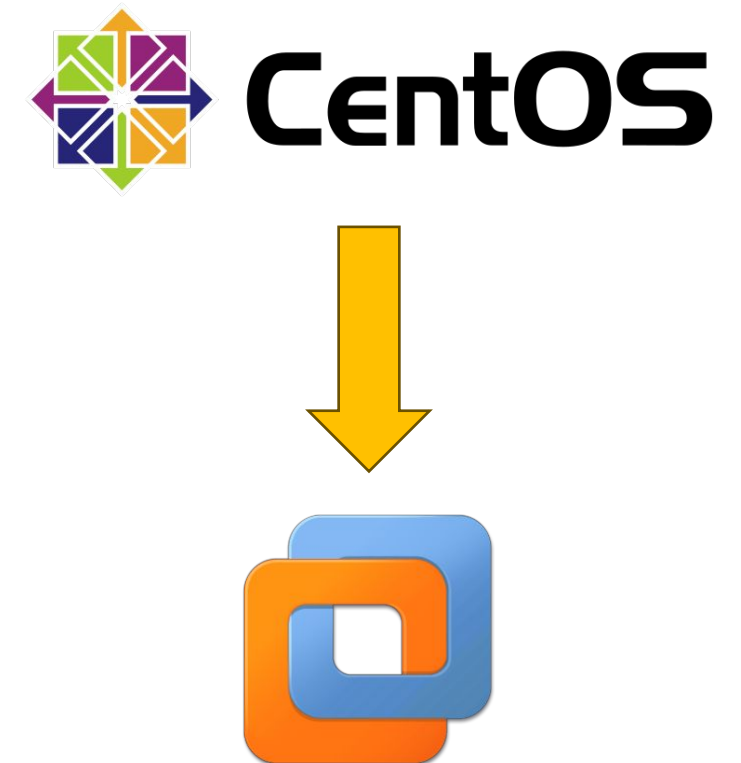
VM Fusion Interface



## ❖ Installation: CentOS



**CentOS:** Community-driven free software effort focused around the goal of providing a rich base platform for open source communities to build upon.





# Hadoop

## ❖ Installation: CentOS

### Download

CentOS Stream

9 8

| Architectures       | Packages | Others                       |
|---------------------|----------|------------------------------|
| x86_64              | RPMs     | Cloud   Containers   Vagrant |
| ARM64 (aarch64)     | RPMs     | Cloud   Containers   Vagrant |
| IBM Power (ppc64le) | RPMs     | Cloud   Containers   Vagrant |
| IBM Z (s390x)       | RPMs     | Cloud   Containers   Vagrant |

**Documentation**

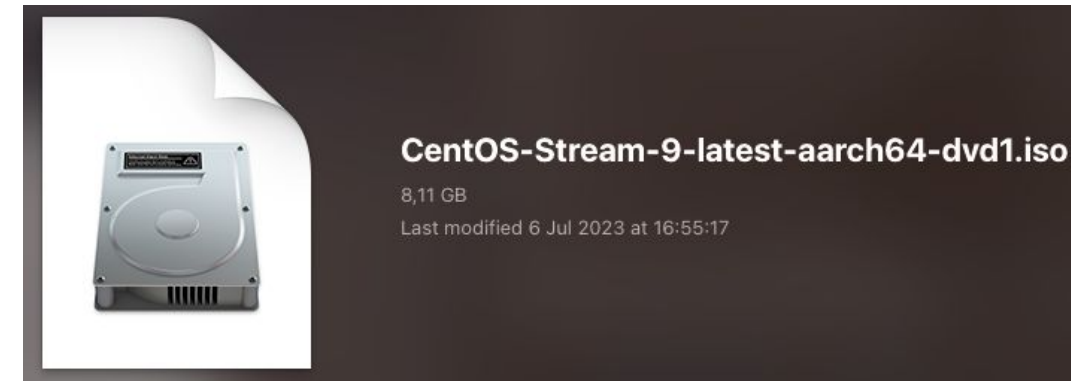
[Release Notes](#) | [Release Email](#) | [Website](#)

**End-of-life**

End of RHEL9 full support phase.

### Choose version:

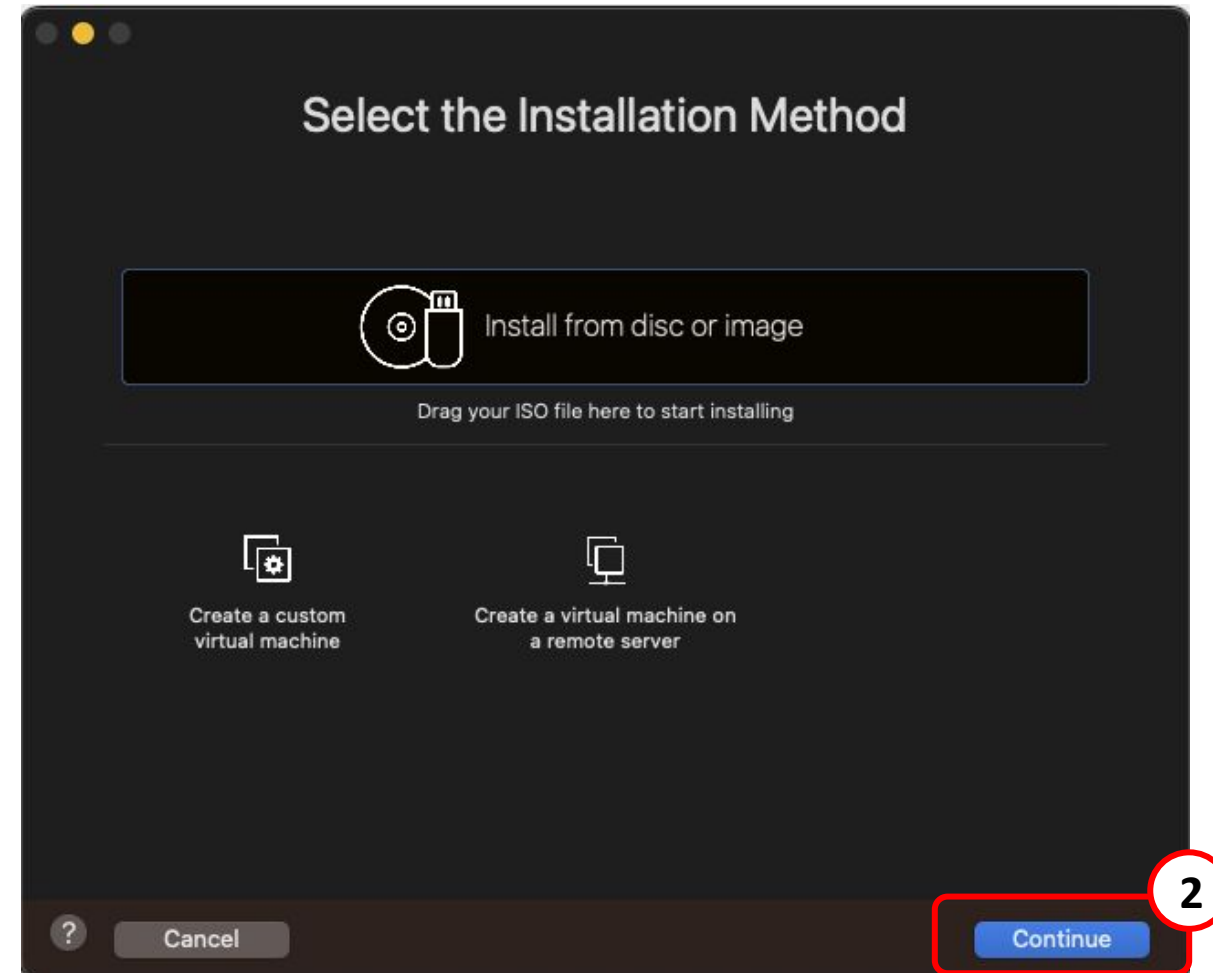
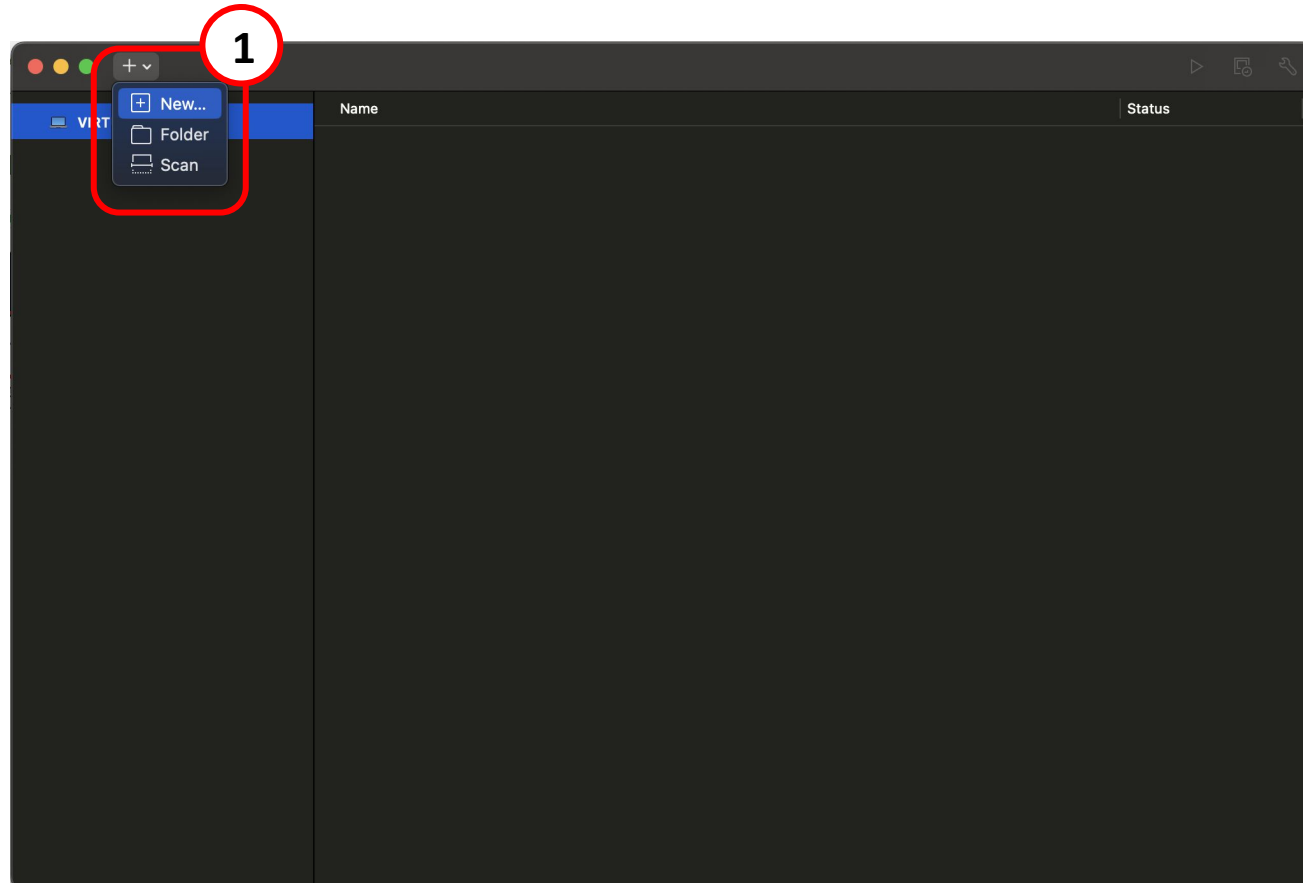
- **x86\_64**: For Windows/Ubuntu/Mac Intel
- **ARM64 (aarch64)**: For Mac M1/M2



Download file .iso of CentOS: [Link](#)

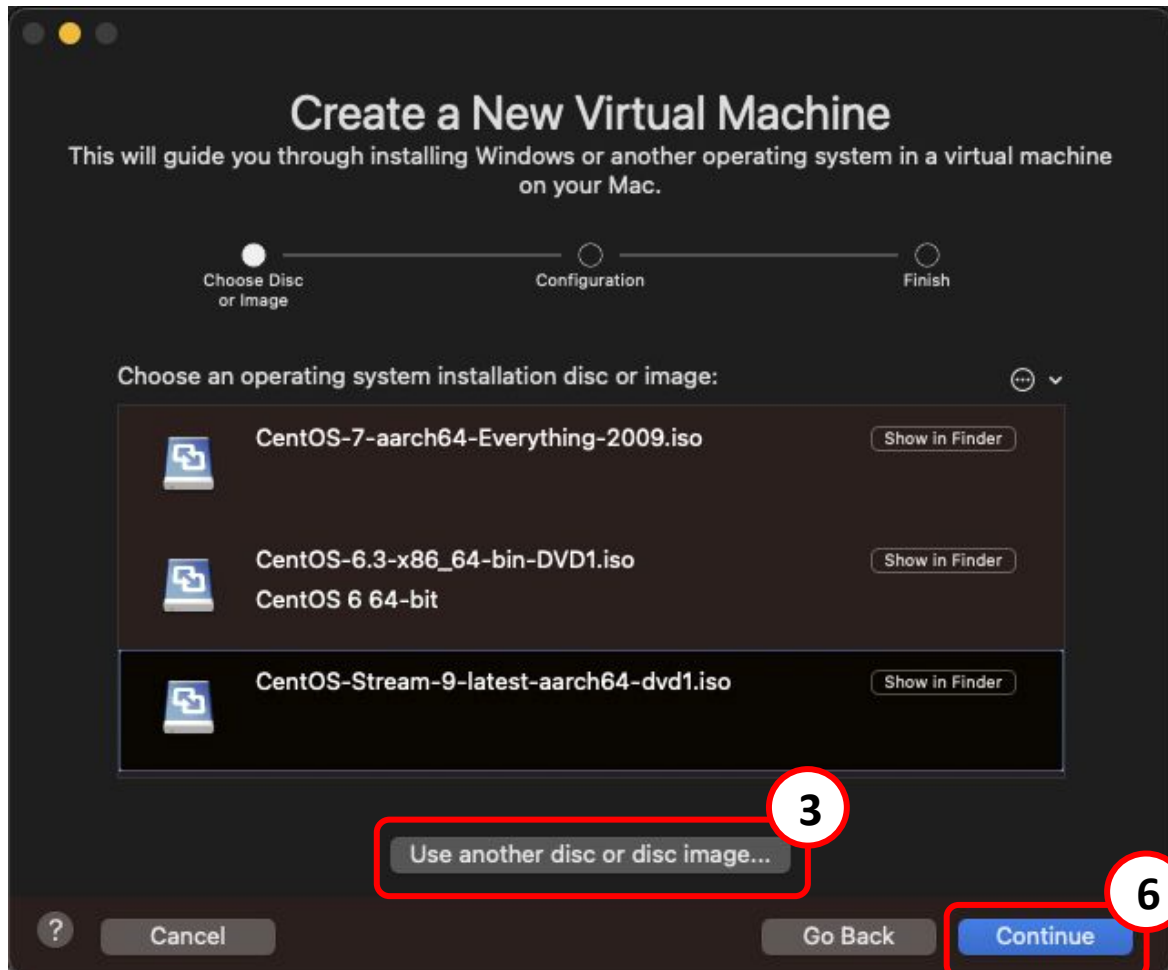
# Hadoop

## ❖ Installation: Run CentOS in VMware



# Hadoop

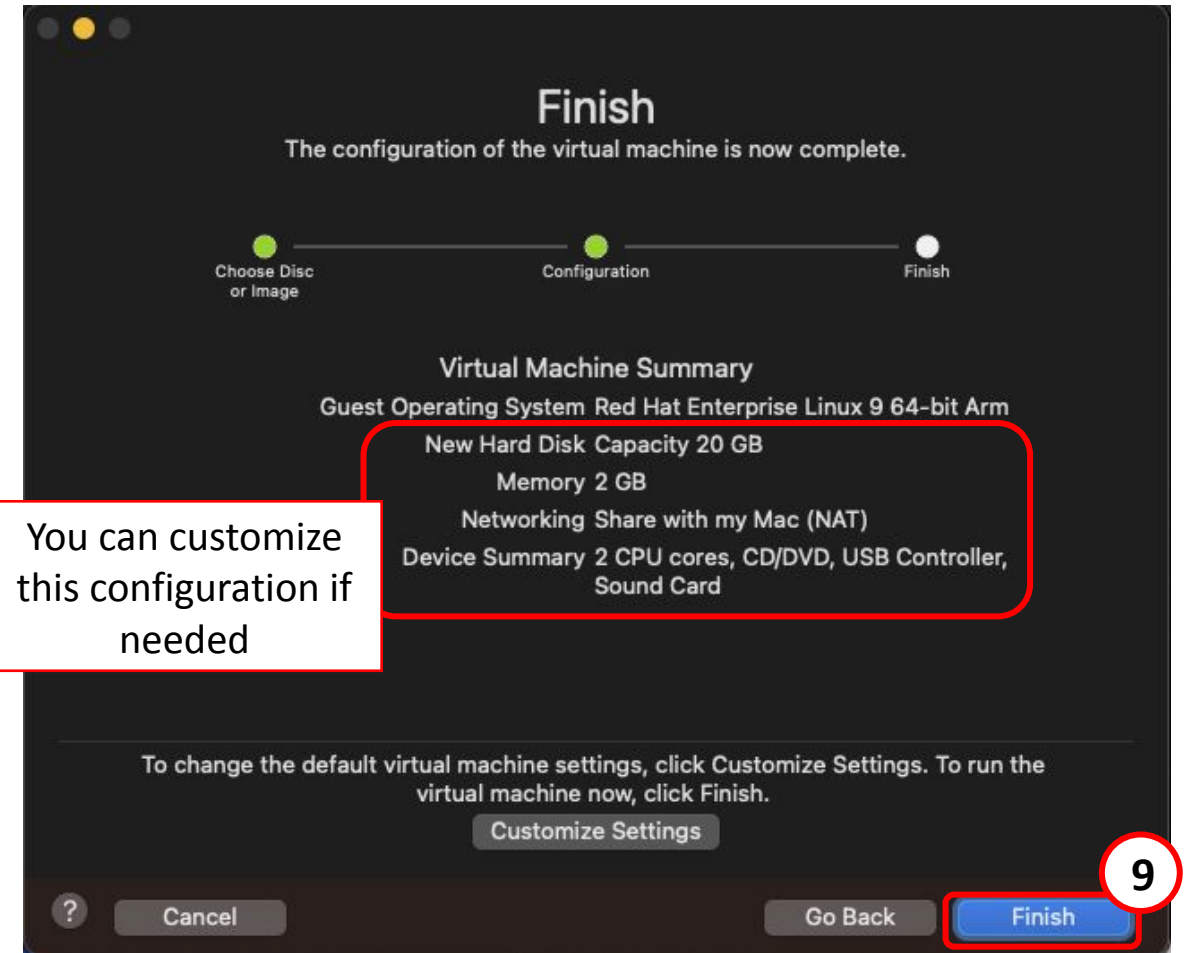
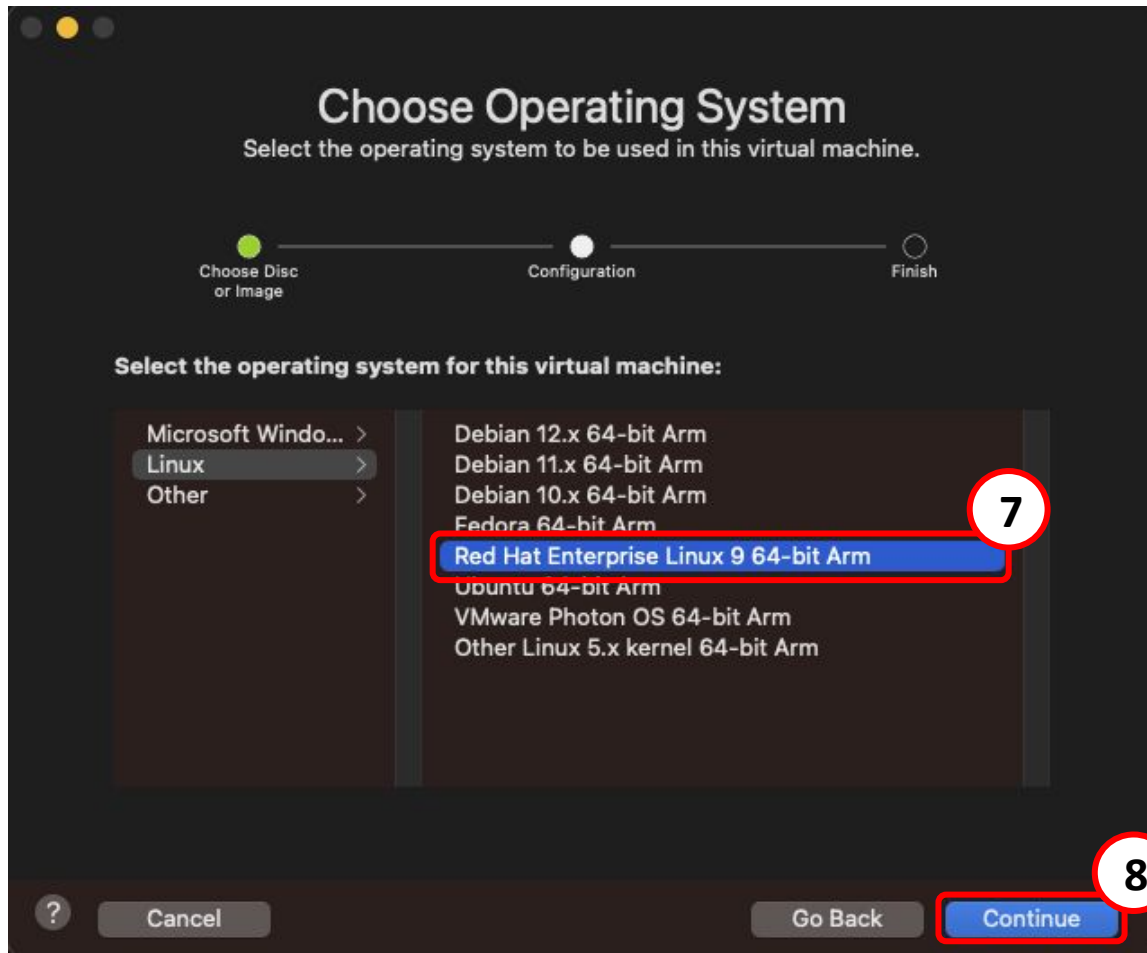
## ❖ Installation: Run CentOS in VMware



Choose CentOS .iso file

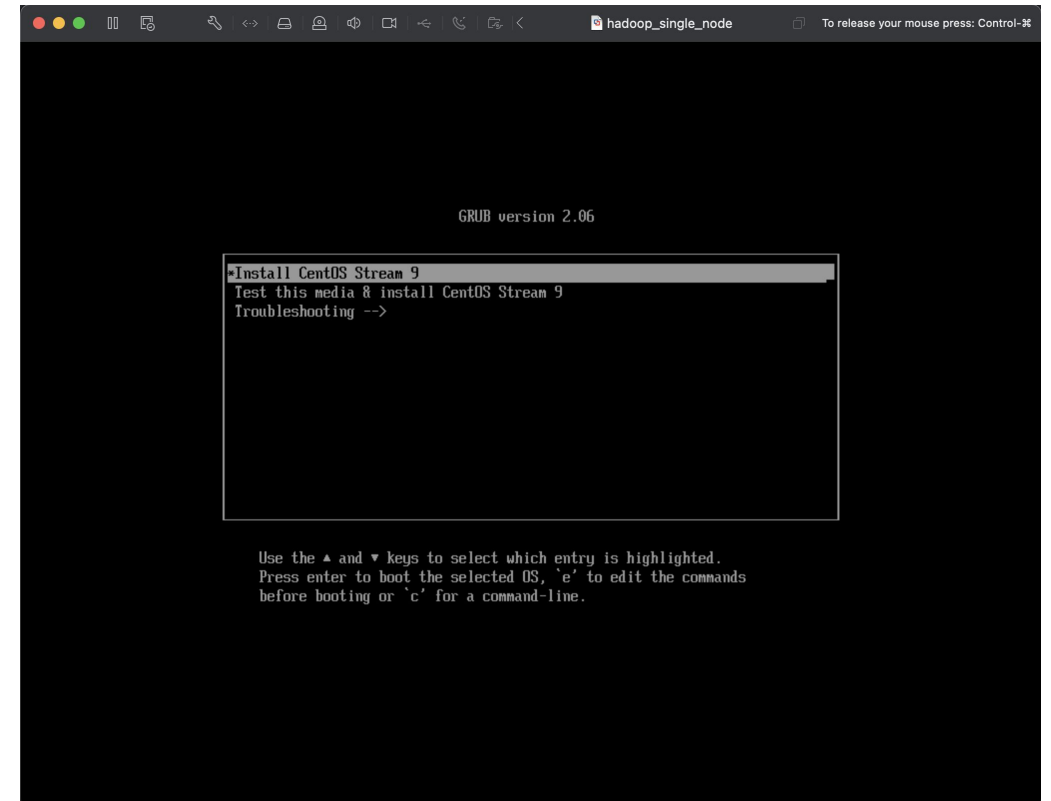
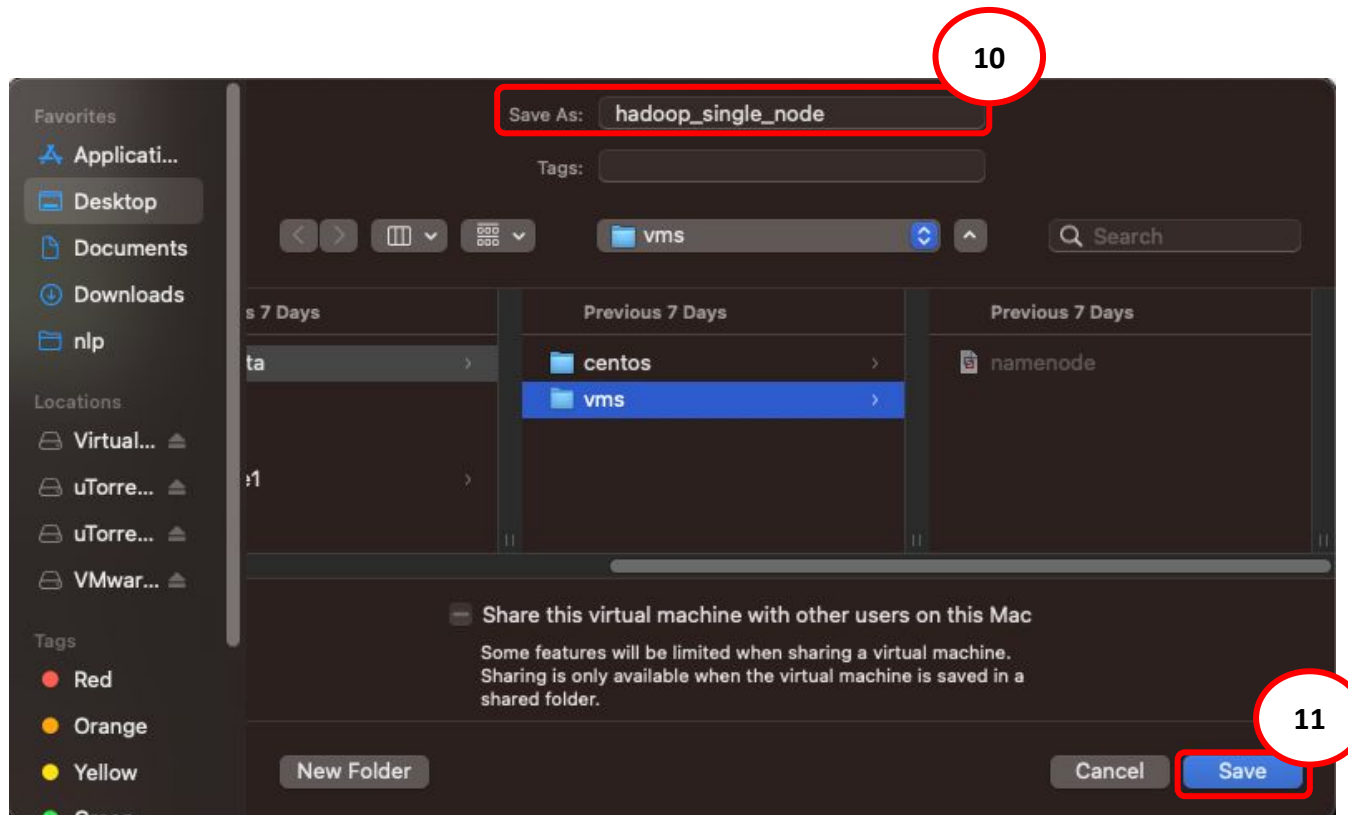
# Hadoop

## ❖ Installation: Run CentOS in VMware



# Hadoop

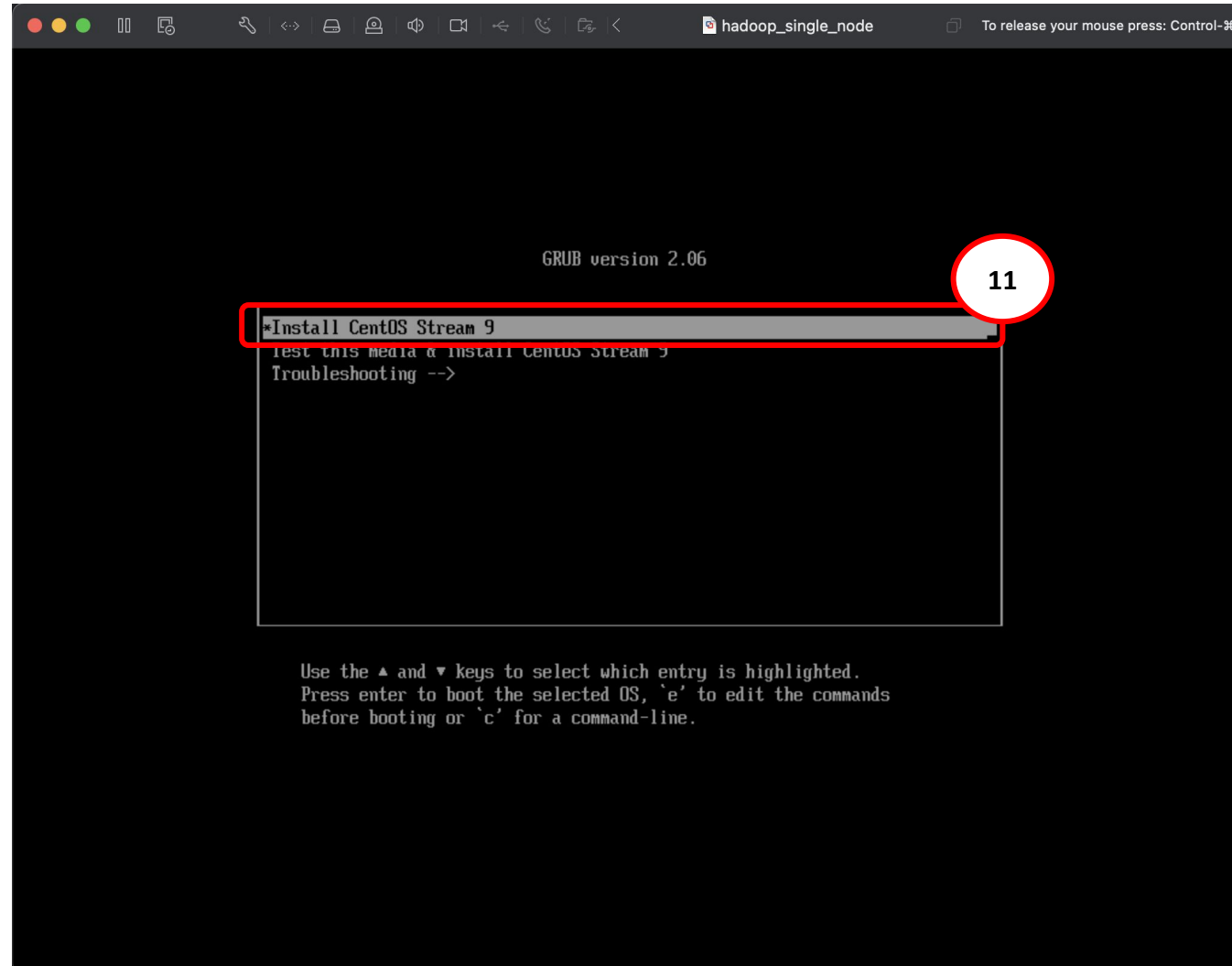
## ❖ Installation: Run CentOS in VMware



Choose where to save the new virtual machine

# Hadoop

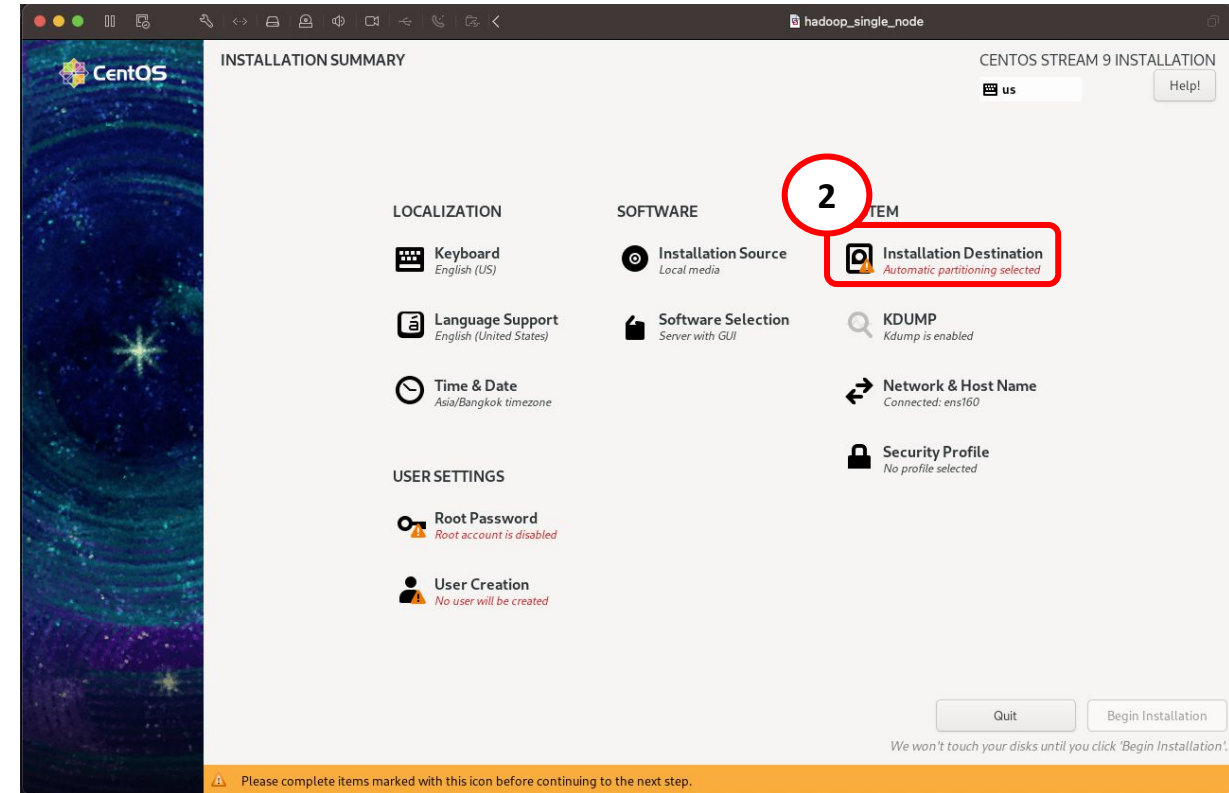
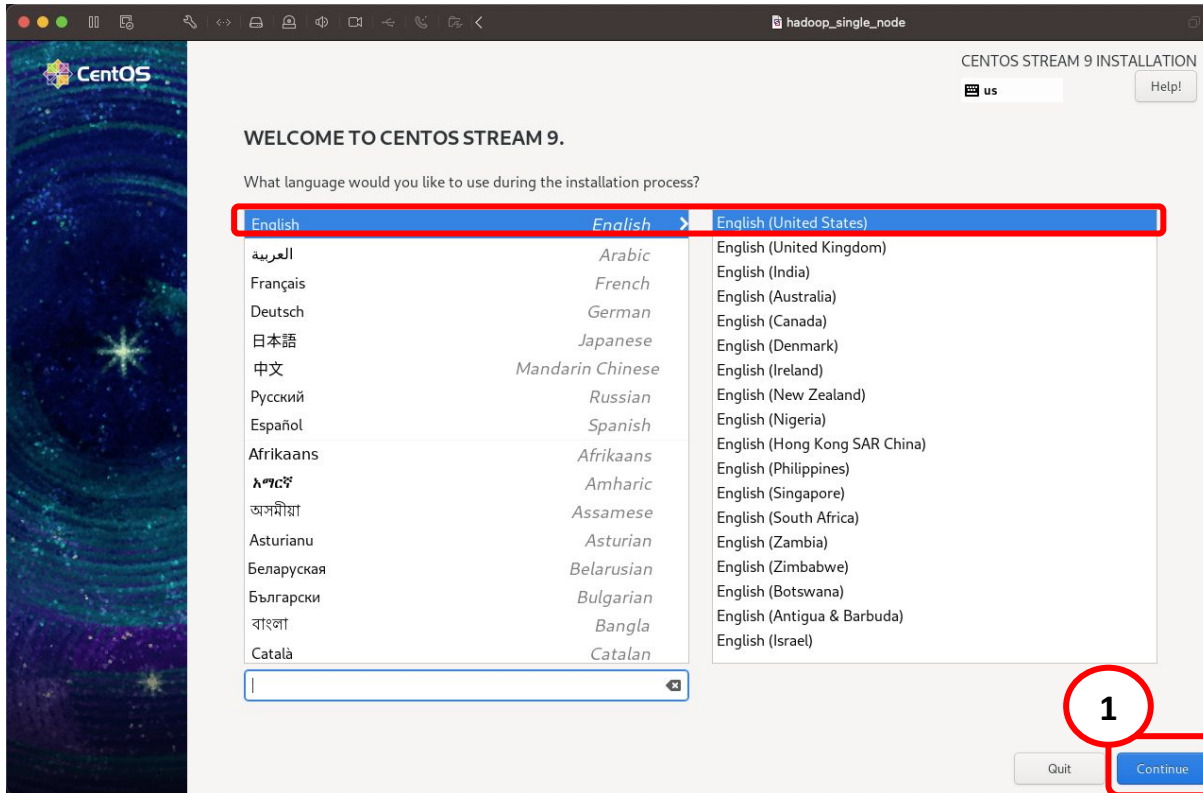
## ❖ Installation: Run CentOS in VMware



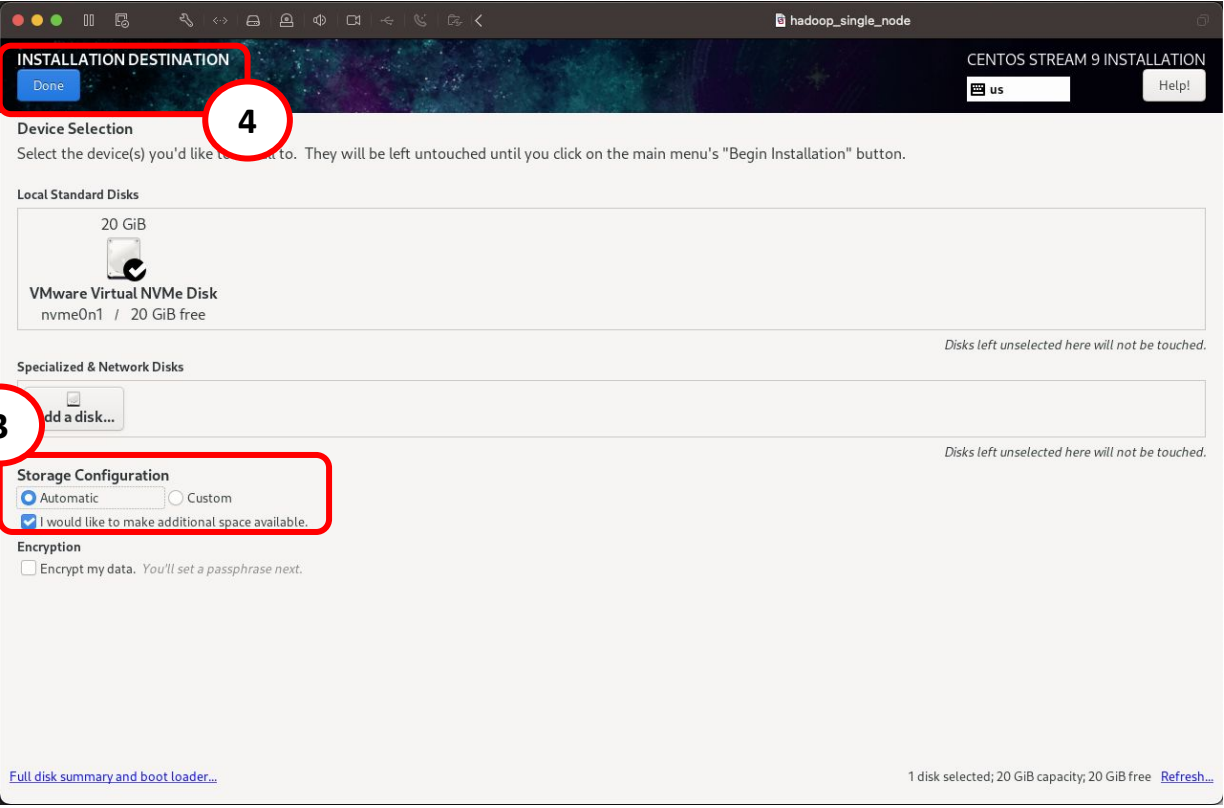


# Hadoop

## ❖ Installation: Setup CentOS



## ❖ Installation: Setup CentOS



The screenshot shows the 'INSTALLATION DESTINATION' window for CentOS Stream 9. It features a 'Done' button at the top left. The 'Device Selection' section includes instructions and a list of 'Local Standard Disks' with a '20 GiB VMware Virtual NVMe Disk' (nvme0n1) showing 20 GiB free space. A 'Specialized & Network Disks' section has an 'Add a disk...' button. The 'Storage Configuration' section has 'Automatic' selected and a checked option 'I would like to make additional space available.'. The 'Encryption' section has 'Encrypt my data' unchecked. A status bar at the bottom indicates '1 disk selected; 20 GiB capacity; 20 GiB free' with a 'Refresh...' link.

INSTALLATION DESTINATION

Done

CENTOS STREAM 9 INSTALLATION

us Help!

Device Selection

Select the device(s) you'd like to install to. They will be left untouched until you click on the main menu's "Begin Installation" button.

Local Standard Disks

20 GiB

VMware Virtual NVMe Disk  
nvme0n1 / 20 GiB free

Specialized & Network Disks

Add a disk...

Storage Configuration

☒ Automatic ☐ Custom

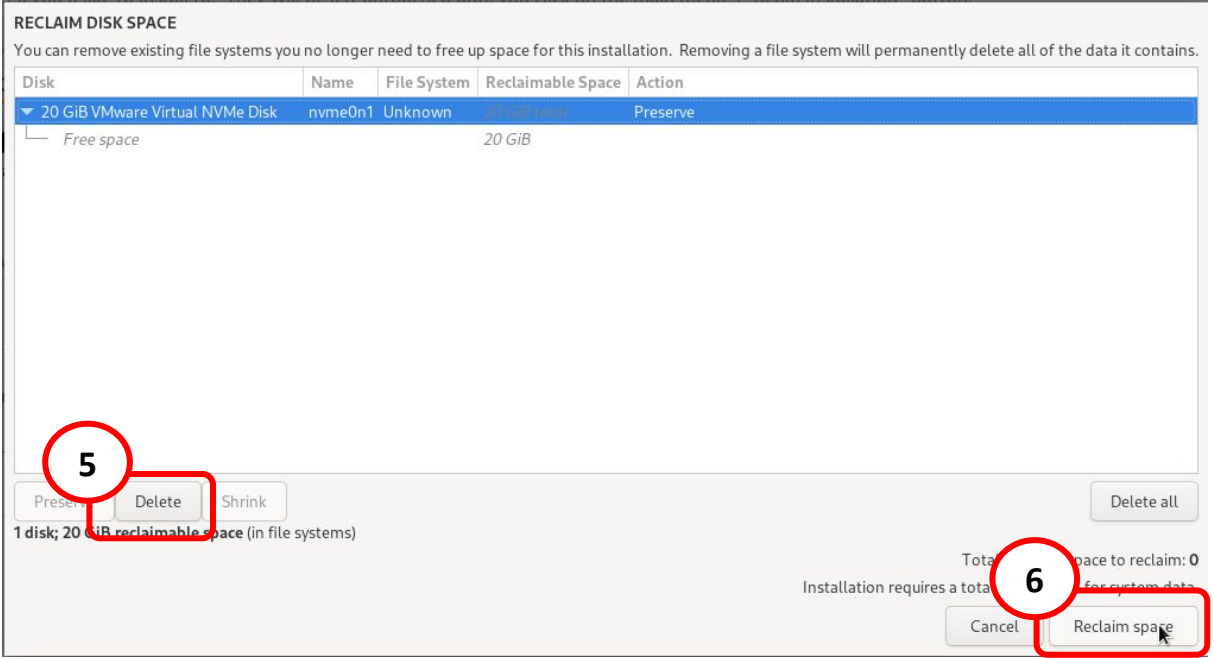
☒ I would like to make additional space available.

Encryption

☐ Encrypt my data. You'll set a passphrase next.

Full disk summary and boot loader...

1 disk selected; 20 GiB capacity; 20 GiB free [Refresh...](#)



The screenshot shows the 'RECLAIM DISK SPACE' window. It contains a table with columns: Disk, Name, File System, Reclaimable Space, and Action. The table lists '20 GiB VMware Virtual NVMe Disk' (nvme0n1, Unknown) with 20 GiB reclaimable space. Below the table, there are buttons for 'Preserve', 'Delete', 'Shrink', and 'Delete all'. A summary line states '1 disk; 20 GiB reclaimable space (in file systems)'. At the bottom right, it shows 'Total space to reclaim: 0' and 'Installation requires a total of 0 space for system data', with 'Cancel' and 'Reclaim space' buttons.

RECLAIM DISK SPACE

You can remove existing file systems you no longer need to free up space for this installation. Removing a file system will permanently delete all of the data it contains.

| Disk                            | Name    | File System | Reclaimable Space | Action   |
|---------------------------------|---------|-------------|-------------------|----------|
| 20 GiB VMware Virtual NVMe Disk | nvme0n1 | Unknown     | 20 GiB            | Preserve |

Free space 20 GiB

Preserve Delete Shrink Delete all

1 disk; 20 GiB reclaimable space (in file systems)

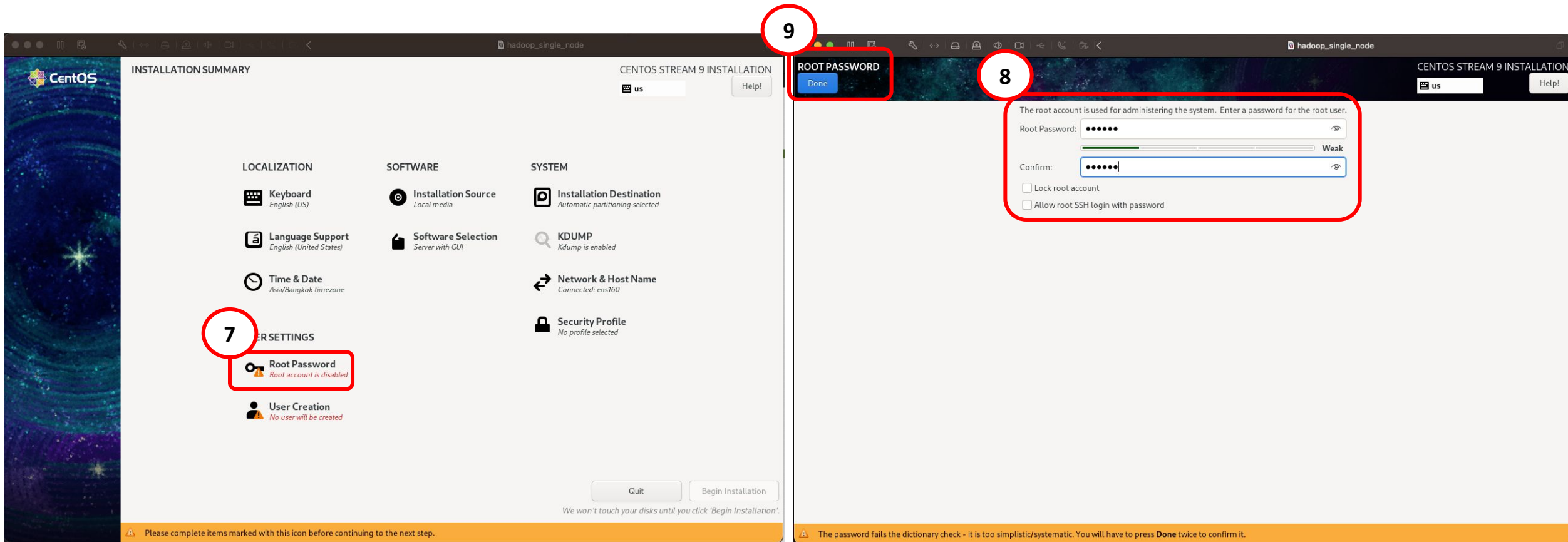
Total space to reclaim: 0

Installation requires a total of 0 space for system data

Cancel Reclaim space

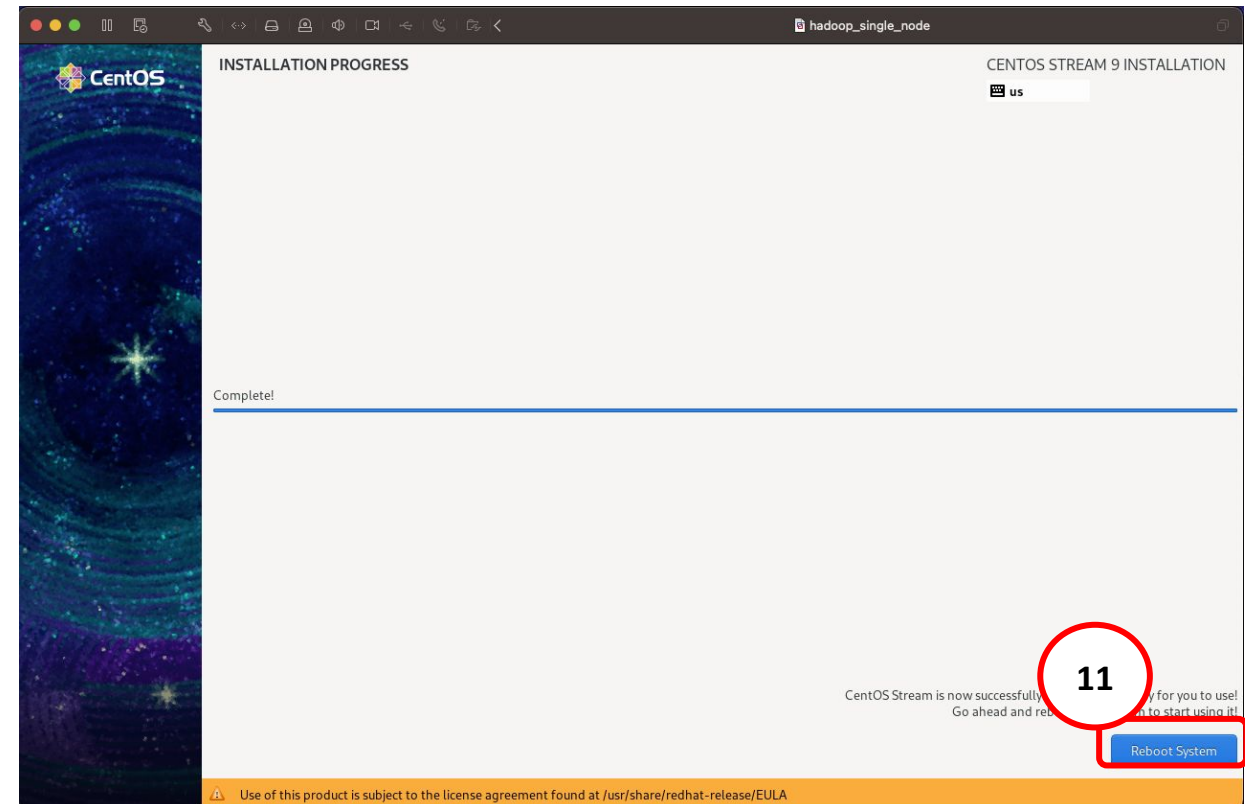
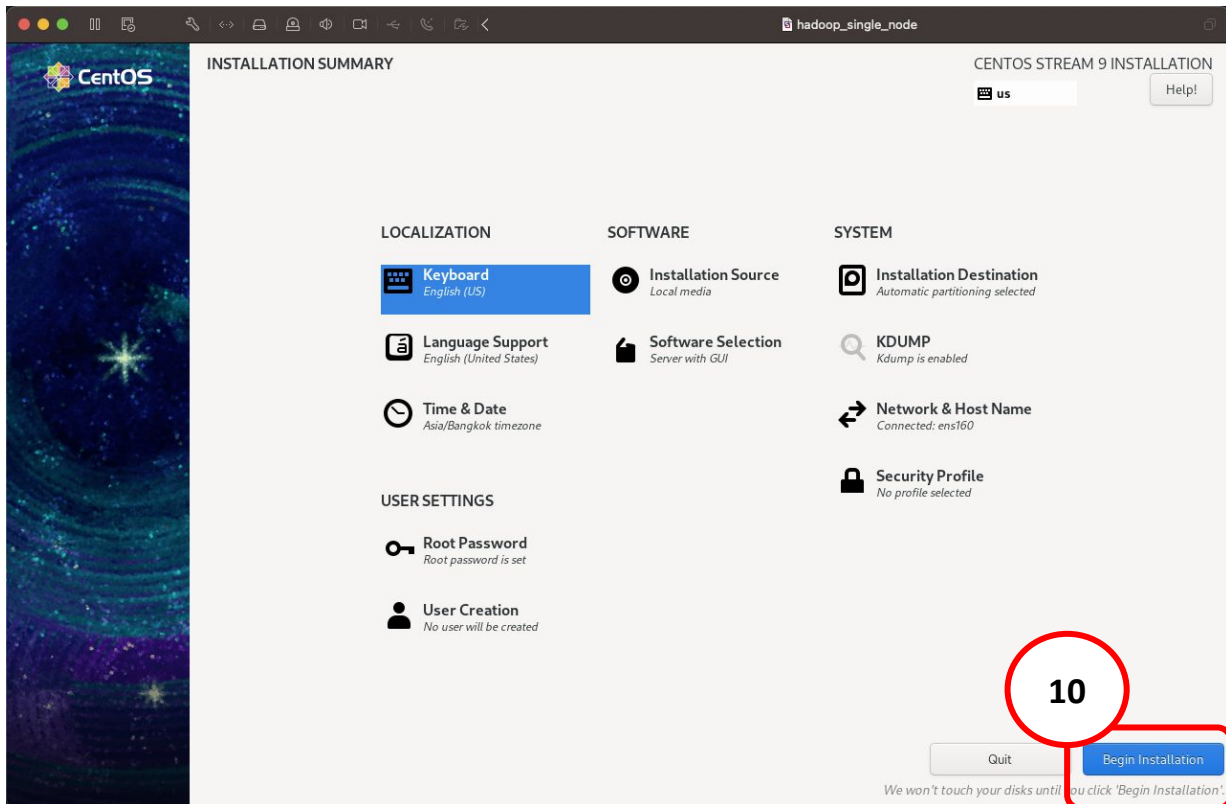
# Hadoop

## ◆ Installation: Setup CentOS



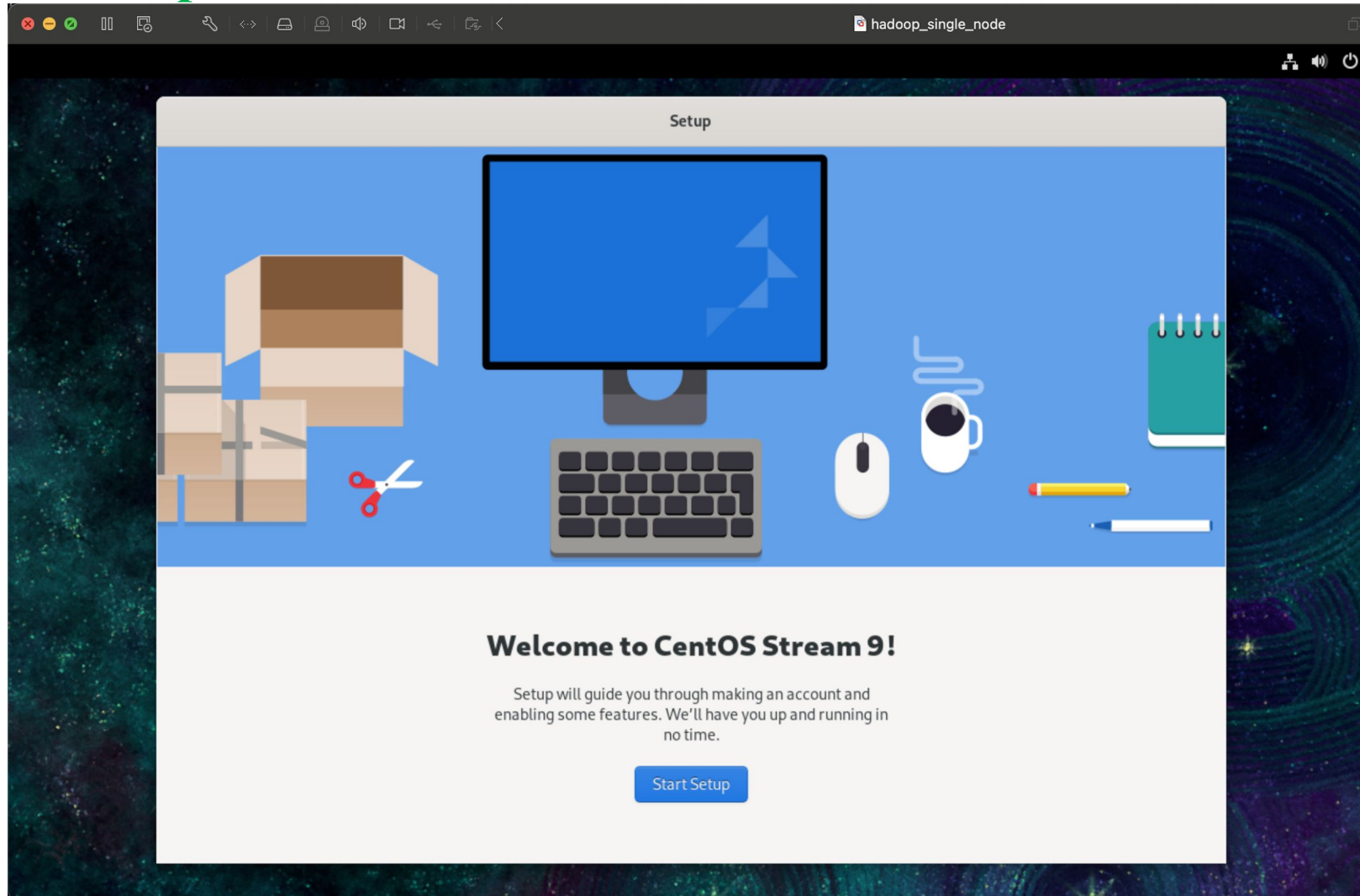
# Hadoop

## ◆ Installation: Setup CentOS



# Hadoop

## ❖ Installation: Setup CentOS



## ◆ Installation: Setup CentOS

[Previous](#)

About You

[Next](#)



**About You**

We need a few details to complete setup.

Full Name

Username

This will be used to name your home folder and can't be changed.

[Enterprise Login](#)

[Previous](#)

Password

[Next](#)



**Set a Password**

Be careful not to lose your password.

Password

This is a weak password. Try to add more letters, numbers and punctuation.

Confirm



# Hadoop

## ❖ Installation: Hadoop Requirements

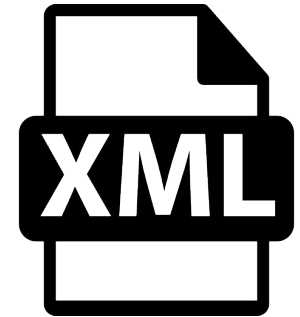
1. Install Java



2. Install Hadoop



3. Set up Hadoop



# Hadoop

## ❖ Installation: Java



**Java:** A high-level, class-based, object-oriented programming language that is designed to have as few implementation dependencies as possible.

### Install Java 8 on CentOS:

1. `sudo yum update -y`
2. `sudo yum install java-11-openjdk-devel -y`

```
aivn@localhost:~  
[aivn@localhost ~]$ java -version  
openjdk version "11.0.20.1" 2023-08-24 LTS  
OpenJDK Runtime Environment (Red_Hat-11.0.20.1.1-2) (build 11.0.20.1+1-LTS)  
OpenJDK 64-Bit Server VM (Red_Hat-11.0.20.1.1-2) (build 11.0.20.1+1-LTS, mixed mode, sharing)  
[aivn@localhost ~]$
```

# Hadoop

## ❖ Installation: Config ssh

```
aivn@localhost:~  
[aivn@localhost ~]$ ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa  
Generating public/private rsa key pair.  
Created directory '/home/aivn/.ssh'.  
Your identification has been saved in /home/aivn/.ssh/id_rsa  
Your public key has been saved in /home/aivn/.ssh/id_rsa.pub  
The key fingerprint is:  
SHA256:uPJLiVZDYYIbdlRATYHfG+2lPIl7XGUfKdlhqfJ+onM aivn@localhost.localdomain  
The key's randomart image is:  
+---[RSA 3072]-----+  
|  +=**      .|  
|  + oo..    +|  
|  . + ... .  = o|  
|  . ...o ...+o+|  
|    + S* +oo...|  
|    o +o * .. .|  
|    + +  o o.   |  
|    . +  . o. E .|  
|    o. . .+ o   |  
+----[SHA256]-----+  
[aivn@localhost ~]$
```

```
aivn@localhost:~ — ssh localhost  
[aivn@localhost ~]$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys  
[aivn@localhost ~]$ chmod 0600 ~/.ssh/authorized_keys  
[aivn@localhost ~]$ ssh localhost  
The authenticity of host 'localhost (:::1)' can't be established.  
ED25519 key fingerprint is SHA256:hLsA7L850MS4eyT+i+dsSnw9iDrE7ErypPzwTor0Xig.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.  
Activate the web console with: systemctl enable --now cockpit.socket  
  
Last login: Fri Aug  2 23:40:23 2024  
[aivn@localhost ~]$
```

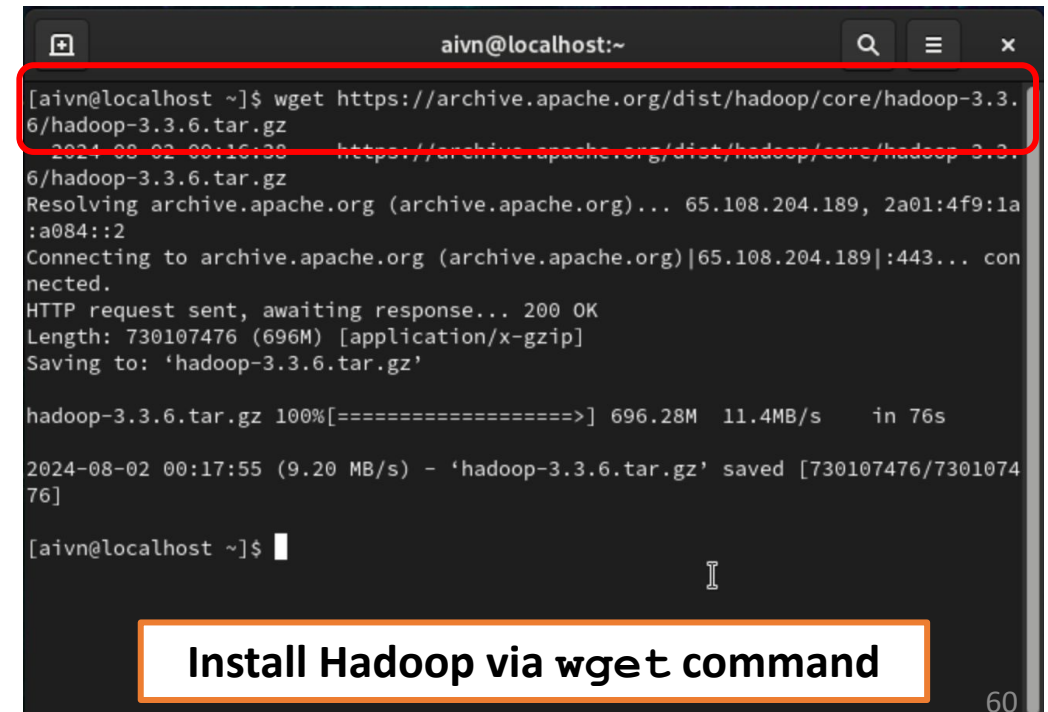
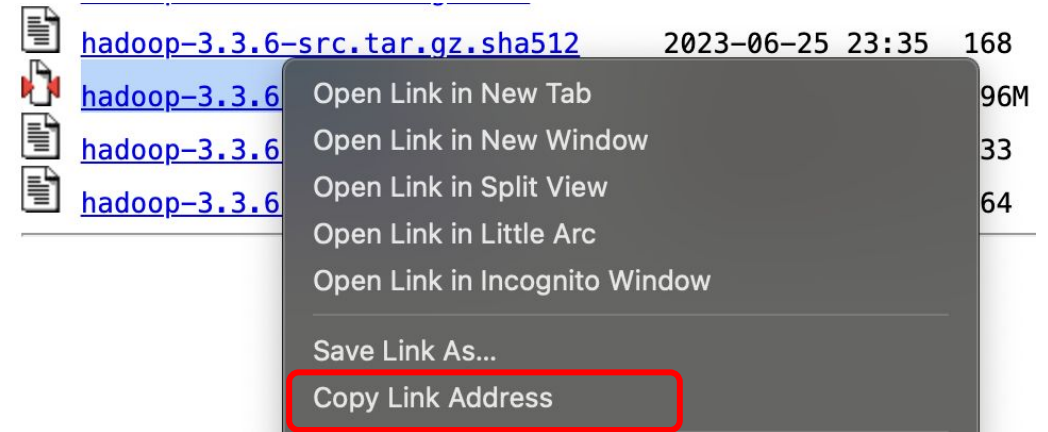
# Hadoop

## ◆ Installation: Hadoop

### Index of /dist/hadoop/core/hadoop-3.3.6

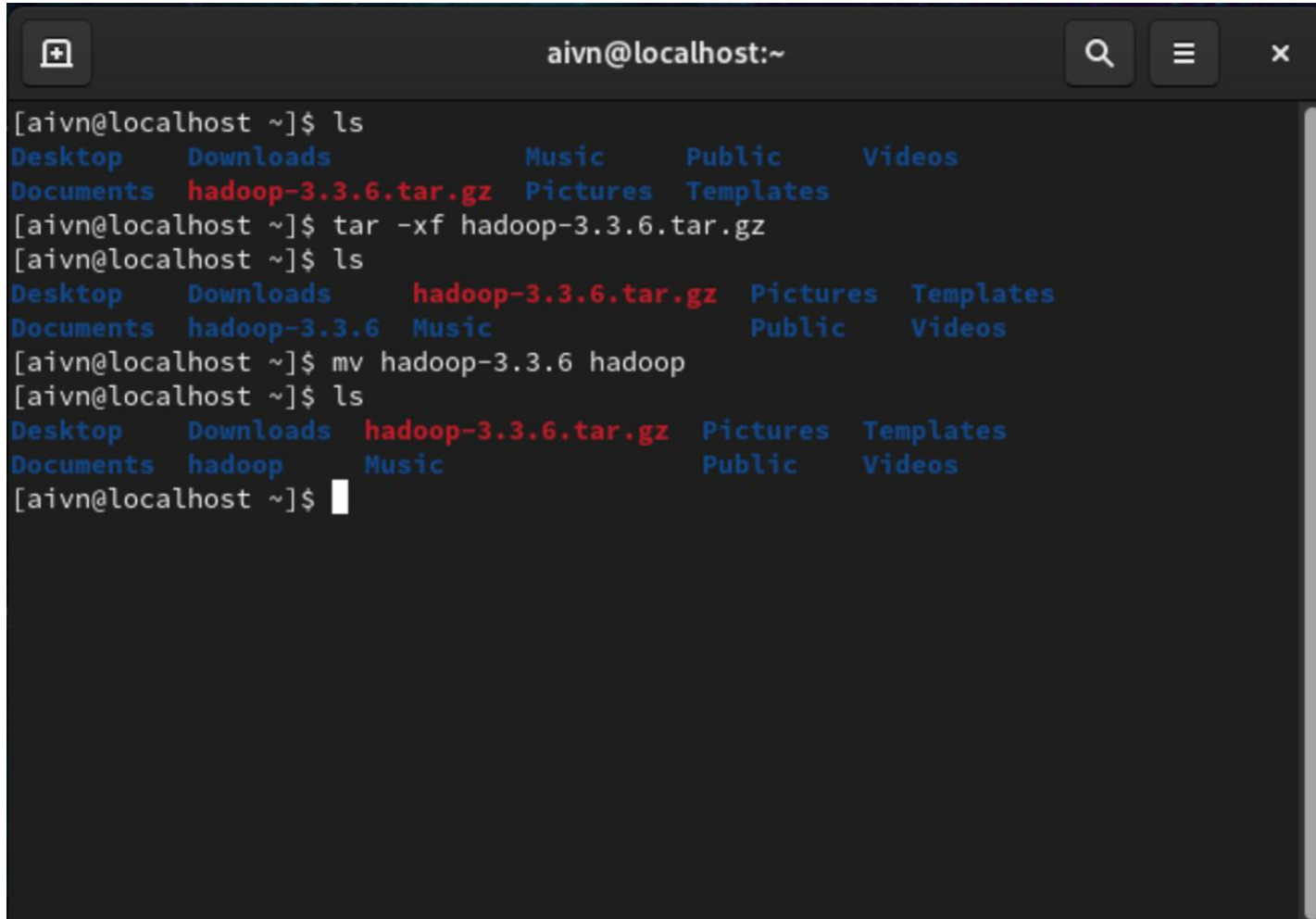
| Name                                               | Last modified    | Size | Description |
|----------------------------------------------------|------------------|------|-------------|
| <a href="#">Parent Directory</a>                   |                  | -    |             |
| <a href="#">CHANGELOG.md</a>                       | 2023-06-25 23:35 | 22K  |             |
| <a href="#">CHANGELOG.md.asc</a>                   | 2023-06-25 23:35 | 833  |             |
| <a href="#">CHANGELOG.md.sha512</a>                | 2023-06-25 23:35 | 153  |             |
| <a href="#">RELEASENOTES.md</a>                    | 2023-06-25 23:35 | 1.4K |             |
| <a href="#">RELEASENOTES.md.asc</a>                | 2023-06-25 23:35 | 833  |             |
| <a href="#">RELEASENOTES.md.sha512</a>             | 2023-06-25 23:35 | 156  |             |
| <a href="#">hadoop-3.3.6-aarch64.tar.gz</a>        | 2023-06-25 23:35 | 710M |             |
| <a href="#">hadoop-3.3.6-aarch64.tar.gz.asc</a>    | 2023-06-25 23:35 | 833  |             |
| <a href="#">hadoop-3.3.6-aarch64.tar.gz.sha512</a> | 2023-06-25 23:35 | 164  |             |
| <a href="#">hadoop-3.3.6-rat.txt</a>               | 2023-06-25 23:35 | 2.0M |             |
| <a href="#">hadoop-3.3.6-rat.txt.asc</a>           | 2023-06-25 23:35 | 833  |             |
| <a href="#">hadoop-3.3.6-rat.txt.sha512</a>        | 2023-06-25 23:35 | 165  |             |
| <a href="#">hadoop-3.3.6-site.tar.gz</a>           | 2023-06-25 23:35 | 41M  |             |
| <a href="#">hadoop-3.3.6-site.tar.gz.asc</a>       | 2023-06-25 23:35 | 833  |             |
| <a href="#">hadoop-3.3.6-site.tar.gz.sha512</a>    | 2023-06-25 23:35 | 169  |             |
| <a href="#">hadoop-3.3.6-src.tar.gz</a>            | 2023-06-25 23:35 | 36M  |             |
| <a href="#">hadoop-3.3.6-src.tar.gz.asc</a>        | 2023-06-25 23:35 | 833  |             |
| <a href="#">hadoop-3.3.6-src.tar.gz.sha512</a>     | 2023-06-25 23:35 | 168  |             |
| <a href="#">hadoop-3.3.6.tar.gz</a>                | 2023-06-25 23:35 | 696M |             |
| <a href="#">hadoop-3.3.6.tar.gz.asc</a>            | 2023-06-25 23:35 | 833  |             |
| <a href="#">hadoop-3.3.6.tar.gz.sha512</a>         | 2023-06-25 23:35 | 164  |             |

Hadoop 3.3.6  
download  
page



# Hadoop

## ❖ Installation: Hadoop

A terminal window titled 'aivn@localhost:~' showing the installation of Hadoop. The user runs 'ls' to see the current directory contents, then 'tar -xf hadoop-3.3.6.tar.gz' to extract the files. After another 'ls', they run 'mv hadoop-3.3.6 hadoop' to rename the directory. A final 'ls' confirms the new directory name.

```
[aivn@localhost ~]$ ls
Desktop  Downloads  Music      Public     Videos
Documents hadoop-3.3.6.tar.gz Pictures Templates

[aivn@localhost ~]$ tar -xf hadoop-3.3.6.tar.gz
[aivn@localhost ~]$ ls
Desktop  Downloads  hadoop-3.3.6.tar.gz Pictures Templates
Documents hadoop-3.3.6 Music      Public     Videos

[aivn@localhost ~]$ mv hadoop-3.3.6 hadoop
[aivn@localhost ~]$ ls
Desktop  Downloads  hadoop-3.3.6.tar.gz Pictures Templates
Documents hadoop      Music      Public     Videos

[aivn@localhost ~]$
```

- Extract downloaded file via tar command:

```
tar -xf hadoop-3.3.6.tar.gz
```

- Rename Hadoop folder:

```
mv hadoop-3.3.6 hadoop
```

# Hadoop

## ❖ Installation: Config Hadoop

### 1. Check Java configuration

```
aivn@localhost:~ — ssh localhost
[aivn@localhost ~]$ sudo update-alternatives --config java
[sudo] password for aivn:

There is 1 program that provides 'java'.

  Selection    Command
-----
++ 1          java-11-openjdk.aarch64 (/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.aarch64/bin/java)

Enter to keep the current selection[+], or type selection number:
[aivn@localhost ~]$
```

### 2. Enter ~/.bashrc script

```
aivn@localhost:~
[aivn@localhost ~]$ sudo vi ~/.bashrc
```



# Hadoop

## ❖ Installation: Hadoop

### 3. Set up .bashrc script

```
aivn@localhost:~ — sudo vi .bashrc
# .bashrc

# Source global definitions
if [ -f /etc/bashrc ]; then
    . /etc/bashrc
fi

# User specific environment
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]
then
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
fi
export PATH

# Uncomment the following line if you don't like systemctl's auto-paging feature
:
# export SYSTEMD_PAGER=

# User specific aliases and functions
if [ -d ~/.bashrc.d ]; then
    for rc in ~/.bashrc.d/*; do
        if [ -f "$rc" ]; then
            . "$rc"
        fi
    done
fi
```

### Paste the code below:

```
export HDFS_NAMENODE_USER="aivn"
export HDFS_DATANODE_USER="aivn"
export HDFS_SECONDARYNAMENODE_USER="aivn"
export YARN_RESOURCEMANAGER_USER="aivn"
export YARN_NODEMANAGER_USER="aivn"
export
JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.a
arch64
export HADOOP_HOME=/home/aivn/hadoop
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export
HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"
```

# Hadoop

## ❖ Installation: Hadoop

```
aivn@localhost:~  
# .bashrc  
  
# Source global definitions  
if [ -f /etc/bashrc ]; then  
    . /etc/bashrc  
fi  
  
export HDFS_NAMENODE_USER="aivn"  
export HDFS_DATANODE_USER="aivn"  
export HDFS_SECONDARYNAMENODE_USER="aivn"  
export YARN_RESOURCEMANAGER_USER="aivn"  
export YARN_NODEMANAGER_USER="aivn"  
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.aarch64  
export HADOOP_HOME=/home/aivn/hadoop  
export HADOOP_MAPRED_HOME=$HADOOP_HOME  
export HADOOP_COMMON_HOME=$HADOOP_HOME  
export HADOOP_HDFS_HOME=$HADOOP_HOME  
export YARN_HOME=$HADOOP_HOME  
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native  
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin  
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"  
  
export SPARK_HOME=/home/aivn/spark
```

```
aivn@localhost:~  
[aivn@localhost ~]$ source .bashrc  
[aivn@localhost ~]$
```

3. Set up .bashrc script then run it

# Hadoop

## ❖ Installation: Hadoop

### 4. Config hadoop-env.sh

```
aivn@localhost:~  
[aivn@localhost ~]$ sudo vi hadoop/etc/hadoop/hadoop-env.sh
```

```
aivn@localhost:~  
# Generic settings for HADOOP  
###  
  
# Technically, the only required environment variable is JAVA_HOME.  
# All others are optional. However, the defaults are probably not  
# preferred. Many sites configure these options outside of Hadoop,  
# such as in /etc/profile.d  
  
# The java implementation to use. By default, this environment  
# variable is REQUIRED on ALL platforms except OS X!  
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.aarch64  
export HDFS_NAMENODE_USER="aivn"  
export HDFS_DATANODE_USER="aivn"  
export HDFS_SECONDARYNAMENODE_USER="aivn"  
export YARN_RESOURCEMANAGER_USER="aivn"  
export YARN_NODEMANAGER_USER="aivn"  
  
# Location of Hadoop. By default, Hadoop will attempt to determine  
# this location based upon its execution path.  
# export HADOOP_HOME=  
  
# Location of Hadoop's configuration information. i.e., where this  
# file is living. If this is not defined, Hadoop will attempt to  
<adoop/etc/hadoop/hadoop-env.sh" 435L, 16873B 61,31 10%
```

## ❖ Installation: Hadoop

### 4. Config hadoop-env.sh

```
aivn@localhost:~  
# Generic settings for HADOOP  
###  
  
# Technically, the only required environment variable is JAVA_HOME.  
# All others are optional. However, the defaults are probably not  
# preferred. Many sites configure these options outside of Hadoop,  
# such as in /etc/profile.d  
  
# The java implementation to use. By default, this environment  
# variable is REQUIRED on ALL platforms except OS X!  
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.aarch64  
export HDFS_NAMENODE_USER="aivn"  
export HDFS_DATANODE_USER="aivn"  
export HDFS_SECONDARYNAMENODE_USER="aivn"  
export YARN_RESOURCEMANAGER_USER="aivn"  
export YARN_NODEMANAGER_USER="aivn"  
  
# Location of Hadoop. By default, Hadoop will attempt to determine  
# this location based upon its execution path.  
# export HADOOP_HOME=  
  
# Location of Hadoop's configuration information. i.e., where this  
# file is living. If this is not defined, Hadoop will attempt to  
<adoop/etc/hadoop/hadoop-env.sh" 435L, 16873B 61,31 10%
```

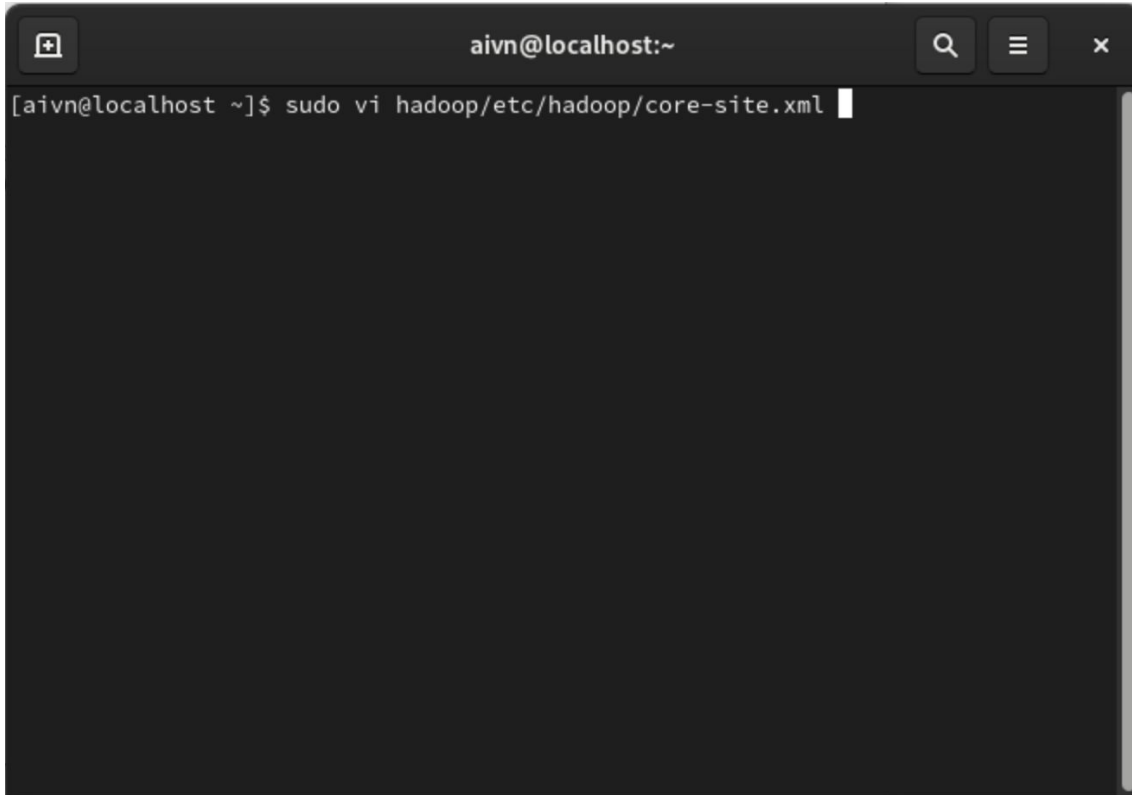
- export  
JAVA\_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.aarch64
- export HDFS\_NAMENODE\_USER="aivn"
- export HDFS\_DATANODE\_USER="aivn"
- export  
HDFS\_SECONDARYNAMENODE\_USER="aivn"
- export YARN\_RESOURCEMANAGER\_USER="aivn"
- export YARN\_NODEMANAGER\_USER="aivn"

# Hadoop

## ❖ Installation: Hadoop

### 5. Config core-site.xml

Paste the code below:

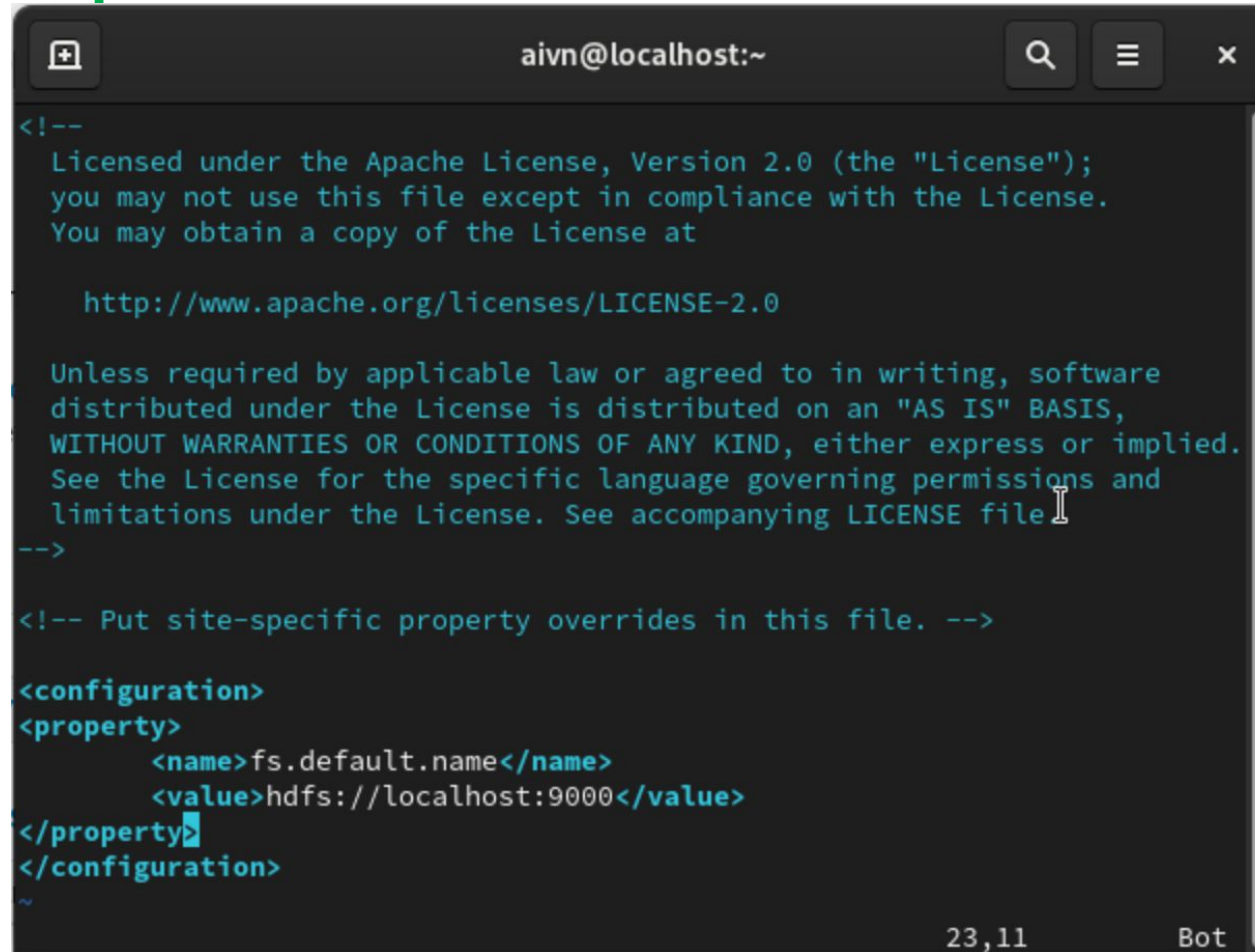
A terminal window with a dark background. The title bar shows 'aivn@localhost:~'. The command prompt shows '[aivn@localhost ~]\$ sudo vi hadoop/etc/hadoop/core-site.xml' with a cursor at the end of the line.

```
aivn@localhost:~  
[aivn@localhost ~]$ sudo vi hadoop/etc/hadoop/core-site.xml
```

```
<configuration>  
<property>  
  <name>fs.default.name</name>  
  <value>hdfs://localhost:9000</value>  
</property>  
</configuration>
```

# Hadoop

## ❖ Installation: Hadoop



```
<!--
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You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>fs.default.name</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
~
```

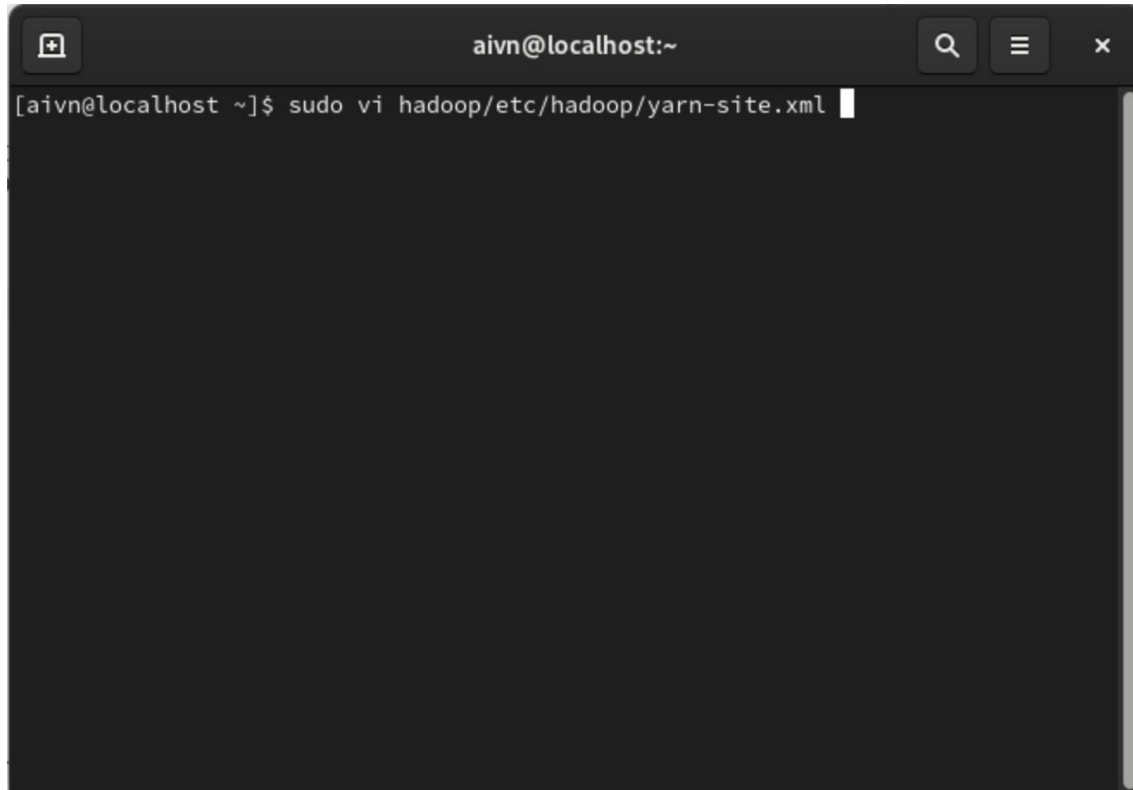
### 5. Config core-site.xml



# Hadoop

## ❖ Installation: Hadoop

### 6. Config yarn-site.xml



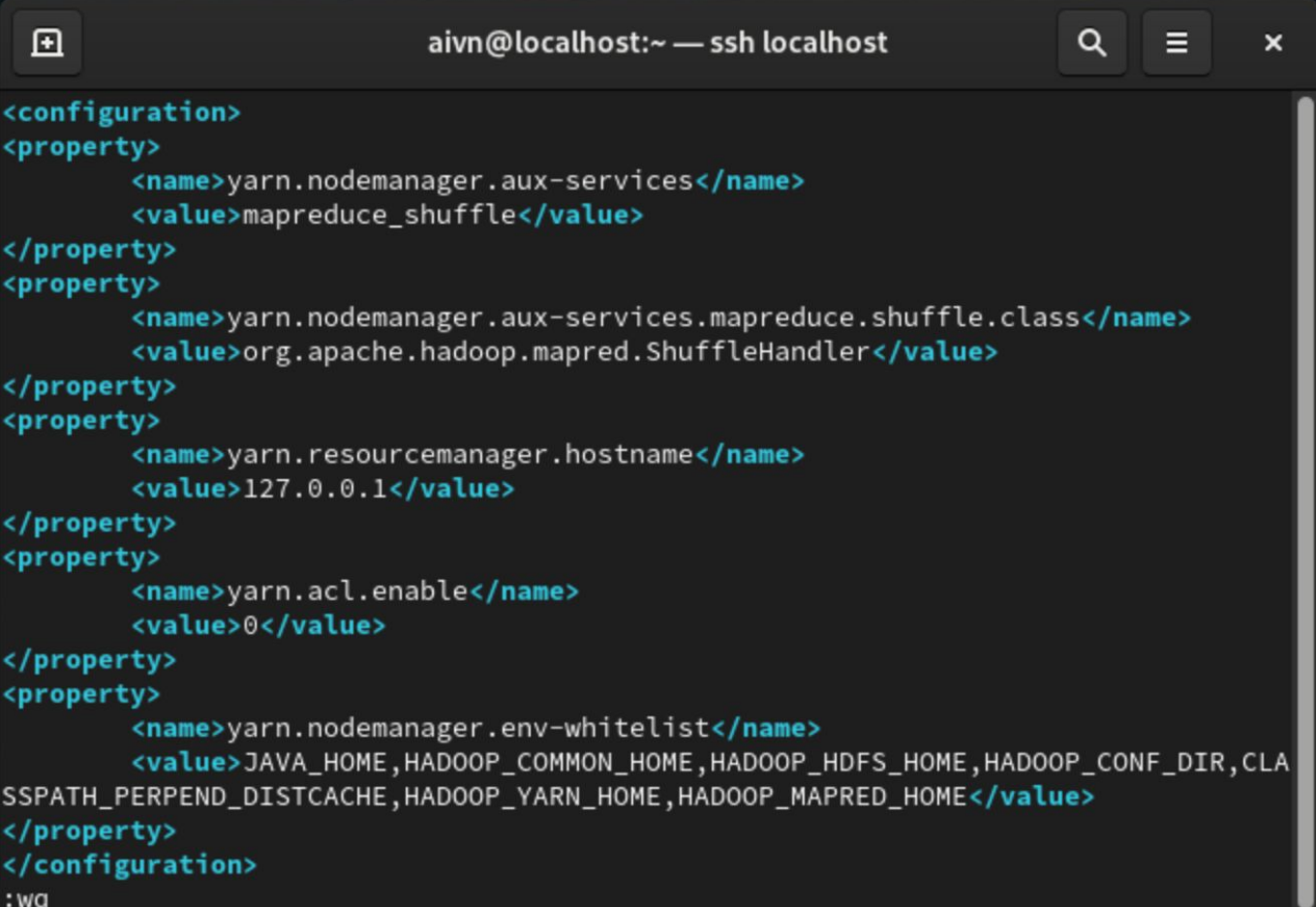
A terminal window with a dark background. The title bar shows 'aivn@localhost:~'. The command prompt is '[aivn@localhost ~]\$' and the command entered is 'sudo vi hadoop/etc/hadoop/yarn-site.xml'. A cursor is visible at the end of the command line.

### Paste the code below:

```
<configuration>
<property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
</property>
<property>
  <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
  <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
<property>
  <name>yarn.resourcemanager.hostname</name>
  <value>127.0.0.1</value>
</property>
<property>
  <name>yarn.acl.enable</name>
  <value>0</value>
</property>
<property>
  <name>yarn.nodemanager.env-whitelist</name>
  <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PERPEND_DISTCACHE,HADOOP_YARN_HOME,HADOOP_MAPRED_HOME</value>
</property>
</configuration>
```

# Hadoop

## ❖ Installation: Hadoop



A terminal window titled "aivn@localhost:~ — ssh localhost" displays the configuration of the `yarn-site.xml` file. The configuration is shown as XML code with syntax highlighting. The properties being configured are:

- `yarn.nodemanager.aux-services` with value `mapreduce_shuffle`
- `yarn.nodemanager.aux-services.mapreduce.shuffle.class` with value `org.apache.hadoop.mapred.ShuffleHandler`
- `yarn.resourcemanager.hostname` with value `127.0.0.1`
- `yarn.acl.enable` with value `0`
- `yarn.nodemanager.env-whitelist` with value `JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PERPEND_DISTCACHE,HADOOP_YARN_HOME,HADOOP_MAPRED_HOME`

The terminal shows the end of the configuration block with `</configuration>` and the cursor is at the bottom with `:wq`.

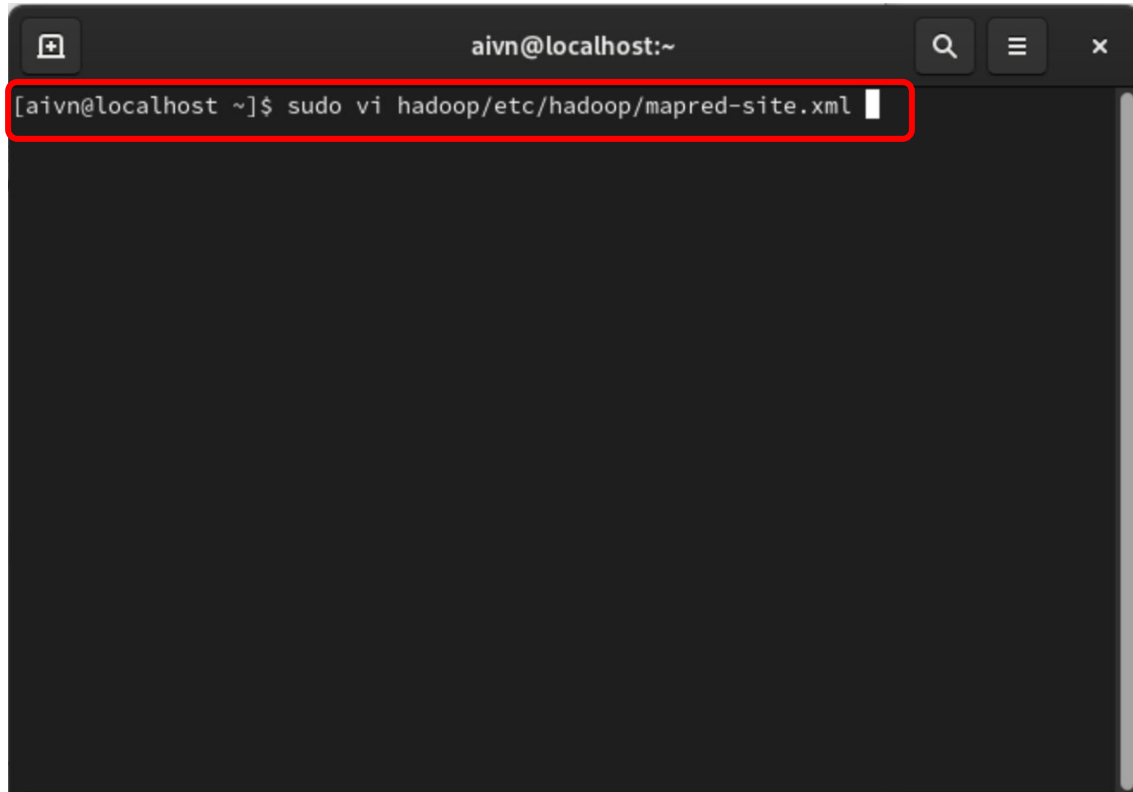
### 6. Config yarn-site.xml

# Hadoop

## ❖ Installation: Hadoop

### 7. Config mapred-site.xml

Paste the code below:

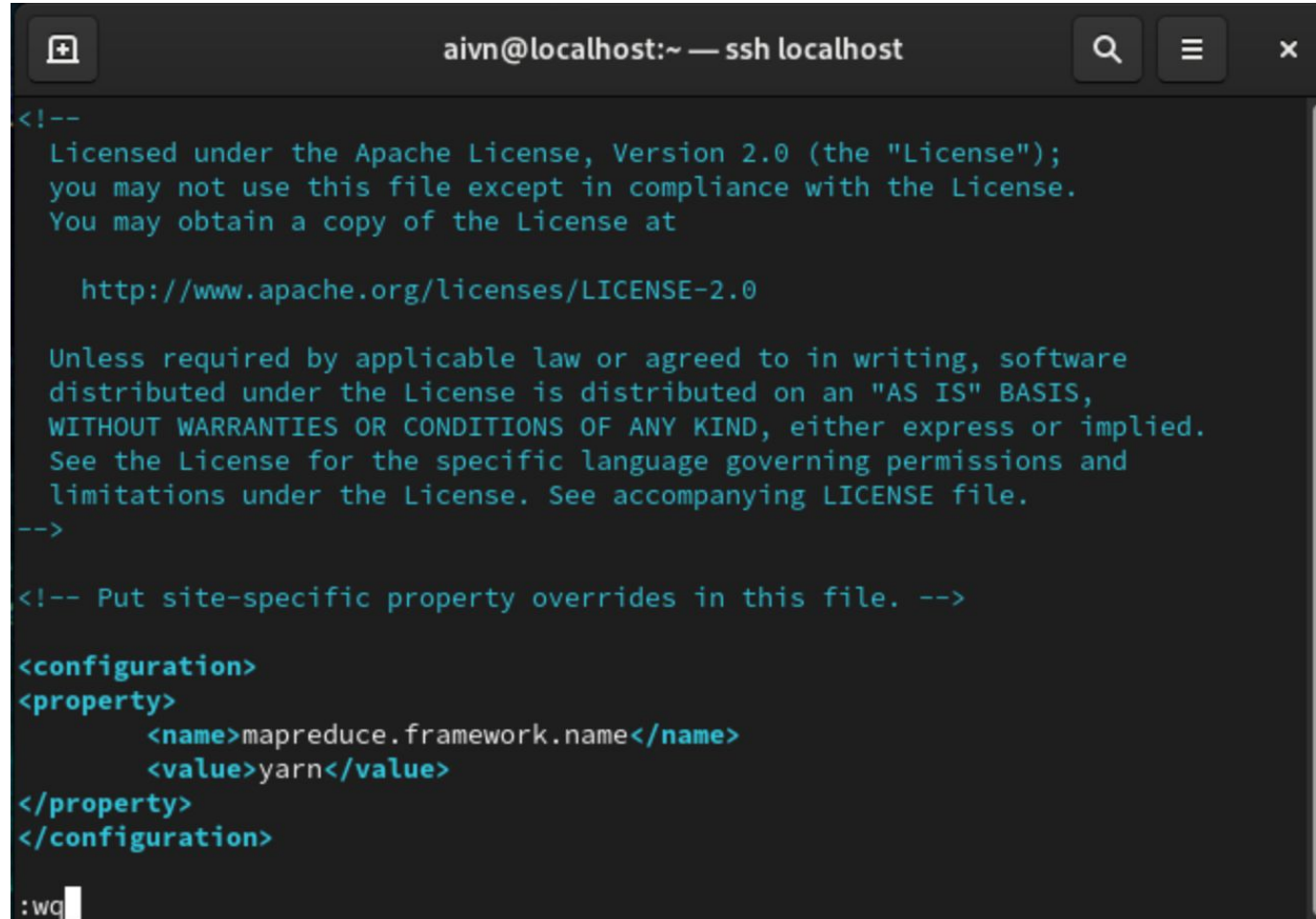
A terminal window with a dark background. The title bar shows 'aivn@localhost:~'. The command prompt is '[aivn@localhost ~]\$ sudo vi hadoop/etc/hadoop/mapred-site.xml'. The command is highlighted with a red box.

```
aivn@localhost:~  
[aivn@localhost ~]$ sudo vi hadoop/etc/hadoop/mapred-site.xml
```

```
<configuration>  
<property>  
    <name>mapreduce.framework.name</name>  
    <value>yarn</value>  
</property>  
</configuration>
```

# Hadoop

## ❖ Installation: Hadoop



```
aivn@localhost:~ — ssh localhost
<!--
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    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>mapreduce.framework.name</name>
  <value>yarn</value>
</property>
</configuration>

:wq
```

### 7. Config mapred-site.xml

# Hadoop

## ❖ Installation: Hadoop

### 8.1. Create namenode and datanode folder

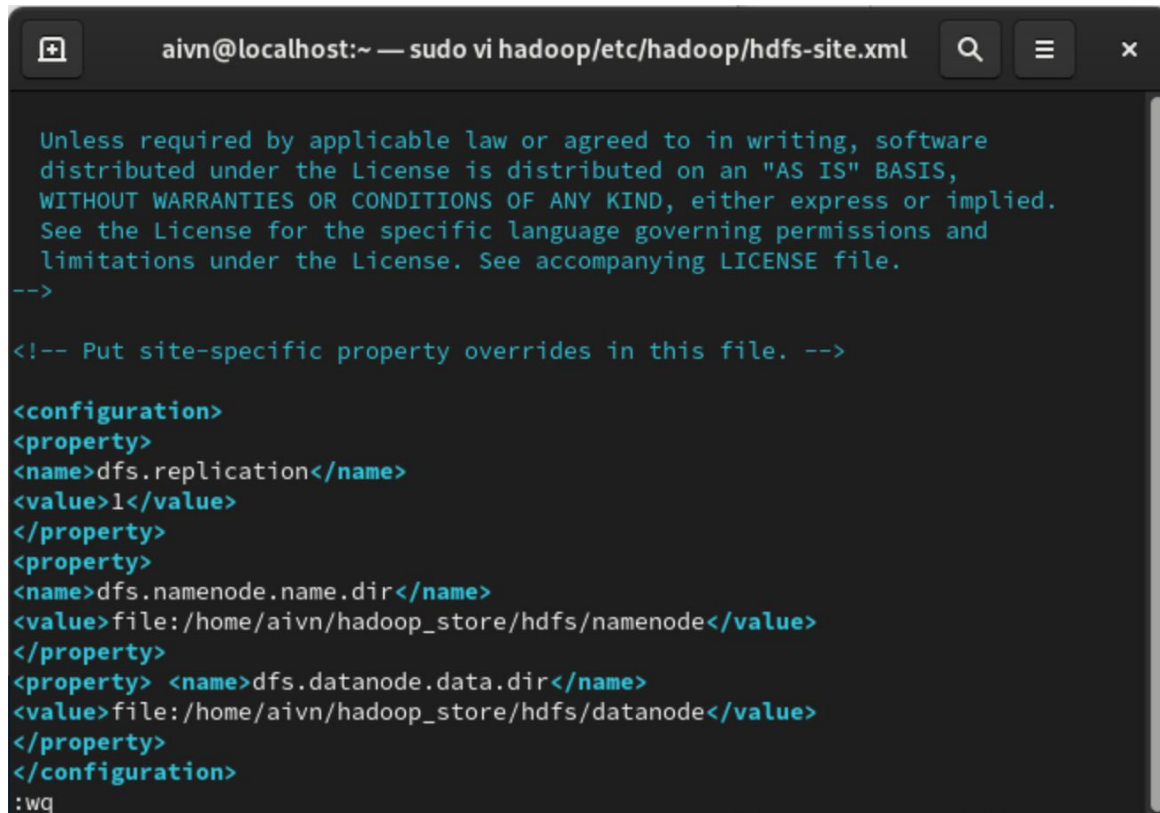
```
aivn@localhost:~  
[aivn@localhost ~]$ mkdir -p hadoop_store/hdfs/namenode  
[aivn@localhost ~]$ mkdir -p hadoop_store/hdfs/datanode  
[aivn@localhost ~]$ ls hadoop_store/hdfs/  
datanode namenode  
[aivn@localhost ~]$
```

### 8.2. Config hdfs-site.xml

```
aivn@localhost:~  
[aivn@localhost ~]$ sudo vi hadoop/etc/hadoop/hdfs-site.xml
```

# Hadoop

## ❖ Installation: Hadoop



The screenshot shows a terminal window with the title bar "aivn@localhost:~ — sudo vi hadoop/etc/hadoop/hdfs-site.xml". The terminal content displays the XML configuration for HDFS. It starts with a license notice, followed by a comment to put site-specific property overrides in this file. The configuration block contains three properties: `dfs.replication` set to `1`, `dfs.namenode.name.dir` set to `file:/home/aivn/hadoop_store/hdfs/namenode`, and `dfs.datanode.data.dir` set to `file:/home/aivn/hadoop_store/hdfs/datanode`. The cursor is at the end of the last line.

```
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>file:/home/aivn/hadoop_store/hdfs/namenode</value>
</property>
<property> <name>dfs.datanode.data.dir</name>
<value>file:/home/aivn/hadoop_store/hdfs/datanode</value>
</property>
</configuration>
:wq
```

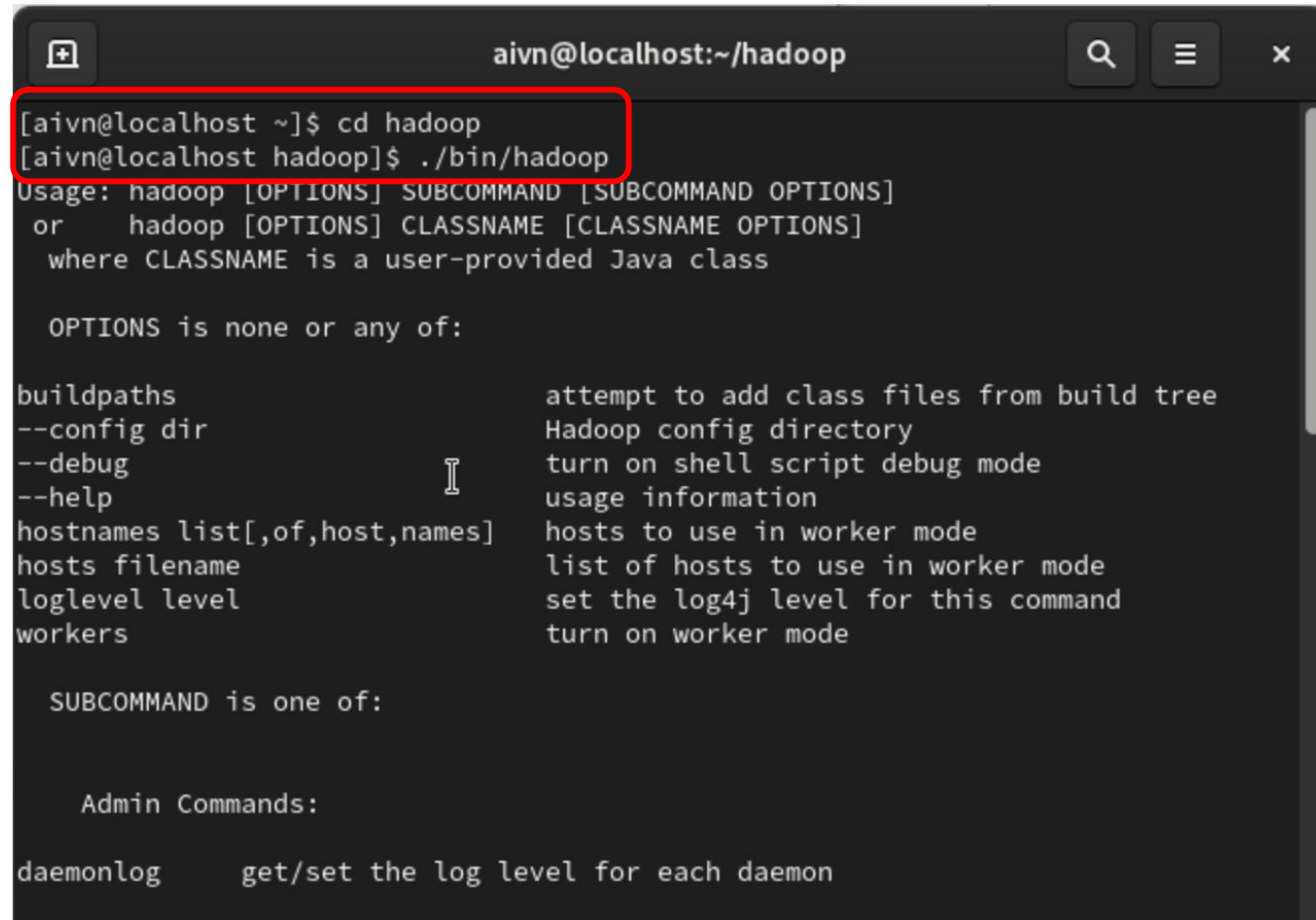
Paste the code below:

```
<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>file:/home/aivn/hadoop_store/hdfs/na
menode</value>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>file:/home/aivn/hadoop_store/hdfs/da
tanode</value>
</property>
</configuration>
```



# Hadoop

## ❖ Check Hadoop



```
aivn@localhost:~/hadoop
[aivn@localhost ~]$ cd hadoop
[aivn@localhost hadoop]$ ./bin/hadoop
Usage: hadoop [OPTIONS] SUBCOMMAND [SUBCOMMAND OPTIONS]
or      hadoop [OPTIONS] CLASSNAME [CLASSNAME OPTIONS]
       where CLASSNAME is a user-provided Java class

OPTIONS is none or any of:

buildpaths          attempt to add class files from build tree
--config dir        Hadoop config directory
--debug             turn on shell script debug mode
--help              usage information
hostnames list[,of,host,names] hosts to use in worker mode
hosts filename      list of hosts to use in worker mode
loglevel level       set the log4j level for this command
workers             turn on worker mode

SUBCOMMAND is one of:

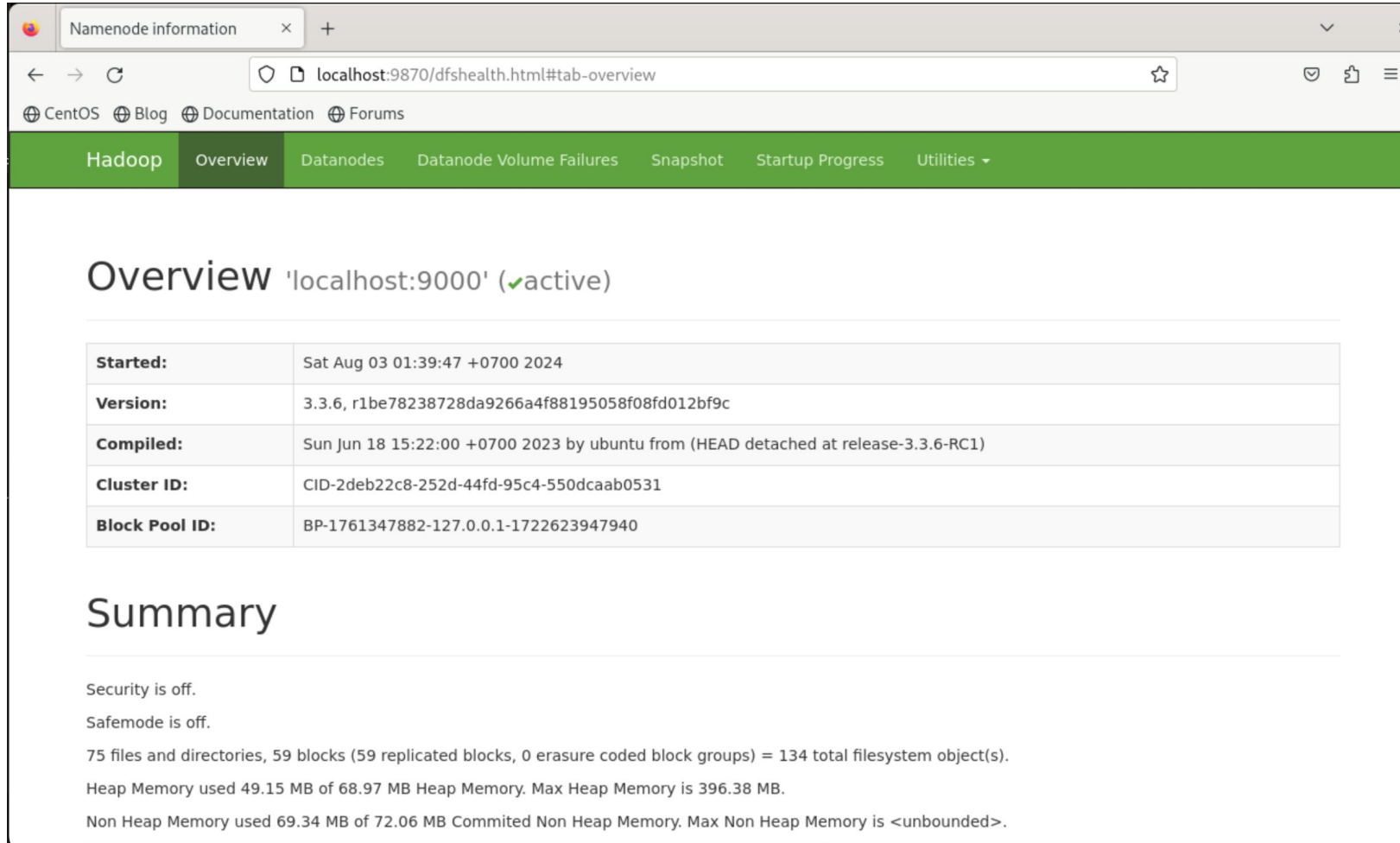
Admin Commands:

daemonlog           get/set the log level for each daemon
```

Check Hadoop Installation

# Hadoop

## ❖ Check Hadoop



The screenshot shows a web browser window with the title 'Namenode information'. The address bar shows 'localhost:9870/dfshealth.html#tab-overview'. The browser has tabs for CentOS, Blog, Documentation, and Forums. The page has a green navigation bar with links: Hadoop, Overview, Datanodes, Datanode Volume Failures, Snapshot, Startup Progress, and Utilities. The main content area is titled 'Overview 'localhost:9000' (✓active)'. Below this is a table with the following information:

|                       |                                                                                    |
|-----------------------|------------------------------------------------------------------------------------|
| <b>Started:</b>       | Sat Aug 03 01:39:47 +0700 2024                                                     |
| <b>Version:</b>       | 3.3.6, r1be78238728da9266a4f88195058f08fd012bf9c                                   |
| <b>Compiled:</b>      | Sun Jun 18 15:22:00 +0700 2023 by ubuntu from (HEAD detached at release-3.3.6-RC1) |
| <b>Cluster ID:</b>    | CID-2deb22c8-252d-44fd-95c4-550dcaab0531                                           |
| <b>Block Pool ID:</b> | BP-1761347882-127.0.0.1-1722623947940                                              |

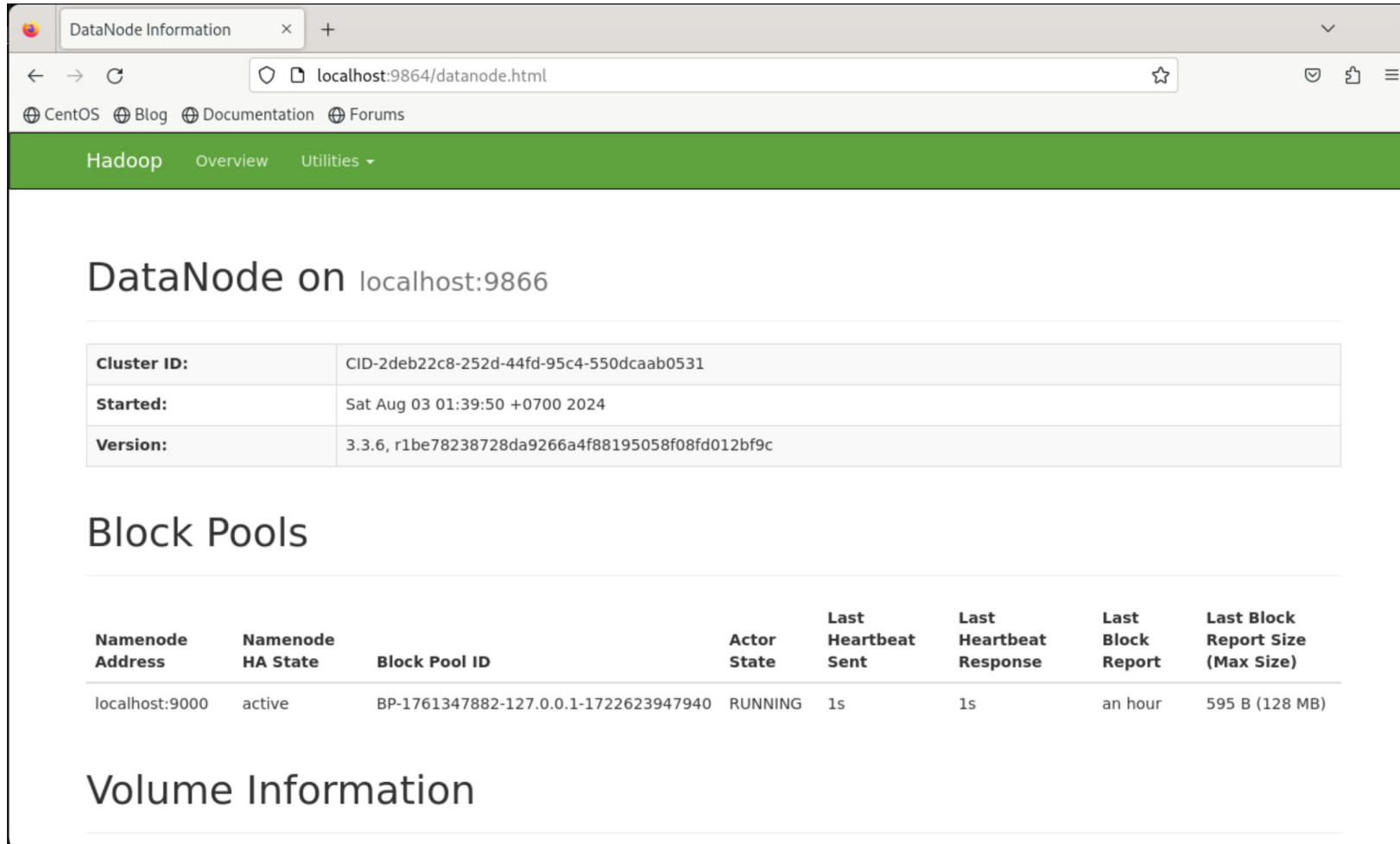
Below the table is a 'Summary' section with the following text:

Security is off.  
Safemode is off.  
75 files and directories, 59 blocks (59 replicated blocks, 0 erasure coded block groups) = 134 total filesystem object(s).  
Heap Memory used 49.15 MB of 68.97 MB Heap Memory. Max Heap Memory is 396.38 MB.  
Non Heap Memory used 69.34 MB of 72.06 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

**localhost:9870** to check namenode

# Hadoop

## ◆ Check Hadoop



The screenshot shows a web browser window with the address bar displaying `localhost:9864/datanode.html`. The page title is "DataNode Information". Below the browser window, the page content is displayed. It starts with a green navigation bar containing "Hadoop", "Overview", and "Utilities". The main heading is "DataNode on localhost:9866". Below this, there is a table with three rows: "Cluster ID", "Started", and "Version". The "Cluster ID" is "CID-2deb22c8-252d-44fd-95c4-550dcaab0531", "Started" is "Sat Aug 03 01:39:50 +0700 2024", and "Version" is "3.3.6, r1be78238728da9266a4f88195058f08fd012bf9c". Below this table, the heading "Block Pools" is shown. Under "Block Pools", there is a table with eight columns: "Namenode Address", "Namenode HA State", "Block Pool ID", "Actor State", "Last Heartbeat Sent", "Last Heartbeat Response", "Last Block Report", and "Last Block Report Size (Max Size)". The table has one row of data: "localhost:9000", "active", "BP-1761347882-127.0.0.1-1722623947940", "RUNNING", "1s", "1s", "an hour", and "595 B (128 MB)". Below the "Block Pools" table, the heading "Volume Information" is shown.

DataNode on localhost:9866

|             |                                                  |
|-------------|--------------------------------------------------|
| Cluster ID: | CID-2deb22c8-252d-44fd-95c4-550dcaab0531         |
| Started:    | Sat Aug 03 01:39:50 +0700 2024                   |
| Version:    | 3.3.6, r1be78238728da9266a4f88195058f08fd012bf9c |

Block Pools

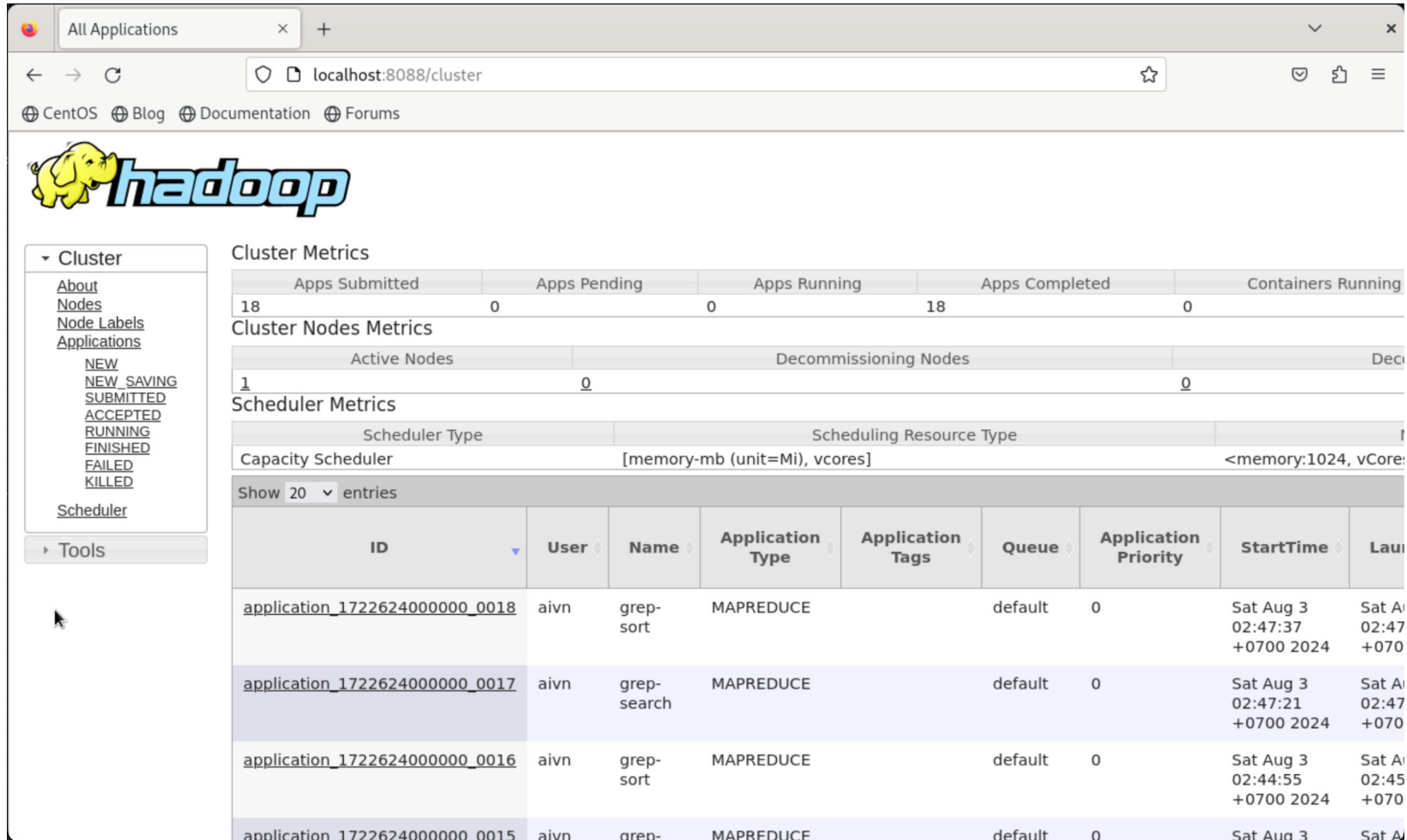
| Namenode Address | Namenode HA State | Block Pool ID                         | Actor State | Last Heartbeat Sent | Last Heartbeat Response | Last Block Report | Last Block Report Size (Max Size) |
|------------------|-------------------|---------------------------------------|-------------|---------------------|-------------------------|-------------------|-----------------------------------|
| localhost:9000   | active            | BP-1761347882-127.0.0.1-1722623947940 | RUNNING     | 1s                  | 1s                      | an hour           | 595 B (128 MB)                    |

Volume Information

**localhost:9864** to check datanode

# Hadoop

## ❖ Check Hadoop



The screenshot displays the Hadoop web interface at `localhost:8088/cluster`. The interface includes a sidebar with navigation links and several metrics tables.

**Cluster Metrics**

| Apps Submitted | Apps Pending | Apps Running | Apps Completed | Containers Running |
|----------------|--------------|--------------|----------------|--------------------|
| 18             | 0            | 0            | 18             | 0                  |

**Cluster Nodes Metrics**

| Active Nodes | Decommissioning Nodes |
|--------------|-----------------------|
| 1            | 0                     |

**Scheduler Metrics**

| Scheduler Type     | Scheduling Resource Type     |
|--------------------|------------------------------|
| Capacity Scheduler | [memory-mb (unit=M), vcores] |

Show 20 entries

| ID                                             | User | Name        | Application Type | Application Tags | Queue   | Application Priority | StartTime                     | LaunchedTime                  |
|------------------------------------------------|------|-------------|------------------|------------------|---------|----------------------|-------------------------------|-------------------------------|
| <a href="#">application_1722624000000_0018</a> | aivn | grep-sort   | MAPREDUCE        |                  | default | 0                    | Sat Aug 3 02:47:37 +0700 2024 | Sat Aug 3 02:47:37 +0700 2024 |
| <a href="#">application_1722624000000_0017</a> | aivn | grep-search | MAPREDUCE        |                  | default | 0                    | Sat Aug 3 02:47:21 +0700 2024 | Sat Aug 3 02:47:21 +0700 2024 |
| <a href="#">application_1722624000000_0016</a> | aivn | grep-sort   | MAPREDUCE        |                  | default | 0                    | Sat Aug 3 02:44:55 +0700 2024 | Sat Aug 3 02:44:55 +0700 2024 |
| <a href="#">application_1722624000000_0015</a> | aivn | grep-sort   | MAPREDUCE        |                  | default | 0                    | Sat Aug 3 02:44:55 +0700 2024 | Sat Aug 3 02:44:55 +0700 2024 |

**localhost:8088** to check datanode

## ❖ Example: Search text

### grep Command

Last Updated: 2023-03-24

#### Purpose

Searches for a pattern in a file.

#### Syntax

```
grep [-E|-F][-i][-h][-H][-L][-r|-R][-s][-u][-v][-w][-x][-y][[-b][-n]][[-c|-l|-q]][-p[Separator]]{[-e PatternList ...]}[-f PatternFile ...]| PatternList ...}[ File ...]
```

Pattern to search: "fs[a-z.]"

#### Description

The **grep** command searches for the pattern specified by the *Pattern* parameter and writes each matching line to standard output. The patterns are limited regular expressions in the style of the **ed** or **egrep** command. The **grep** command uses a compact non-deterministic algorithm.

# Hadoop

## ❖ Example: Search text

### 1. Initialize namenode

```
aivn@localhost:~/hadoop
[aivn@localhost hadoop]$ ./bin/hdfs namenode -format
WARNING: /home/aivn/hadoop/logs does not exist. Creating.
2024-08-02 01:53:52,909 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG:  host = localhost.localdomain/127.0.0.1
STARTUP_MSG:  args = [-format]
STARTUP_MSG:  version = 3.3.6
STARTUP_MSG:  classpath = /home/aivn/hadoop/etc/hadoop:/home/aivn/hadoop/share/
hadoop/common/lib/netty-resolver-dns-native-macos-4.1.89.Final-osx-aarch_64.jar:
/home/aivn/hadoop/share/hadoop/common/lib/netty-codec-socks-4.1.89.Final.jar:/ho
me/aivn/hadoop/share/hadoop/common/lib/kerb-identity-1.0.1.jar:/home/aivn/hadoop
/share/hadoop/common/lib/kerby-asn1-1.0.1.jar:/home/aivn/hadoop/share/hadoop/com
mon/lib/commons-math3-3.1.1.jar:/home/aivn/hadoop/share/hadoop/common/lib/jetty-
xml-9.4.51.v20230217.jar:/home/aivn/hadoop/share/hadoop/common/lib/kerb-crypto-1
.0.1.jar:/home/aivn/hadoop/share/hadoop/common/lib/netty-codec-xml-4.1.89.Final
.jar:/home/aivn/hadoop/share/hadoop/common/lib/netty-transport-sctp-4.1.89.Final
.jar:/home/aivn/hadoop/share/hadoop/common/lib/commons-collections-3.2.2.jar:/hom
e/aivn/hadoop/share/hadoop/common/lib/jsp-api-2.1.jar:/home/aivn/hadoop/share/ha
dooop/common/lib/commons-codec-1.15.jar:/home/aivn/hadoop/share/hadoop/common/lib
/jackson-mapper-asl-1.9.13.jar:/home/aivn/hadoop/share/hadoop/common/lib/reload4
j-1.2.22.jar:/home/aivn/hadoop/share/hadoop/common/lib/jersey-server-1.19.4.jar:
/home/aivn/hadoop/share/hadoop/common/lib/jsch-0.1.55.jar:/home/aivn/hadoop/shar
e/hadoop/common/lib/kerby-util-1.0.1.jar:/home/aivn/hadoop/share/hadoop/common/l
```

### 2. Start namenode and datanode daemon

```
aivn@localhost:~/hadoop
[aivn@localhost hadoop]$ sudo ./sbin/start-all.sh
Starting namenodes on [localhost]
Last login: Fri Aug  2 10:45:09 +07 2024 on pts/1
Starting datanodes
Last login: Fri Aug  2 10:45:15 +07 2024 on pts/1
Starting secondary namenodes [localhost.localdomain]
Last login: Fri Aug  2 10:45:17 +07 2024 on pts/1
2024-08-02 10:45:24,631 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Starting resourcemanager
Last login: Fri Aug  2 10:45:21 +07 2024 on pts/1
Starting nodemanagers
Last login: Fri Aug  2 10:45:25 +07 2024 on pts/1
[aivn@localhost hadoop]$
```



# Hadoop

## ❖ Example: Search text

### 3. Create input folder and move all .xml file into input

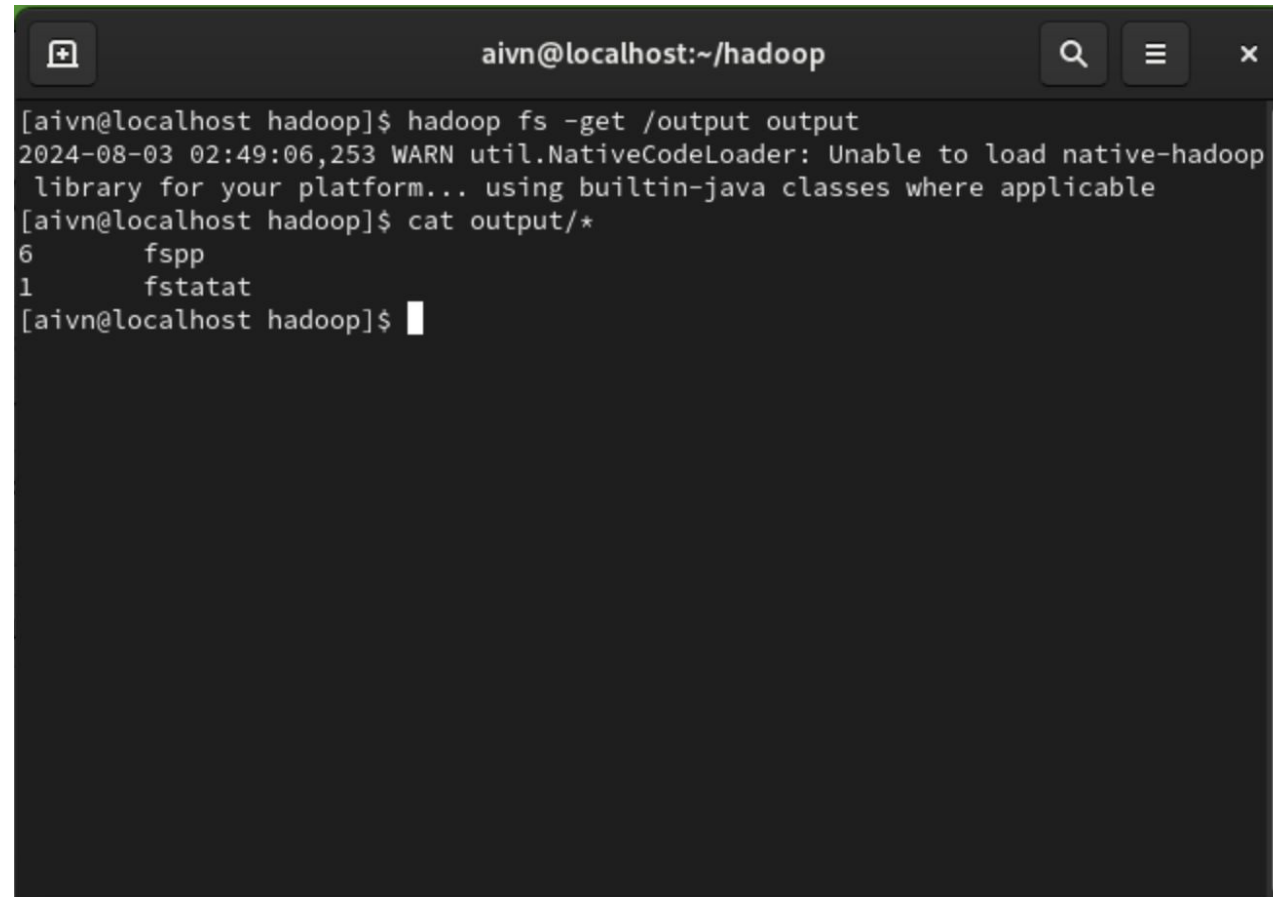
```
aivn@localhost:~/hadoop
[aivn@localhost hadoop]$ hadoop fs -mkdir /input
2024-08-03 01:56:31,117 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
[aivn@localhost hadoop]$ hadoop fs -ls /
2024-08-03 01:56:40,468 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 1 items
drwxr-xr-x  - aivn supergroup          0 2024-08-03 01:56 /input
[aivn@localhost hadoop]$
```

### 4. Perform grep with string pattern “fs[a-z.]+”

```
aivn@localhost:~/hadoop
[aivn@localhost hadoop]$ ./bin/hadoop jar share/hadoop/mapreduce/hadoop-mapred
e-examples-3.3.6.jar grep /input /output 'fs[a-z.]+'
2024-08-03 02:47:20,270 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
2024-08-03 02:47:20,588 INFO client.DefaultNoHARMFailoverProxyProvider: Connecti
ng to ResourceManager at /127.0.0.1:8032
2024-08-03 02:47:20,808 INFO mapreduce.JobResourceUploader: Disabling Erasure Co
ding for path: /tmp/hadoop-yarn/staging/aivn/.staging/job_1722624000000_0017
2024-08-03 02:47:20,982 INFO input.FileInputFormat: Total input files to process
: 1
2024-08-03 02:47:21,048 INFO mapreduce.JobSubmitter: number of splits:1
2024-08-03 02:47:21,149 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_1722624000000_0017
2024-08-03 02:47:21,149 INFO mapreduce.JobSubmitter: Executing with tokens: []
2024-08-03 02:47:21,252 INFO conf.Configuration: resource-types.xml not found
2024-08-03 02:47:21,252 INFO resource.ResourceUtils: Unable to find 'resource-ty
pes.xml'.
2024-08-03 02:47:21,290 INFO impl.YarnClientImpl: Submitted application applicat
ion_1722624000000_0017
2024-08-03 02:47:21,315 INFO mapreduce.Job: The url to track the job: http://loc
alhost:8088/proxy/application_1722624000000_0017/
2024-08-03 02:47:21,315 INFO mapreduce.Job: Running job: job_1722624000000_0017
2024-08-03 02:47:25,883 INFO mapreduce.Job: Job job_1722624000000_0017 running i
n uber mode : false
```

# Hadoop

## ❖ Example: Search text



```
aivn@localhost:~/hadoop
[aivn@localhost hadoop]$ hadoop fs -get /output output
2024-08-03 02:49:06,253 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
[aivn@localhost hadoop]$ cat output/*
6      fspp
1      fstatat
[aivn@localhost hadoop]$
```

5. Get output and print results

## ❖ Example: Other default applications

```
Valid program names are:
  aggregatewordcount: An Aggregate based map/reduce program that counts the words in the input files.
  aggregatewordhist: An Aggregate based map/reduce program that computes the histogram of the words in the input files.
  bbb: A map/reduce program that uses Bailey-Borwein-Plouffe to compute exact digits of Pi.
  .
  dbcount: An example job that count the pageview counts from a database.
  distbbp: A map/reduce program that uses a BBP-type formula to compute exact bits of Pi.
  grep: A map/reduce program that counts the matches of a regex in the input.
  join: A job that effects a join over sorted, equally partitioned datasets
  multifilewc: A job that counts words from several files.
  pentomino: A map/reduce tile laying program to find solutions to pentomino problems.
  pi: A map/reduce program that estimates Pi using a quasi-Monte Carlo method.
  randomtextwriter: A map/reduce program that writes 10GB of random textual data per node.
  randomwriter: A map/reduce program that writes 10GB of random data per node.
  secondarysort: An example defining a secondary sort to the reduce.
  sort: A map/reduce program that sorts the data written by the random writer.
  sudoku: A sudoku solver.
  teragen: Generate data for the terasort
  terasort: Run the terasort
  teravalidate: Checking results of terasort
  wordcount: A map/reduce program that counts the words in the input files.
  wordmean: A map/reduce program that counts the average length of the words in the input files.
  wordmedian: A map/reduce program that counts the median length of the words in the input files.
  wordstandarddeviation: A map/reduce program that counts the standard deviation of the length of the words in the input files.
```

# QUIZ

# Spark

## ◆ Introduction



**Apache Spark:** an open-source unified analytics engine for **large-scale data processing**. Spark is an alternative replacement for Hadoop MapReduce.



## ❖ Why Spark?



Fast

Batch Processing

Stores Data on Disk

Written in Java

Low Cost



100x Faster than MapReduce

Real-time Processing

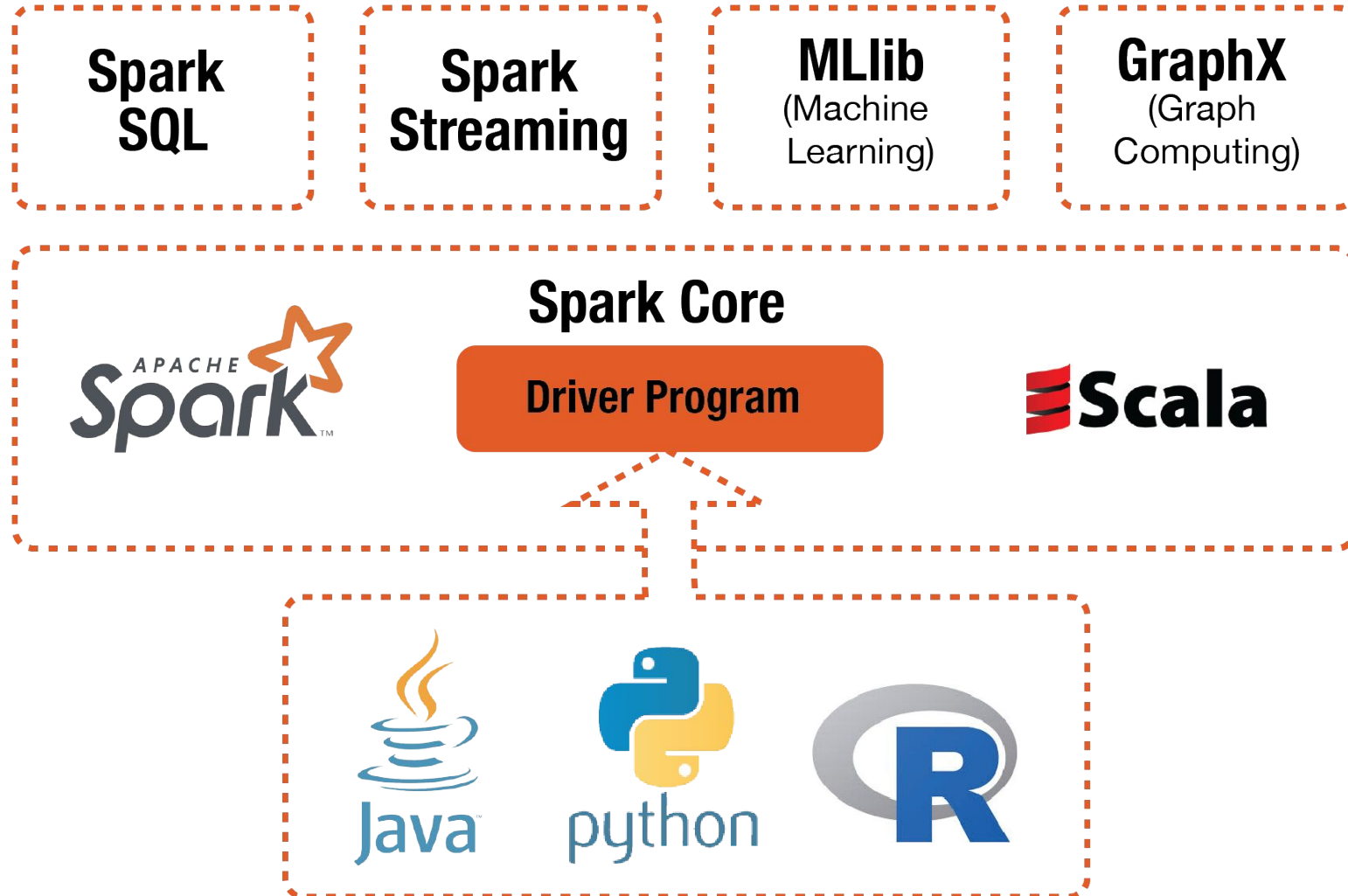
Stores Data Memory

Written in Scala

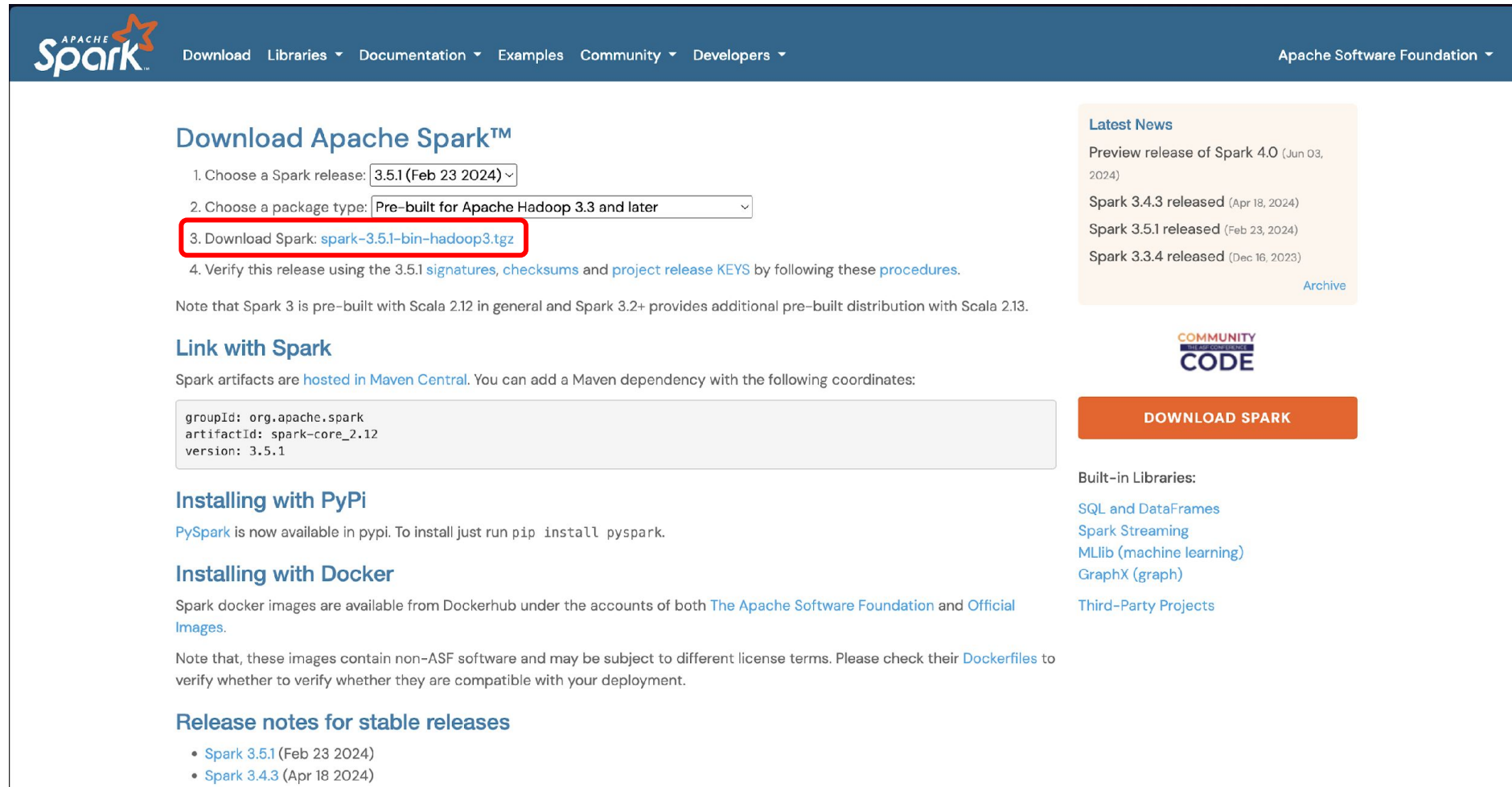
High Cost

# Spark

## ◆ Spark Modules



## ❖ Installation: Get spark



The screenshot shows the Apache Spark download page. The navigation bar at the top includes links for Download, Libraries, Documentation, Examples, Community, and Developers, along with the Apache Software Foundation logo. The main heading is "Download Apache Spark™". Below this, there are four steps: 1. Choose a Spark release (3.5.1 (Feb 23 2024)), 2. Choose a package type (Pre-built for Apache Hadoop 3.3 and later), 3. Download Spark (spark-3.5.1-bin-hadoop3.tgz), and 4. Verify this release using the 3.5.1 signatures, checksums and project release KEYS by following these procedures. The third step is highlighted with a red box. A note mentions that Spark 3 is pre-built with Scala 2.12 and Spark 3.2+ provides additional pre-built distribution with Scala 2.13. Below the steps, there is a section "Link with Spark" showing Maven coordinates: groupId: org.apache.spark, artifactId: spark-core\_2.12, version: 3.5.1. There is also a "DOWNLOAD SPARK" button. On the right side, there is a "Latest News" section with links to Spark 4.0 preview release, Spark 3.4.3 released, Spark 3.5.1 released, and Spark 3.3.4 released. Below this is a "COMMUNITY CODE" logo and a "Built-in Libraries" section with links to SQL and DataFrames, Spark Streaming, MLlib (machine learning), and GraphX (graph). There is also a link to "Third-Party Projects".

Download Apache Spark™

1. Choose a Spark release: 3.5.1 (Feb 23 2024) ▾
2. Choose a package type: Pre-built for Apache Hadoop 3.3 and later ▾
3. Download Spark: [spark-3.5.1-bin-hadoop3.tgz](#)
4. Verify this release using the 3.5.1 [signatures](#), [checksums](#) and [project release KEYS](#) by following these [procedures](#).

Note that Spark 3 is pre-built with Scala 2.12 in general and Spark 3.2+ provides additional pre-built distribution with Scala 2.13.

### Link with Spark

Spark artifacts are [hosted in Maven Central](#). You can add a Maven dependency with the following coordinates:

```
groupId: org.apache.spark  
artifactId: spark-core_2.12  
version: 3.5.1
```

### Installing with PyPi

[PySpark](#) is now available in pypi. To install just run `pip install pyspark`.

### Installing with Docker

Spark docker images are available from Dockerhub under the accounts of both [The Apache Software Foundation](#) and [Official Images](#).

Note that, these images contain non-ASF software and may be subject to different license terms. Please check their [Dockerfiles](#) to verify whether to verify whether they are compatible with your deployment.

### Release notes for stable releases

- [Spark 3.5.1](#) (Feb 23 2024)
- [Spark 3.4.3](#) (Apr 18 2024)

Latest News

- Preview release of Spark 4.0 (Jun 03, 2024)
- Spark 3.4.3 released (Apr 18, 2024)
- Spark 3.5.1 released (Feb 23, 2024)
- Spark 3.3.4 released (Dec 16, 2023)

[Archive](#)

COMMUNITY CODE

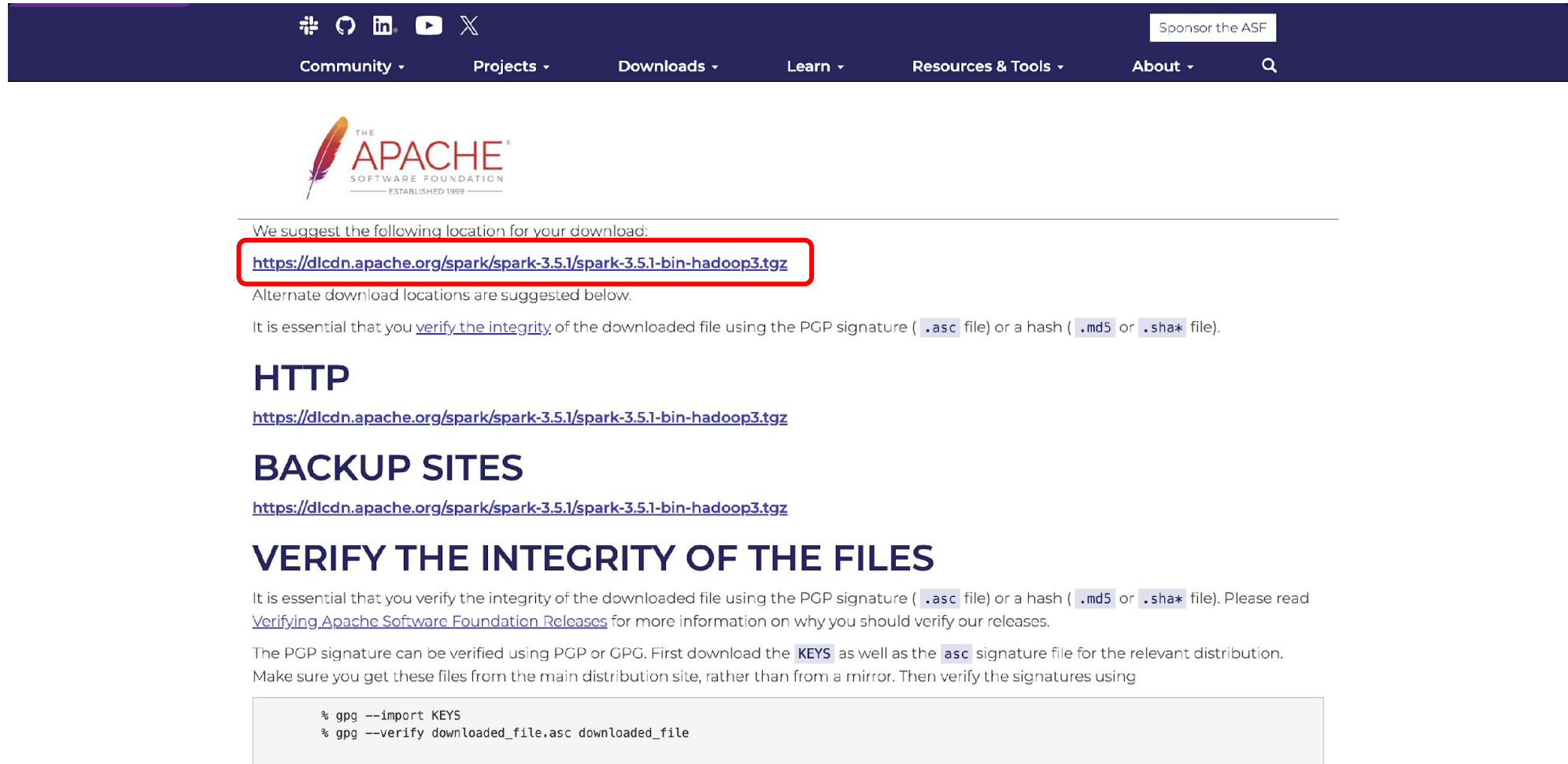
DOWNLOAD SPARK

Built-in Libraries:

- [SQL and DataFrames](#)
- [Spark Streaming](#)
- [MLlib \(machine learning\)](#)
- [GraphX \(graph\)](#)

[Third-Party Projects](#)

## ❖ Installation: Get spark



The screenshot shows the Apache Spark download page. At the top is a dark blue navigation bar with social media icons (GitHub, YouTube, LinkedIn, etc.) and a 'Sponsor the ASF' button. Below the navigation bar is the Apache Software Foundation logo. The main content area has a heading 'We suggest the following location for your download:' followed by a red-bordered box containing the URL <https://dlcdn.apache.org/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz>. Below this, it says 'Alternate download locations are suggested below.' and 'It is essential that you [verify the integrity](#) of the downloaded file using the PGP signature ( `.asc` file) or a hash ( `.md5` or `.sha*` file).'. There are three sections: 'HTTP' with the same URL, 'BACKUP SITES' with the same URL, and 'VERIFY THE INTEGRITY OF THE FILES'. The verification section explains the importance of verifying the integrity of the downloaded file using the PGP signature or a hash, and provides instructions on how to verify the integrity using GPG. It mentions downloading the 'KEYS' as well as the '.asc' signature file for the relevant distribution. It also states that the files should be downloaded from the main distribution site, rather than from a mirror. At the bottom, there is a code block with the following commands:

```
% gpg --import KEYS
% gpg --verify downloaded_file.asc downloaded_file
```

## ❖ Installation: Get spark

```
aivn@localhost:~  
[aivn@localhost ~]$ wget https://dlcdn.apache.org/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz  
--2024-08-02 22:54:27-- https://dlcdn.apache.org/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz  
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644  
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connecte  
d.  
HTTP request sent, awaiting response... 200 OK  
Length: 400446614 (382M) [application/x-gzip]  
Saving to: 'spark-3.5.1-bin-hadoop3.tgz'  
  
spark-3.5.1-bin-had 100%[=====>] 381.90M  8.90MB/s   in 41s  
  
2024-08-02 22:55:08 (9.32 MB/s) - 'spark-3.5.1-bin-hadoop3.tgz' saved [400446614  
/400446614]  
  
[aivn@localhost ~]$
```

```
aivn@localhost:~  
[aivn@localhost ~]$ tar -xvf spark-3.5.1-bin-hadoop3.tgz spark-3.5.1-bin-hadoop3  
spark-3.5.1-bin-hadoop3/  
spark-3.5.1-bin-hadoop3/sbin/  
spark-3.5.1-bin-hadoop3/sbin/spark-config.sh  
spark-3.5.1-bin-hadoop3/sbin/stop-slave.sh  
spark-3.5.1-bin-hadoop3/sbin/stop-mesos-dispatcher.sh  
spark-3.5.1-bin-hadoop3/sbin/start-workers.sh  
spark-3.5.1-bin-hadoop3/sbin/start-slaves.sh  
spark-3.5.1-bin-hadoop3/sbin/start-all.sh  
spark-3.5.1-bin-hadoop3/sbin/stop-all.sh  
spark-3.5.1-bin-hadoop3/sbin/workers.sh  
spark-3.5.1-bin-hadoop3/sbin/start-mesos-dispatcher.sh  
spark-3.5.1-bin-hadoop3/sbin/spark-daemon.sh  
spark-3.5.1-bin-hadoop3/sbin/decommission-worker.sh  
spark-3.5.1-bin-hadoop3/sbin/slaves.sh  
spark-3.5.1-bin-hadoop3/sbin/stop-mesos-shuffle-service.sh  
spark-3.5.1-bin-hadoop3/sbin/stop-history-server.sh  
spark-3.5.1-bin-hadoop3/sbin/stop-worker.sh  
spark-3.5.1-bin-hadoop3/sbin/decommission-slave.sh  
spark-3.5.1-bin-hadoop3/sbin/stop-thriftserver.sh  
spark-3.5.1-bin-hadoop3/sbin/start-worker.sh  
spark-3.5.1-bin-hadoop3/sbin/stop-slaves.sh  
spark-3.5.1-bin-hadoop3/sbin/start-connect-server.sh  
spark-3.5.1-bin-hadoop3/sbin/start-thriftserver.sh
```

## ❖ Installation: Get spark

```
aivn@localhost:~  
[aivn@localhost ~]$ mv spark-3.5.1-bin-hadoop3 spark  
[aivn@localhost ~]$
```

```
aivn@localhost:~  
[aivn@localhost ~]$ ls  
Desktop    hadoop          Music    spark          Videos  
Documents  hadoop-3.3.6.tar.gz Pictures  spark-3.5.1-bin-hadoop3.tgz  
Downloads  hadoop_store    Public   Templates  
[aivn@localhost ~]$
```

## ❖ Installation: Get spark

```
aivn@localhost:~  
export YARN_NODEMANAGER_USER="aivn"  
export JAVA_HOME=/usr/lib/jvm/java-1.8.0-openjdk-1.8.0.362.b09-4.el9.aarch64  
export HADOOP_HOME=/home/aivn/hadoop  
export HADOOP_MAPRED_HOME=$HADOOP_HOME  
export HADOOP_COMMON_HOME=$HADOOP_HOME  
export HADOOP_HDFS_HOME=$HADOOP_HOME  
export YARN_HOME=$HADOOP_HOME  
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native  
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin  
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"  
  
export SPARK_HOME=/home/aivn/spark  
export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin  
export PYSPARK_PYTHON=/usr/bin/python3  
export SPARK_LOCAL_IP=localhost  
  
# User specific environment  
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]  
then  
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"  
fi  
export PATH
```

- export  
SPARK\_HOME=/home/aivn/spark-3.5.1-bin-hadoop3
- export  
PATH=\$PATH:\$SPARK\_HOME/bin:\$SPARK\_HOME/sbin
- export PYSPARK\_PYTHON=/usr/bin/python3
- export SPARK\_LOCAL\_IP=localhost



## ❖ Installation: Start spark

```
aivn@localhost:~  
[aivn@localhost ~]$ sudo vi .bashrc  
[aivn@localhost ~]$ source .bashrc  
[aivn@localhost ~]$
```

```
aivn@localhost:~/spark  
[aivn@localhost spark]$ ./sbin/start-all.sh  
starting org.apache.spark.deploy.master.Master, logging to /home/aivn/spark/logs  
/spark-aivn-org.apache.spark.deploy.master.Master-1-localhost.out  
localhost: starting org.apache.spark.deploy.worker.Worker, logging to /home/aivn  
/spark/logs/spark-aivn-org.apache.spark.deploy.worker.Worker-1-localhost.out  
[aivn@localhost spark]$ jps  
74657 Jps  
74514 Master  
57573 DataNode  
58202 NodeManager  
74605 Worker  
57421 NameNode  
57838 SecondaryNameNode  
58062 ResourceManager  
[aivn@localhost spark]$
```

## ❖ Spark: localhost

The screenshot shows the Spark Master web interface in a browser. The address bar shows 'localhost:8080'. The page title is 'Spark Master at spark://localhost:7077'. The interface displays the following information:

- URL: spark://localhost:7077
- Alive Workers: 1
- Cores in use: 2 Total, 0 Used
- Memory in use: 1024.0 MiB Total, 0.0 B Used
- Resources in use:
- Applications: 0 Running, 0 Completed
- Drivers: 0 Running, 0 Completed
- Status: ALIVE

Below this information, there are three expandable sections:

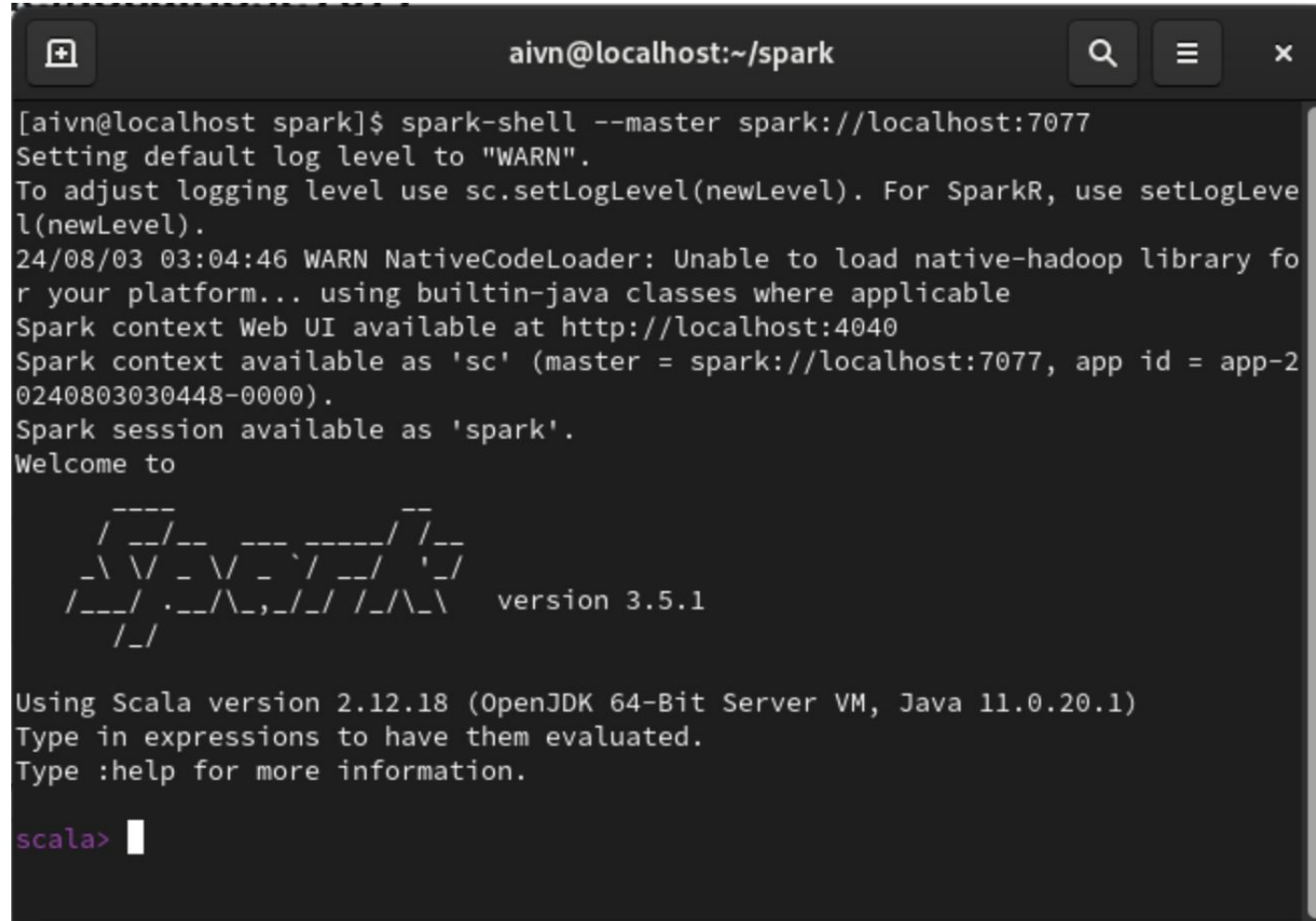
- Workers (1)**: A table showing one worker.
- Running Applications (0)**: A table showing no running applications.
- Completed Applications (0)**: A table showing no completed applications.

| Worker Id                             | Address         | State | Cores      | Memory                  | Resources |
|---------------------------------------|-----------------|-------|------------|-------------------------|-----------|
| worker-20240803030205-127.0.0.1-40449 | 127.0.0.1:40449 | ALIVE | 2 (0 Used) | 1024.0 MiB (0.0 B Used) |           |

| Application ID | Name | Cores | Memory per Executor | Resources Per Executor | Submitted Time | User | State | Duration |
|----------------|------|-------|---------------------|------------------------|----------------|------|-------|----------|
|----------------|------|-------|---------------------|------------------------|----------------|------|-------|----------|

| Application ID | Name | Cores | Memory per Executor | Resources Per Executor | Submitted Time | User | State | Duration |
|----------------|------|-------|---------------------|------------------------|----------------|------|-------|----------|
|----------------|------|-------|---------------------|------------------------|----------------|------|-------|----------|

## ❖ Spark: spark session

A terminal window titled 'aivn@localhost:~/spark' with search, menu, and close icons. It shows the execution of 'spark-shell --master spark://localhost:7077'. The output includes log messages about the log level, native-hadoop library, and Spark context. It displays a ASCII art logo for Spark version 3.5.1 and information about the Scala version (2.12.18) and OpenJDK (64-Bit Server VM, Java 11.0.20.1). The prompt 'scala>' is shown at the bottom.

```
aivn@localhost:~/spark
[aivn@localhost spark]$ spark-shell --master spark://localhost:7077
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
24/08/03 03:04:46 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://localhost:4040
Spark context available as 'sc' (master = spark://localhost:7077, app id = app-20240803030448-0000).
Spark session available as 'spark'.
Welcome to

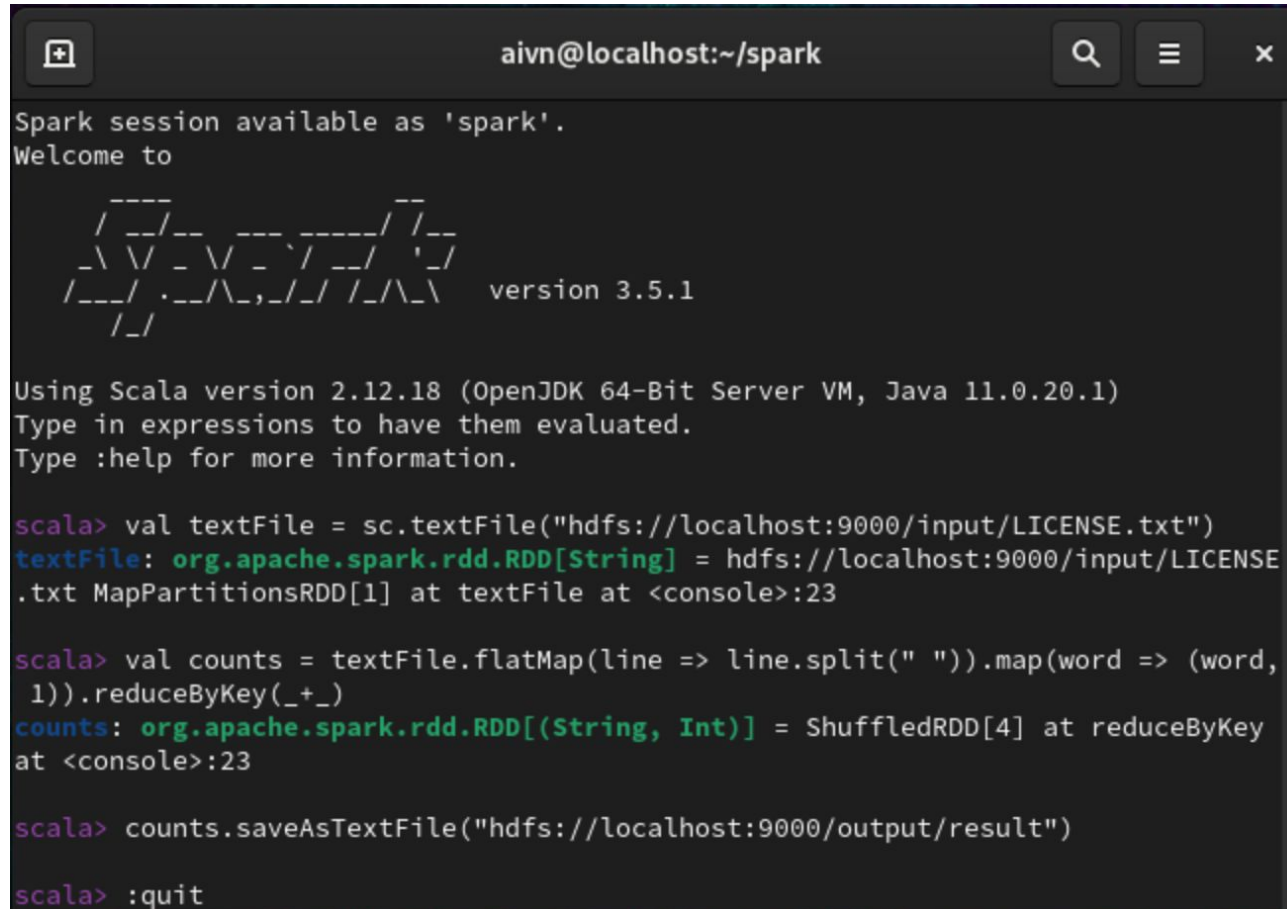
  _--_
 /_ _ \_/_ _ _ _ _/_/_ _
/_ _ \_/_ _ _/_ _ _/_ _ \_/_ _
/_ _ \_/_ _ _/_ _ _/_ _ \_/_ _
/_ _ \_/_ _ _/_ _ _/_ _ \_/_ _

version 3.5.1

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 11.0.20.1)
Type in expressions to have them evaluated.
Type :help for more information.

scala> 
```

## ❖ Spark: Word count



```
aivn@localhost:~/spark

Spark session available as 'spark'.
Welcome to

  ____
 /  __ \   _ __   ___
 \  /  >  / __>  / _ \
  /  /  > /  >  /  __/
 /_____/ /_/  /_/  \___/
version 3.5.1

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 11.0.20.1)
Type in expressions to have them evaluated.
Type :help for more information.

scala> val textFile = sc.textFile("hdfs://localhost:9000/input/LICENSE.txt")
textFile: org.apache.spark.rdd.RDD[String] = hdfs://localhost:9000/input/LICENSE
.txt MapPartitionsRDD[1] at textFile at <console>:23

scala> val counts = textFile.flatMap(line => line.split(" ")).map(word => (word,
1)).reduceByKey(_+_ )
counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[4] at reduceByKey
at <console>:23

scala> counts.saveAsTextFile("hdfs://localhost:9000/output/result")

scala> :quit
```

- `val textFile =  
sc.textFile("hdfs://localhost:9000/input/LICENSE.txt")`
- `val counts =  
textFile.flatMap(line =>  
line.split(" ")).map(word =>  
(word, 1)).reduceByKey(_+_ )`
- `counts.saveAsTextFile("hdfs://localhost:9000/output/result")`

## ❖ Spark: Word count

```
aivn@localhost:~/spark

[aivn@localhost spark]$ hadoop fs -ls /output/result/*
2024-08-03 03:10:47,655 WARN util.NativeCodeLoader: Unable to load native-hadoop
  library for your platform... using builtin-java classes where applicable
-rw-r--r--    3 aivn supergroup          0 2024-08-03 03:08 /output/result/_SUCCE
SS
-rw-r--r--    3 aivn supergroup      4732 2024-08-03 03:08 /output/result/part-0
0000
-rw-r--r--    3 aivn supergroup      6484 2024-08-03 03:08 /output/result/part-0
0001
^[[A[aivn@localhost spark]$ hadoop fs -cat /output/result/part-00000
2024-08-03 03:11:03,969 WARN util.NativeCodeLoader: Unable to load native-hadoop
  library for your platform... using builtin-java classes where applicable
(notice,,1)
(Unless,3)
(under,10)
(Contributions),1)
(offer,1)
(NON-INFRINGEMENT,,1)
(agree,1)
(its,3)
(event,1)
(intentionally,2)
(Grant,2)
(have,2)
```

## ❖ Break-quiz

**1. What is Apache Spark primarily used for?**

- A) Web development.
- B) Real-time data processing.
- C) Desktop application development.
- D) Database management.

**2. Which of the following is a core component of Apache Spark?**

- A) Spark SQL.
- B) Spark Mapper.
- C) Spark Developer.
- D) Spark Coder.

**3. Which language is not supported natively by Spark?**

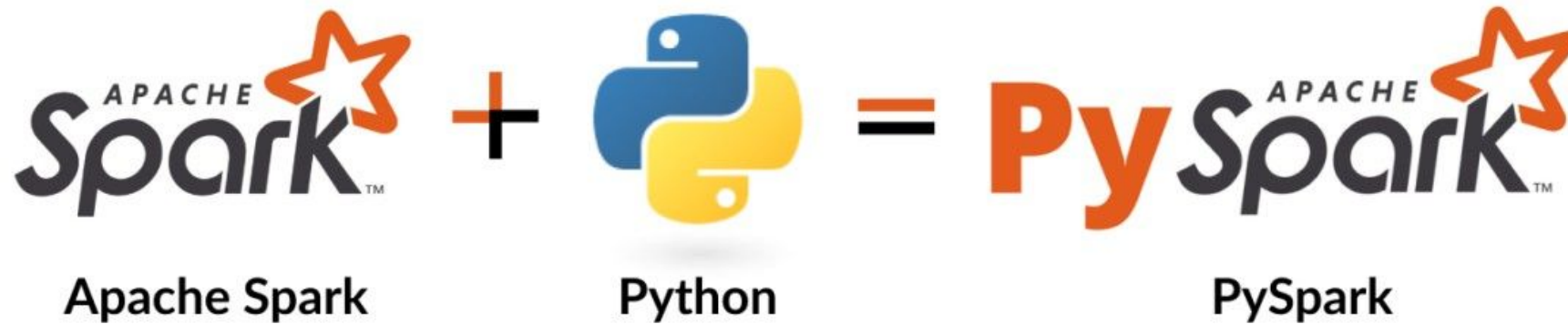
- A) Java.
- B) Python.
- C) Scala.
- D) Ruby.

**4. Which Spark component is used for machine learning?**

- A) Spark SQL.
- B) Spark Streaming.
- C) Spark Mllib.
- D) Spark GraphX.

# Spark

◆ How about use Spark with Python instead of Scala ?





## ◆ PySpark Introduction



**PySpark:** The Python API for Apache Spark. It enables to perform real-time, large-scale data processing in a distributed environment using Python.



**Description:** Perform some basic operation of Spark with Flights.csv dataset. Download link: [here](#).

## ◆ PySpark Installation

### ▼ Install pyspark

```
[ ] 1 !pip install pyspark==3.4.1
```

```
Requirement already satisfied: pyspark==3.4.1 in /usr/local/lib/python3.10/dist-packages (3.4.1)
```

```
Requirement already satisfied: py4j==0.10.9.7 in /usr/local/lib/python3.10/dist-packages (from pyspark==3.4.1) (0.10.9.7)
```

## ❖ PySpark Usage

### ▼ Init pyspark session

```
[ ] 1 from pyspark.sql import SparkSession

[ ] 1 spark = SparkSession.builder.appName('Example').getOrCreate()

[ ] 1 print(spark)

<pyspark.sql.session.SparkSession object at 0x7efffec7190>
```

**SparkSession:** Represents the connection to a Spark cluster and provides a way to interact with various Spark APIs. It allows to create DataFrames, execute SQL queries, and perform distributed data processing tasks.

## ❖ PySpark Usage: Create DataFrame

```
1 data = [  
2     {'name': 'Rin', 'age': 20, 'gender': 'F'},  
3     {'name': 'Thang', 'age': 22, 'gender': 'M'},  
4     {'name': 'Saitama', 'age': 25, 'gender': 'M'},  
5     {'name': 'Edward', 'age': 31, 'gender': 'M'},  
6     {'name': 'Violet', 'age': 17, 'gender': 'F'}  
7 ]
```

```
1 df = spark.createDataFrame(  
2     data=data  
3 )
```

```
1 df.printSchema()
```

```
root  
|-- age: long (nullable = true)  
|-- gender: string (nullable = true)  
|-- name: string (nullable = true)
```

```
1 df.show()
```

| age | gender | name    |
|-----|--------|---------|
| 20  | F      | Rin     |
| 22  | M      | Thang   |
| 25  | M      | Saitama |
| 31  | M      | Edward  |
| 17  | F      | Violet  |

## ❖ PySpark Usage: Operations

```
1 # Select
2 df.select('age').show()
```

|     |
|-----|
|     |
| age |
|     |
| 20  |
| 22  |
| 25  |
| 31  |
| 17  |
|     |

**select():** Choose columns to show in table

```
1 # Filter
2 df.filter(df.gender == 'F').show()
```

|     |        |        |
|-----|--------|--------|
|     |        |        |
| age | gender | name   |
|     |        |        |
| 20  | F      | Rin    |
| 17  | F      | Violet |
|     |        |        |

**filter():** Extract rows satisfied a criteria



## ❖ PySpark Example: Working with csv file

**Description:** Perform some basic operation of Spark with Flights.csv dataset. Download link: [here](#).

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute |
|------|-------|-----|----------|-----------|----------|-----------|---------|---------|--------|--------|------|----------|----------|------|--------|
| 2014 | 12    | 8   | 658      | -7        | 935      | -5        | VX      | N846VA  | 1780   | SEA    | LAX  | 132      | 954      | 6    | 58     |
| 2014 | 1     | 22  | 1040     | 5         | 1505     | 5         | AS      | N559AS  | 851    | SEA    | HNL  | 360      | 2677     | 10   | 40     |
| 2014 | 3     | 9   | 1443     | -2        | 1652     | 2         | VX      | N847VA  | 755    | SEA    | SFO  | 111      | 679      | 14   | 43     |
| 2014 | 4     | 9   | 1705     | 45        | 1839     | 34        | WN      | N360SW  | 344    | PDX    | SJC  | 83       | 569      | 17   | 5      |
| 2014 | 3     | 9   | 754      | -1        | 1015     | 1         | AS      | N612AS  | 522    | SEA    | BUR  | 127      | 937      | 7    | 54     |

## ❖ PySpark Example: Working with csv file

```
1 from pyspark.sql import SparkSession
```

```
1 spark = SparkSession.builder.appName('Example').getOrCreate()
```

```
1 flights_filepath = './flights_small.csv'  
2 flights_df = spark.read.csv(flights_filepath, header=True)
```

```
1 print(flights_df.printSchema())  
2 print(flights_df.show(5))
```

```
root  
|-- year: string (nullable = true)  
|-- month: string (nullable = true)  
|-- day: string (nullable = true)  
|-- dep_time: string (nullable = true)  
|-- dep_delay: string (nullable = true)  
|-- arr_time: string (nullable = true)  
|-- arr_delay: string (nullable = true)  
|-- carrier: string (nullable = true)  
|-- tailnum: string (nullable = true)  
|-- flight: string (nullable = true)  
|-- origin: string (nullable = true)  
|-- dest: string (nullable = true)  
|-- air_time: string (nullable = true)  
|-- distance: string (nullable = true)  
|-- hour: string (nullable = true)  
|-- minute: string (nullable = true)
```

None

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute |
|------|-------|-----|----------|-----------|----------|-----------|---------|---------|--------|--------|------|----------|----------|------|--------|
| 2014 | 12    | 8   | 658      | -7        | 935      | -5        | VX      | N846VA  | 1780   | SEA    | LAX  | 132      | 954      | 6    | 58     |
| 2014 | 1     | 22  | 1040     | 5         | 1505     | 5         | AS      | N559AS  | 851    | SEA    | HNL  | 360      | 2677     | 10   | 40     |
| 2014 | 3     | 9   | 1443     | -2        | 1652     | 2         | VX      | N847VA  | 755    | SEA    | SFO  | 111      | 679      | 14   | 43     |
| 2014 | 4     | 9   | 1705     | 45        | 1839     | 34        | WN      | N360SW  | 344    | PDX    | SJC  | 83       | 569      | 17   | 5      |
| 2014 | 3     | 9   | 754      | -1        | 1015     | 1         | AS      | N612AS  | 522    | SEA    | BUR  | 127      | 937      | 7    | 54     |

only showing top 5 rows

## ❖ PySpark Example: Working with csv file

### ▼ Create new column: Flight Duration

```
[ ] 1 flights_df = flights_df.withColumn('duration_hours', flights_df.air_time / 60)
    2 print(flights_df.show())
```

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute | duration_hours     |
|------|-------|-----|----------|-----------|----------|-----------|---------|---------|--------|--------|------|----------|----------|------|--------|--------------------|
| 2014 | 12    | 8   | 658      | -7        | 935      | -5        | VX      | N846VA  | 1780   | SEA    | LAX  | 132      | 954      | 6    | 58     | 2.2                |
| 2014 | 1     | 22  | 1040     | 5         | 1505     | 5         | AS      | N559AS  | 851    | SEA    | HNL  | 360      | 2677     | 10   | 40     | 6.0                |
| 2014 | 3     | 9   | 1443     | -2        | 1652     | 2         | VX      | N847VA  | 755    | SEA    | SFO  | 111      | 679      | 14   | 43     | 1.85               |
| 2014 | 4     | 9   | 1705     | 45        | 1839     | 34        | WN      | N360SW  | 344    | PDX    | SJC  | 83       | 569      | 17   | 5      | 1.3833333333333333 |
| 2014 | 3     | 9   | 754      | -1        | 1015     | 1         | AS      | N612AS  | 522    | SEA    | BUR  | 127      | 937      | 7    | 54     | 2.1166666666666667 |
| 2014 | 1     | 15  | 1037     | 7         | 1352     | 2         | WN      | N646SW  | 48     | PDX    | DEN  | 121      | 991      | 10   | 37     | 2.0166666666666666 |
| 2014 | 7     | 2   | 847      | 42        | 1041     | 51        | WN      | N422WN  | 1520   | PDX    | OAK  | 90       | 543      | 8    | 47     | 1.5                |
| 2014 | 5     | 12  | 1655     | -5        | 1842     | -18       | VX      | N361VA  | 755    | SEA    | SFO  | 98       | 679      | 16   | 55     | 1.6333333333333333 |
| 2014 | 4     | 19  | 1236     | -4        | 1508     | -7        | AS      | N309AS  | 490    | SEA    | SAN  | 135      | 1050     | 12   | 36     | 2.25               |
| 2014 | 11    | 19  | 1812     | -3        | 2352     | -4        | AS      | N564AS  | 26     | SEA    | ORD  | 198      | 1721     | 18   | 12     | 3.3                |
| 2014 | 11    | 8   | 1653     | -2        | 1924     | -1        | AS      | N323AS  | 448    | SEA    | LAX  | 130      | 954      | 16   | 53     | 2.1666666666666665 |
| 2014 | 8     | 3   | 1120     | 0         | 1415     | 2         | AS      | N305AS  | 656    | SEA    | PHX  | 154      | 1107     | 11   | 20     | 2.5666666666666667 |
| 2014 | 10    | 30  | 811      | 21        | 1038     | 29        | AS      | N433AS  | 608    | SEA    | LAS  | 127      | 867      | 8    | 11     | 2.1166666666666667 |
| 2014 | 11    | 12  | 2346     | -4        | 217      | -28       | AS      | N765AS  | 121    | SEA    | ANC  | 183      | 1448     | 23   | 46     | 3.05               |
| 2014 | 10    | 31  | 1314     | 89        | 1544     | 111       | AS      | N713AS  | 306    | SEA    | SFO  | 129      | 679      | 13   | 14     | 2.15               |
| 2014 | 1     | 29  | 2009     | 3         | 2159     | 9         | UA      | N27205  | 1458   | PDX    | SFO  | 90       | 550      | 20   | 9      | 1.5                |
| 2014 | 12    | 17  | 2015     | 50        | 2150     | 41        | AS      | N626AS  | 368    | SEA    | SMF  | 76       | 605      | 20   | 15     | 1.2666666666666666 |
| 2014 | 8     | 11  | 1017     | -3        | 1613     | -7        | WN      | N8634A  | 827    | SEA    | MDW  | 216      | 1733     | 10   | 17     | 3.6                |
| 2014 | 1     | 13  | 2156     | -9        | 607      | -15       | AS      | N597AS  | 24     | SEA    | BOS  | 290      | 2496     | 21   | 56     | 4.833333333333333  |
| 2014 | 6     | 5   | 1733     | -12       | 1945     | -10       | OO      | N215AG  | 3488   | PDX    | BUR  | 111      | 817      | 17   | 33     | 1.85               |

only showing top 20 rows

## ❖ PySpark Example: Working with csv file

### ▼ Assign tables

```
[ ] 1 flights_df.createOrReplaceTempView('flights')
```

```
[ ] 1 spark.catalog.listDatabases()
```

```
[Database(name='default', catalog='spark_catalog', description='default database', locationUri='file:/content/spark-warehouse')]
```

```
[ ] 1 spark.catalog.listTables()
```

```
[Table(name='airports', catalog=None, namespace=[], description=None, tableType='TEMPORARY', isTemporary=True),  
Table(name='flights', catalog=None, namespace=[], description=None, tableType='TEMPORARY', isTemporary=True)]
```

### Assign DataFrame to Database



## ❖ PySpark Example: Working with csv file

```
1 flights_table = spark.table('flights')
2 print(flights_table.show())
```

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute | duration_hours     |
|------|-------|-----|----------|-----------|----------|-----------|---------|---------|--------|--------|------|----------|----------|------|--------|--------------------|
| 2014 | 12    | 8   | 658      | -7        | 935      | -5        | VX      | N846VA  | 1780   | SEA    | LAX  | 132      | 954      | 6    | 58     | 2.2                |
| 2014 | 1     | 22  | 1040     | 5         | 1505     | 5         | AS      | N559AS  | 851    | SEA    | HNL  | 360      | 2677     | 10   | 40     | 6.0                |
| 2014 | 3     | 9   | 1443     | -2        | 1652     | 2         | VX      | N847VA  | 755    | SEA    | SFO  | 111      | 679      | 14   | 43     | 1.85               |
| 2014 | 4     | 9   | 1705     | 45        | 1839     | 34        | WN      | N360SW  | 344    | PDX    | SJC  | 83       | 569      | 17   | 5      | 1.3833333333333333 |
| 2014 | 3     | 9   | 754      | -1        | 1015     | 1         | AS      | N612AS  | 522    | SEA    | BUR  | 127      | 937      | 7    | 54     | 2.1666666666666667 |
| 2014 | 1     | 15  | 1037     | 7         | 1352     | 2         | WN      | N646SW  | 48     | PDX    | DEN  | 121      | 991      | 10   | 37     | 2.0166666666666666 |
| 2014 | 7     | 2   | 847      | 42        | 1041     | 51        | WN      | N422WN  | 1520   | PDX    | OAK  | 90       | 543      | 8    | 47     | 1.5                |
| 2014 | 5     | 12  | 1655     | -5        | 1842     | -18       | VX      | N361VA  | 755    | SEA    | SFO  | 98       | 679      | 16   | 55     | 1.6333333333333333 |
| 2014 | 4     | 19  | 1236     | -4        | 1508     | -7        | AS      | N309AS  | 490    | SEA    | SAN  | 135      | 1050     | 12   | 36     | 2.25               |
| 2014 | 11    | 19  | 1812     | -3        | 2352     | -4        | AS      | N564AS  | 26     | SEA    | ORD  | 198      | 1721     | 18   | 12     | 3.3                |
| 2014 | 11    | 8   | 1653     | -2        | 1924     | -1        | AS      | N323AS  | 448    | SEA    | LAX  | 130      | 954      | 16   | 53     | 2.1666666666666665 |
| 2014 | 8     | 3   | 1120     | 0         | 1415     | 2         | AS      | N305AS  | 656    | SEA    | PHX  | 154      | 1107     | 11   | 20     | 2.5666666666666667 |
| 2014 | 10    | 30  | 811      | 21        | 1038     | 29        | AS      | N433AS  | 608    | SEA    | LAS  | 127      | 867      | 8    | 11     | 2.1166666666666667 |
| 2014 | 11    | 12  | 2346     | -4        | 217      | -28       | AS      | N765AS  | 121    | SEA    | ANC  | 183      | 1448     | 23   | 46     | 3.05               |
| 2014 | 10    | 31  | 1314     | 89        | 1544     | 111       | AS      | N713AS  | 306    | SEA    | SFO  | 129      | 679      | 13   | 14     | 2.15               |
| 2014 | 1     | 29  | 2009     | 3         | 2159     | 9         | UA      | N27205  | 1458   | PDX    | SFO  | 90       | 550      | 20   | 9      | 1.5                |
| 2014 | 12    | 17  | 2015     | 50        | 2150     | 41        | AS      | N626AS  | 368    | SEA    | SMF  | 76       | 605      | 20   | 15     | 1.2666666666666666 |
| 2014 | 8     | 11  | 1017     | -3        | 1613     | -7        | WN      | N8634A  | 827    | SEA    | MDW  | 216      | 1733     | 10   | 17     | 3.6                |
| 2014 | 1     | 13  | 2156     | -9        | 607      | -15       | AS      | N597AS  | 24     | SEA    | BOS  | 290      | 2496     | 21   | 56     | 4.833333333333333  |
| 2014 | 6     | 5   | 1733     | -12       | 1945     | -10       | OO      | N215AG  | 3488   | PDX    | BUR  | 111      | 817      | 17   | 33     | 1.85               |

only showing top 20 rows

Call the datatable

## ❖ PySpark Example: Working with csv file

```
1 flights_table.filter(flights_table.duration_hours > 5).show()
```

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute | duration_hours    |
|------|-------|-----|----------|-----------|----------|-----------|---------|---------|--------|--------|------|----------|----------|------|--------|-------------------|
| 2014 | 1     | 22  | 1040     | 5         | 1505     | 5         | AS      | N559AS  | 851    | SEA    | HNL  | 360      | 2677     | 10   | 40     | 6.0               |
| 2014 | 12    | 4   | 954      | -6        | 1348     | -17       | HA      | N395HA  | 29     | SEA    | OGG  | 333      | 2640     | 9    | 54     | 5.55              |
| 2014 | 6     | 7   | 1823     | -7        | 2112     | -28       | AS      | N512AS  | 815    | SEA    | LIH  | 335      | 2701     | 18   | 23     | 5.583333333333333 |
| 2014 | 4     | 30  | 801      | 1         | 1757     | 90        | AS      | N407AS  | 18     | SEA    | MCO  | 342      | 2554     | 8    | 1      | 5.7               |
| 2014 | 4     | 1   | 1010     | -5        | 1258     | -17       | HA      | N381HA  | 25     | PDX    | HNL  | 328      | 2603     | 10   | 10     | 5.466666666666667 |
| 2014 | 7     | 27  | 925      | 25        | 1232     | 27        | AS      | N513AS  | 875    | SEA    | LIH  | 343      | 2701     | 9    | 25     | 5.716666666666667 |
| 2014 | 8     | 18  | 2116     | -4        | 541      | 18        | B6      | N536JB  | 264    | SEA    | JFK  | 304      | 2422     | 21   | 16     | 5.066666666666667 |
| 2014 | 11    | 3   | 840      | 0         | 1703     | -17       | AS      | N462AS  | 38     | SEA    | FLL  | 307      | 2717     | 8    | 40     | 5.116666666666667 |
| 2014 | 9     | 2   | 705      | -5        | 950      | -15       | AS      | N442AS  | 833    | PDX    | HNL  | 326      | 2603     | 7    | 5      | 5.433333333333334 |
| 2014 | 12    | 28  | 1705     | -10       | 2046     | -42       | AS      | N517AS  | 863    | PDX    | OGG  | 324      | 2562     | 17   | 5      | 5.4               |
| 2014 | 4     | 10  | 829      | -1        | 1128     | -7        | AS      | N589AS  | 861    | SEA    | OGG  | 332      | 2640     | 8    | 29     | 5.533333333333333 |
| 2014 | 9     | 19  | 1039     | -1        | 1335     | -6        | AS      | N431AS  | 843    | SEA    | KOA  | 332      | 2688     | 10   | 39     | 5.533333333333333 |
| 2014 | 9     | 28  | 657      | -3        | 1557     | 34        | DL      | N704X   | 2588   | SEA    | JFK  | 311      | 2422     | 6    | 57     | 5.183333333333334 |
| 2014 | 10    | 13  | 853      | -2        | 1217     | 16        | AS      | N517AS  | 879    | SEA    | LIH  | 358      | 2701     | 8    | 53     | 5.966666666666667 |
| 2014 | 5     | 2   | 1022     | -3        | 1408     | 38        | AS      | N508AS  | 843    | SEA    | KOA  | 378      | 2688     | 10   | 22     | 6.3               |
| 2014 | 3     | 30  | 1646     | 6         | 1938     | -17       | AS      | N537AS  | 867    | SEA    | OGG  | 332      | 2640     | 16   | 46     | 5.533333333333333 |
| 2014 | 5     | 29  | 705      | -3        | 1547     | 15        | DL      | N3750D  | 400    | PDX    | JFK  | 311      | 2454     | 7    | 5      | 5.183333333333334 |
| 2014 | 3     | 22  | 1007     | -8        | 1305     | -20       | HA      | N389HA  | 25     | PDX    | HNL  | 338      | 2603     | 10   | 7      | 5.633333333333334 |
| 2014 | 8     | 16  | 1032     | -8        | 1315     | -20       | HA      | N395HA  | 25     | PDX    | HNL  | 304      | 2603     | 10   | 32     | 5.066666666666667 |
| 2014 | 4     | 4   | 1008     | -7        | 1248     | -27       | HA      | N385HA  | 25     | PDX    | HNL  | 320      | 2603     | 10   | 8      | 5.333333333333333 |

only showing top 20 rows

Perform filtering: Using function or SQL Query

## ◆ PySpark Example: Working with csv file

```
1 spark.sql('SELECT * FROM flights WHERE duration_hours > 5').show()
```

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute | duration_hours    |
|------|-------|-----|----------|-----------|----------|-----------|---------|---------|--------|--------|------|----------|----------|------|--------|-------------------|
| 2014 | 1     | 22  | 1040     | 5         | 1505     | 5         | AS      | N559AS  | 851    | SEA    | HNL  | 360      | 2677     | 10   | 40     | 6.0               |
| 2014 | 12    | 4   | 954      | -6        | 1348     | -17       | HA      | N395HA  | 29     | SEA    | OGG  | 333      | 2640     | 9    | 54     | 5.55              |
| 2014 | 6     | 7   | 1823     | -7        | 2112     | -28       | AS      | N512AS  | 815    | SEA    | LIH  | 335      | 2701     | 18   | 23     | 5.583333333333333 |
| 2014 | 4     | 30  | 801      | 1         | 1757     | 90        | AS      | N407AS  | 18     | SEA    | MCO  | 342      | 2554     | 8    | 1      | 5.7               |
| 2014 | 4     | 1   | 1010     | -5        | 1258     | -17       | HA      | N381HA  | 25     | PDX    | HNL  | 328      | 2603     | 10   | 10     | 5.466666666666667 |
| 2014 | 7     | 27  | 925      | 25        | 1232     | 27        | AS      | N513AS  | 875    | SEA    | LIH  | 343      | 2701     | 9    | 25     | 5.716666666666667 |
| 2014 | 8     | 18  | 2116     | -4        | 541      | 18        | B6      | N536JB  | 264    | SEA    | JFK  | 304      | 2422     | 21   | 16     | 5.066666666666667 |
| 2014 | 11    | 3   | 840      | 0         | 1703     | -17       | AS      | N462AS  | 38     | SEA    | FLL  | 307      | 2717     | 8    | 40     | 5.116666666666667 |
| 2014 | 9     | 2   | 705      | -5        | 950      | -15       | AS      | N442AS  | 833    | PDX    | HNL  | 326      | 2603     | 7    | 5      | 5.433333333333334 |
| 2014 | 12    | 28  | 1705     | -10       | 2046     | -42       | AS      | N517AS  | 863    | PDX    | OGG  | 324      | 2562     | 17   | 5      | 5.4               |
| 2014 | 4     | 10  | 829      | -1        | 1128     | -7        | AS      | N589AS  | 861    | SEA    | OGG  | 332      | 2640     | 8    | 29     | 5.533333333333333 |
| 2014 | 9     | 19  | 1039     | -1        | 1335     | -6        | AS      | N431AS  | 843    | SEA    | KOA  | 332      | 2688     | 10   | 39     | 5.533333333333333 |
| 2014 | 9     | 28  | 657      | -3        | 1557     | 34        | DL      | N704X   | 2588   | SEA    | JFK  | 311      | 2422     | 6    | 57     | 5.183333333333334 |
| 2014 | 10    | 13  | 853      | -2        | 1217     | 16        | AS      | N517AS  | 879    | SEA    | LIH  | 358      | 2701     | 8    | 53     | 5.966666666666667 |
| 2014 | 5     | 2   | 1022     | -3        | 1408     | 38        | AS      | N508AS  | 843    | SEA    | KOA  | 378      | 2688     | 10   | 22     | 6.3               |
| 2014 | 3     | 30  | 1646     | 6         | 1938     | -17       | AS      | N537AS  | 867    | SEA    | OGG  | 332      | 2640     | 16   | 46     | 5.533333333333333 |
| 2014 | 5     | 29  | 705      | -3        | 1547     | 15        | DL      | N3750D  | 400    | PDX    | JFK  | 311      | 2454     | 7    | 5      | 5.183333333333334 |
| 2014 | 3     | 22  | 1007     | -8        | 1305     | -20       | HA      | N389HA  | 25     | PDX    | HNL  | 338      | 2603     | 10   | 7      | 5.633333333333334 |
| 2014 | 8     | 16  | 1032     | -8        | 1315     | -20       | HA      | N395HA  | 25     | PDX    | HNL  | 304      | 2603     | 10   | 32     | 5.066666666666667 |
| 2014 | 4     | 4   | 1008     | -7        | 1248     | -27       | HA      | N385HA  | 25     | PDX    | HNL  | 320      | 2603     | 10   | 8      | 5.333333333333333 |

only showing top 20 rows

Perform filtering: Using function or SQL Query



# Summarization



In this session, we will discuss about:

- Introduction to big data.
- Introduction to Hadoop.
- How to install virtual machine.
- How to install and use Hadoop on virtual machine.
- Introduction to spark and pyspark.

# Question

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